

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S
IDENTIFICATION OF EXPERT WITNESSES**

Pursuant to the Procedural Schedule (Order No. 6), Complainant GlobalFoundries U.S. Inc. (“GlobalFoundries”) identifies the following individuals that GlobalFoundries currently believes may testify on its behalf in this Investigation and their expected areas of expertise, based on the evidence produced to date. As this Investigation is ongoing, GlobalFoundries reserves the right to identify additional expert witnesses and/or expected areas of testimony as further information becomes available, including in response to Respondents’ expert disclosures and expert reports. GlobalFoundries also reserves the right to call any expert witness that a party deposes, any expert identified or called to testify at the hearing by Respondents, and/or any expert for impeachment purposes, even if not identified below. Attached hereto as **Exhibits A-K** are copies of the experts’ curricula vitae.

1. John C. Bravman, Ph.D.

Dr. Bravman is an expert in the fields of electrical engineering and computer engineering, particularly semiconductor devices, manufacturing, processes, design, and transistors. A copy of Dr. Bravman’s curriculum vitae is attached hereto as Exhibit A. Dr. Bravman received a Bachelor of Science in Materials Science and Engineering from Stanford University in 1979 and a Master

of Science and Ph.D. in Materials Science and Engineering from Stanford University in 1984. Dr. Bravman then began teaching as a Professor of Materials Science and Engineering at Stanford University, and served as the Chairman of the Department of Materials Science and Engineering from 1996-1999. He was also the Director of Stanford's Center for Material Research for 1997-1999. From 2001-2010, Dr. Bravman was also a Professor, by courtesy, of Electrical Engineering. In 2010, Dr. Bravman became the President, as well as a Professor of Electrical and Computer Engineering, at Bucknell University. In this Investigation, Dr. Bravman may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Bravman may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

2. Jack C. Lee, Ph.D.

Dr. Jack C. Lee is an expert in the field of semiconductor design and process technology and has forty years of first-hand experience as a researcher, educator, and consultant in this field. A copy of Dr. Lee's curriculum vitae is attached hereto as Exhibit **B**. Dr. Lee has a B.S. degree in Electrical Engineering, with highest honors, in 1980, and an M.S. degree in Electrical Engineering in 1981, both from University of California, Los Angeles. Dr. Lee received a Ph.D. degree in

Electrical Engineering in 1988 from University of California, Berkeley (“UC Berkeley”). After receiving his Ph.D. in August 1988, Dr. Lee joined the faculty at The University of Texas at Austin (“UT Austin”). As a faculty member, Dr. Lee has taught numerous courses in semiconductor device fabrication and design, at both the undergraduate and graduate levels. He has supervised over 40 students who received their doctoral degrees under his guidance. Since September 2000, he has held the Cullen Trust for Higher Education Endowed Professor in Engineering in the Department of Electrical and Computer Engineering at UT Austin. In September 2023, he was promoted to the Professor Emeritus level. His current research and teaching interests encompass semiconductor devices including 3D structures such as FinFETs or tri-gate MOSFETs, high-K gate dielectrics and gate electrodes, fabrication processes, electrical and material characterization, reliability and modeling, packaging technologies, dielectric and passivation processes, display technologies, photovoltaic and solar cell technologies, semiconductor memory applications, and alternative channel materials. In this Investigation, Dr. Lee may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products’ practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Lee may also be called to rebut the testimony of Respondents’ experts or fact witnesses, should any be presented, regarding any of the areas described above.

3. David Kuan-Yu Liu, Ph.D.

Dr. Liu is an expert in the fields of electrical engineering and computer engineering, particularly semiconductor devices, manufacturing, processes, design, and transistors. A copy of Dr. Liu's curriculum vitae is attached hereto as Exhibit C. Dr. David Kuan-Yu Liu received a Bachelor of Science in Electrical Engineering from the University of California, Berkeley, in 1983. He received a Master of Science and Ph.D. in Electrical Engineering from Stanford University in 1985 and 1989, respectively. From 1989 to 1992, he was a member of technical staff at Texas Instruments, Inc. ("TI"). At TI, his job responsibilities included process integration, device modeling, high-voltage complementary metal oxide semiconductor (CMOS) process integration, and investigating novel source-side injection mechanisms for Flash EPROM channel hot-electron programming. From 1992 to 1995, he was a member of the technical staff at Advanced Micro Devices, Inc. ("AMD") where he was a key contributor in optimizing Flash cell and periphery devices in AMD's CMOS-based 0.5 μ m and 0.35 μ m Flash EPROM technology. His job responsibilities at AMD also included process integration, device modeling, and development of triple-well process technology to accommodate x-decoder transistors and high-voltage transistors for negative gate erase operations. While at AMD, he was awarded a Spotlight Award for developing a method of manufacturing a self-aligned source (SAS) etch for a NOR flash memory. He spent the next five years in managerial and director roles at several California-based semiconductor companies, where he was responsible for increasing yield and for leading teams of engineers working to develop next generation memory devices. In 2000, he co-founded Progressant Technologies in Fremont, California. Progressant Technologies developed IP for negative differential resistance transistor technology and was eventually acquired by Synopsys, Inc. From 2000 to 2004, he was a Senior Manager at Xilinx, Incorporated, where he was responsible for developing nonvolatile memory process technology for flash and CPLD product

applications, as well as advanced CMOS process technology (specifically 75nm CMOS technology node, a half node version between 90nm and 65nm). From 2004 to 2007, he was a Senior Scientist at Maxim Integrated Products where he was responsible for developing Embedded Non-volatile Memory process technology for Power Management product applications. Since 2007, he has served as a technical consultant, providing expert advice regarding semiconductor memory technologies, including CMOS process technology, and semiconductor device physics. In this Investigation, Dr. Liu may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Liu may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

4. Joseph C. McAlexander, Ph.D.

Dr. McAlexander is an expert in the fields of electrical engineering and computer engineering, particularly semiconductor manufacturing, design, transistors, and digital circuits. A copy of Dr. McAlexander's curriculum vitae is attached hereto as Exhibit **D**. Dr. McAlexander received his Bachelor of Science in Electrical and Electronics Engineering in 1969 from North Carolina State University. Dr. McAlexander has decades of experience with microcircuit and semiconductor technologies in the areas of circuit design and analysis, device fabrication and

assembly, testing, marketing, control system design and analysis, manufacturing operations and respective areas of quality, reliability, and defect / failure analysis. From 1972 to 1986, he worked at Texas Instruments, Inc. ("TI") as TI's Houston and Singapore operations in various roles, including Design Section Manager and Engineer, Product Engineering Manager, Engineering Operations Manager, Quality/Reliability Assurance and Engineering Manager, and Quality/Reliability Assurance Manager. From 1986 to 1990, Dr. McAlexander was Executive Vice President and Company Officer of EPI Technologies, Inc. From 1991 to 2002, he was a Senior Managing Consultant at Cochran Consulting, Inc. and since 1988, he has served as President of McAlexander Sound, Inc. Dr. McAlexander is also a senior member of the Institute of Electrical and Electronics Engineers, Inc. In this Investigation, Dr. McAlexander may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. McAlexander may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

5. Carla Mulhern, M.Sc.

Ms. Mulhern is an expert in the field of economics, particularly as it relates to intellectual property issues, including those before the U.S. International Trade Commission. A copy of Ms.

Mulhern's curriculum vitae is attached hereto as Exhibit E. She received a bachelor's degree in mathematics from Bucknell University in 1986, a diploma with merit in economics from the London School of Economics and Political science in 1987, and a M.Sc. in Economics from the London School of Economics and Political Science in 1988. Since 1997, she has been a Senior Advisor at Analysis Group, Inc. in Washington, D.C. Ms. Mulhern may be called on to testify regarding matters relating to the economic prong of the domestic industry; importation issues; and/or the remedy and bond sought by GlobalFoundries in the Investigation. Ms. Mulhern may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

6. Massoud Pedram, Ph.D.

Dr. Massoud Pedram is a Professor of Electrical Engineering and the Charles Lee Powell Chair in Electrical Engineering and Computer Science in the Viterbi School of Engineering at the University of Southern California ("USC"). A copy of Dr. Pedram's curriculum vitae is attached hereto as Exhibit F. He is an expert on Very Large Scale Integration ("VLSI") design including parallel computing, distributed processing, high-speed design, power efficient design, digital integrated circuits, and semiconductor memory. He also has expertise in neural networks and graphic processing units, as well as system-on-chip designs targeting applications ranging from high-performance computing to low-power embedded computing. His current research interests are in, among others, computer-aided design methodologies and tools for CMOS circuits and systems, energy-efficient computing, computer architecture and hardware, parallel and distributed computing, neural network acceleration, and integrated circuit and system design including system-on-chip design. He earned his B.S. in Electrical Engineering from the California Institute of Technology in 1986, and his M.S. and Ph.D. in Electrical Engineering and Computer Sciences from the University of California at Berkeley in 1989 and 1991, respectively. In this Investigation,

Dr. Pedram may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Pedram may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

7. Udit Sharma, M.S.

Mr. Sharma is an expert in the field of materials engineering and analysis, including Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Scanning TEM (STEM), Electron Energy Loss Spectroscopy (EELS), Energy Dispersive Spectroscopy (EDS), Scanning Capacitance Microscopy (SCM), and Focused Ion Beam (FIB) techniques. A copy of Mr. Sharma's curriculum vitae is attached hereto as Exhibit G. Mr. Sharma received a Master of Science in Materials Engineering from the University of Alberta, Canada in 2000 and a Bachelor of Technology in Materials Engineering from the Indian Institute of Technology, India in 1997. He also received a Diploma in Biotechnology from Vitesse, National Research Council Canada in partnership with the University of Ottawa and Carleton University in 2003. Since 2022, Mr. Sharma has been a Scientific Fellow at Eurofins EAG Laboratories, where he consults, plans, and manages multi-technique analysis projects. From 2016 to 2022, he served as a Project Manager, Materials Sciences at Eurofins EAG Laboratories, where he was responsible for technical accuracy

of projects and managed accounts. From 2008 to 2016, he was a Senior Engineer at EAG (Evans Analytical Group), where he performed TEM/STEM/EELS/EDS/SEM/EBSD characterization of a wide variety of advanced materials systems and devices, including microelectronic devices, FinFETs, MRAMs, and Si devices. Prior to that, he held positions as an Electron Microscopist at Semiconductor Insights Inc. from 2004 to 2007, where he developed imaging and analysis protocols for TEM, SEM, AFM, and SCM for applications in competitive intelligence and intellectual property support, and as a TEM Analyst at Nortel Networks/Bookham Inc. from 2000 to 2003. In this Investigation, Mr. Sharma may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Mr. Sharma may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

8. S. Shukri Souri, B.A., M.A., M.S., Ph.D.

Dr. Souri is an expert in the fields of electrical engineering and computer engineering, particularly semiconductor manufacturing, design, transistors, and digital circuits. A copy of Dr. Souri's curriculum vitae is attached hereto as Exhibit H. Dr. Souri received his B.A. in Engineering Science from the University of Oxford, his M.S. and Ph.D. in Electrical Engineering from Stanford University, and his M.A. from the University of Oxford. From 1994 to 1997, he was a research

scientist in the corporate research and development division, and then a research engineer in the circuit protection division, at Raychem. He then co-founded arcadiaOne, Inc. in 1999. He also founded Merenga Inc. in 2000. He is currently the Corporate Vice President, and Practice / Office Director and Principal of Electrical Engineering and Computer Science at Exponent, Inc. He is also a member of the Institution of Engineering and Technology, and a senior member of the Institute of Electrical and Electronic Engineers. In this Investigation, Dr. Sourì may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Sourì may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

9. Vivek Subramanian, Ph.D.

Dr. Subramanian is an expert in the fields of electrical engineering and computer engineering, particularly semiconductor manufacturing, design, transistors, and digital circuits. A copy of Dr. Subramanian's curriculum vitae is attached hereto as Exhibit I. Dr. Subramanian received his B.S. in Electrical Engineering from Louisiana State University, and his M.S. and Ph.D. in Electrical Engineering from Stanford University. He currently serves as a Professor in the Department of Electrical Engineering and Computer Sciences at the University of California,

Berkeley, and as Director of the Institute of Electrical and Micro Engineering and the Micromanufacturing Science and Engineering Center at the École Polytechnique Fédérale De Lausanne (EPFL) in Switzerland. His early research focused on the crystallization of amorphous silicon and silicon-germanium films using low thermal budget processes. He served as Chief Technical Advisor to an energy conversion device startup company and was a founding scientific advisor to a printed electronics company. He also served as an independent consultant regarding silicon process technology. In this Investigation, Dr. Subramanian may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Subramanian may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

10. Scott E. Thompson, Ph.D.

Dr. Thompson is an expert in the fields of electrical engineering, computer engineering, and computer science, particularly semiconductor manufacturing, design, transistors, and digital circuits. A copy of Dr. Thompson's curriculum vitae is attached hereto as Exhibit J. Dr. Thompson received from the University of Florida a Bachelor of Electrical Engineering with Highest Honors in 1987, a Master of Science in Electrical Engineering in 1988, and a Ph.D. in Electrical

Engineering in 1992. From 1992 to 2004, he worked at Intel Corporation, where he became an Intel Fellow, the company's highest technical level, and Director of Computer Chip Technology. At Intel, he worked on semiconductor manufacturing, including the transfer and start-up of new manufacturing facilities worldwide. He has developed ultra-low power chip technology for image sensors, system-on-chip devices, RFICs, memory, CPUs, image-sensor processors, and other chips. Since 2004, Dr. Thompson has been a professor in the Electrical and Computer Engineering Department at the University of Florida, where he has conducted research on advanced chip technology and developed graduate-level courses on advanced level semiconductor chip, semiconductor design and packaging technology, and computer chip technology. From 2009 to 2015, he was also the founder and CTO of SuVolta, a low power chip company whose technology was used in an image-sensor processor. He was elected an IEEE Fellow in 2006, and he and his colleagues were the first to publish on a 90-nm logic technology that introduced high levels of strain using SiGe for significant mobility enhancement. In this Investigation, Dr. Thompson may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Thompson may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas

described above.

11. Shaojie Wang, Ph.D.

Dr. Wang is an expert in the field of materials engineering and analysis, including Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), Scanning TEM (STEM), Electron Energy Loss Spectroscopy (EELS), Energy Dispersive Spectroscopy (EDS), Scanning Capacitance Microscopy (SCM), and Focused Ion Beam (FIB) techniques. A copy of Dr. Wang's curriculum vitae is attached hereto as Exhibit K. Dr. Wang received a Ph.D. in Materials Science and Engineering from Lehigh University, a Master of Science in Physics from Indiana University, and a Bachelor of Science in Physics from the University of Science and Technology of China. Since 2012, Dr. Wang has been a Senior Principal Scientist in the TEM Group at Eurofins Nanolab Technologies, where he conducts analytical services using various TEM techniques (TEM/STEM/PED/EDS/EELS/EFTEM) to support a wide range of customers in semiconductor and consumer electronics, interfaces with customers in the development of new materials test plans, and is responsible for coordinating different characterization groups to provide comprehensive analysis. Prior to that, in 2012, he was a Senior Process Engineer in the TD-Contamination Free Manufacturing Group at GlobalFoundries, where he developed Bright Field Inspection, SEM Review, and Automatic Process Inspection recipes and conducted Defect Source Analysis. From 2006 to 2012, he was a Research Assistant in the Glass Research Group at Lehigh University. In this Investigation, Dr. Wang may be called on to testify regarding matters related to, at least, the technical background of the Asserted Patents; the interpretation of the claims in the Asserted Patents; issues relating to the infringement of the Asserted Patents by the Accused Products; the design, structure, function, and operation of the Accused Products; issues relating to the validity of the Asserted Patents; issues relating to the Domestic Industry Products' practice of the Asserted Patents; issues relating to whether GlobalFoundries satisfies the technical prong for the domestic

industry for the Asserted Patents; testing performed on the Domestic Industry Products and the Accused Products; and other issues in connection with the infringement, validity, enforceability, and other technical issues that may arise with respect to the Asserted Patents. Dr. Wang may also be called to rebut the testimony of Respondents' experts or fact witnesses, should any be presented, regarding any of the areas described above.

Dated: August 21, 2026

Respectfully submitted,

/s/ Bert C. Reiser

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*Counsel for Complainant
GlobalFoundries U.S. Inc.*

CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S IDENTIFICATION OF EXPERT WITNESSES** were served on August 21, 2026 as follows:

The Honorable Lisa R. Barton Secretary to the Commission U.S. International Trade Commission 500 E Street, SW, Room 112 Washington, DC 20436	BY EDIS
The Honorable Clark S. Cheney Chief Administrative Law Judge U.S. International Trade Commission 500 E Street, SW, Room 317 Washington, DC 20436	By Email and Box: Cheney1500@usitc.gov
Stephen R. Smith COOLEY LLP 1299 Pennsylvania Ave., N.W. Suite 700 Washington, D.C. 20004 <i>Counsel for Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC</i>	By Email: TowerITC@cooley.com

/s/ Erika J. Weinstein

Erika J. Weinstein

CLEARY GOTTlieb STEEN & HAMILTON LLP

EXHIBIT A

– John C. Bravman –

President, and Professor of Electrical & Computer Engineering, Bucknell University
Bing Centennial Professor Emeritus, Materials Science & Engineering, Stanford University
Freeman–Thornton Vice Provost, Emeritus, Stanford University

Research Interests & Expertise

- Integrated circuit process technology
- IC failure processes and phenomena
- Mechanical behavior of thin films
- High spatial resolution materials analysis
- Microelectronic packaging
- Superconductivity

Education

- | | |
|-------------|---|
| 1975 – 1979 | B.S., Materials Science and Engineering, Stanford University |
| 1979 – 1984 | M.S., Ph.D., Materials Science and Engineering, Stanford University |

Professional Career

- | | |
|----------------|---|
| 1979 – 1984 | Research Engineer, Fairchild Semiconductor Research Laboratory, Palo Alto, CA |
| 1985 – 1991 | Assistant Professor, Materials Science & Engineering, Stanford University |
| 1991 – 1995 | Associate Professor, Materials Science & Engineering, Stanford University |
| 1991 – 1995 | Associate Chairman, Materials Science & Engineering, Stanford University |
| 1992 – 1993 | Associate Dean, School of Engineering, Stanford University |
| 1993 – 2001 | Senior Associate Dean, School of Engineering, Stanford University |
| 1995 – 1997 | Professor, Dept. of Materials Science & Engineering, Stanford University |
| 1997 – 2010 | Bing Centennial Professor, Materials Science & Engineering, Stanford University |
| 1996 – 1999 | Chairman, Dept. of Materials Science & Engineering, Stanford University |
| 1998 – 1999 | Director, Center for Materials Research, Stanford University |
| 1999 – 2010 | Founder and Dean of the Freshman/Sophomore College, Stanford University |
| 1999 – 2010 | Freeman Thornton Vice Provost for Undergraduate Education, Stanford Univ. |
| 2001 | Interim Vice Provost for Student Affairs, Stanford University |
| 2001 – 2010 | Professor, by courtesy, Electrical Engineering, Stanford University |
| 2010 – present | Bing Centennial Professor of Materials Science and Freeman Thornton Vice Provost for Undergraduate Education, Emeritus, Stanford University |
| 2010 – present | President, and Professor of Electrical Engineering, Bucknell University |
| 2012 – present | Board of Directors, Geisinger Health System; Chairman, 2017 – 2023 |
| 2024 – present | Board of Directors, Risant Health |

Honors and Awards

1987	School of Engineering Distinguished Advisor Award, Stanford University
1988	Excellence in Teaching Award, Society of Black Scientists & Engineers
1989	Walter J. Gores Award for Excellence in Teaching, Stanford University's highest award for teaching
1990	Tau Beta Pi Engineering Honor Society Award for Excellence in Undergraduate Engineering Teaching, Stanford University
1991	Bradley Stoughton Award, ASM International, a national teaching award for scientists under the age of 35.
1992 – 1995	Bing Teaching Fellow, Stanford University
1992	Excellence in Teaching Award, Society of Women Engineers, Stanford Univ.
1994	President, Materials Research Society
1995 – 1997	University Fellow, Stanford University
1997	Bing Centennial Professor, Stanford University
2001	Stanford Associates Award for Service to the Alumni Association
2004	Woody White Award, Materials Research Society
2010	Kenneth M. Cuthbertson Award for Exceptional Service, Stanford University's highest award for contributions to the university.
2022	Modern Healthcare's Excellence in Healthcare Governance, Class of 2022
2024	President's Star Award, Posse Foundation

External Service and Scholarly Activities

1986 – 1987	Symposium Organizer, 1987 Fall Meeting of the Materials Research Society; "Workshop on Specimen Preparation for Transmission Electron Microscopy of Materials"
1987 – 1988	Symposium Organizer, 1988 Fall Meeting of the Materials Research Society; "Thin Films: Stresses and Mechanical Properties"
1988 – 1989	Organizing Committee, International Conference on Materials and Mechanisms of Superconductivity – High Temperature Superconductivity
1988 – 1989	Symposium Organizer, 1989 Meeting of the Electron Microscopy Society of America: "Structure and Reactivity of Metal – Semiconductor Interfaces"
1989 – 1990	Conference Chairman, 1990 Spring Meeting of the Materials Research Society
1989 – 1990	Symposium Organizer, 1990 Spring Meeting of the Materials Research Society: "Laser Ablation for Thin Film Synthesis"
1989 – 1990	Symposium Organizer, 1990 Spring Meeting of the Materials Research Society: "Frontiers of Materials Science"
1990 – 1991	Symposium Organizer, 1991 Fall Meeting of the Materials Research Society; "Thin Films: Stresses and Mechanical Properties III"
1990 – 1991	Symposium Organizer, 1991 Fall Meeting of the Materials Research Society; "Workshop on Specimen Preparation for Transmission Electron Microscopy of Materials III"
1991 – 1993	Council Member, Materials Research Society

1991 – 1992	Symposium Organizer, 1992 Spring Meeting of the Materials Research Society: “Structure and Defects in Electronic Oxides”
1991 – 1996	Technical Editorial Board, Materials Research Society Bulletin
1992	Second Vice President, Materials Research Society
1992 – 1998	Editorial Board, Annual Reviews of Materials Science
1992	Guest Editor, July 1992 Issue of the Materials Research Society Bulletin; “Mechanical Properties of Thin Film Materials”
1993	First Vice President, Materials Research Society
1993 – 1995	Selection Committee, Bradley Stoughton Award, ASM International
1993, 2001	Member, National Advisory Board, Materials Science & Technology Programs, Sandia National Laboratories
1994	President, Materials Research Society
1994 – 1995	Conference Organizer, 3 rd International Workshop on Stress Induced Phenomena, Stanford, CA
1995 – 1998	Solid States Sciences Committee of the National Research Council
1996 – 1999	Awards Committee, ASM International
1996 – 1999	Chair, Program Development Subcommittee, Materials Research Society
1996 – 1999	Member, Electronics and Photonics Program National Advisory Board, Sandia National Laboratories
1997 – 1998	Symposium Organizer, 1998 Spring Meeting of the Materials Research Society, “Materials Reliability in Microelectronics VIII”
1997 – 1998	Conference Organizer, 15 th Biennial Conference on National Materials Policy, Washington, D.C.
1998 – 1999	Program Committee, 5 th International Workshop on Stress Induced Phenomena, Stuttgart, Germany
1998 – 1999	Chair, Long Range Planning Committee, Materials Research Society
1998 – 2001	Member, Editorial Board, Applied Physics Reviews
1998 – 2006	Associate Editor, Annual Reviews of Materials Science
2000 – 2004	Chair, Program Committee, Materials Research Society
2001	Member, Middle States Association Accreditation Review Panel, Cornell
2001	Member, Middle States Association Accreditation Review Panel, University of Pennsylvania
2002 – 2008	Visiting Committee, Massachusetts Institute of Technology
2002 – 2006	Member, Director’s Review Board, Chemistry and Materials Science, Lawrence Livermore National Laboratory.
2005 – 2009	Board of Directors, LEAD Program for Underserved Youth
2006 – 2010	Member, Advisory Board, Journal of Engineering Education
2010 – 2013	Member, Board of Directors, Council of Higher Education Accreditation
2010 – 2020	Member, Selection Committee for the Clare Booth Luce Awards, Henry Luce Foundation

2011 – 2012	Chair, Publications Review Committee, Materials Research Society
2012 – 2013	Chair, Middle States Association Accreditation Review Panel, Marist College
2012 – present	Member, Board of Directors, Geisinger Health; Chairman, 2017 – 2023
2015 – 2016	Chair, Middle States Association Accreditation Review Panel, United States Naval Academy
2017 – 2018	Chair, Middle States Association Accreditation Review Panel, Colgate University
2023 – present	Member, Division I Board of Directors, NCAA
2024 – present	Member, United States Council on Competitiveness
2024 – present	Member, Board of Directors, Risant Health

Stanford University Service

1986 – 2010	Freshman Advisor
1991 – 1992	Deans' Advisory Group, Libraries & Information Resources
1991	Faculty Advisor, Freshman Alcohol Responsibility & Management Program
1991 – 1995	Committee on Residential Computing
1991 – 1997	Steering Committee, Continuing Education Program
1992	Faculty Expenditure Certification Process Task Force
1992 – 1995	Faculty Resident Fellow, Yost House Undergraduate Residence
1992	Search Committee, Director of Environmental Health and Safety
1992 – 1996	Chair, University Committee on Health and Safety
1993 – 1999	Chair, C-AAA Subcommittee on University and Departmental Honors
1993	Search Committee, Director of Residential Education
1993 – 1997	Faculty Senate of the Academic Council
1993 – 1999	Steering Committee, Center for Teaching and Learning
1993 – 1994	President's Commission on Undergraduate Education at Stanford; Chair of Subcommittee on Technology in Teaching and Learning
1993 – 1995	C-ASIS Subcom. on Distributed Information & Computing Environment
1994 – 1995	President's Commission on Technology in Teaching and Learning
1994 – 1997	Board of Directors, Faculty Club
1994 – 2003	C-AAA Subcommittee on Academic Standing
1995 – 2004	Faculty Member, Stanford Sophomore College
1995	Search Committee, Dean of Students
1995 – 1996	Faculty Senate Committee on Committees
1995 – 1997	Design Committee for Cultures, Ideas & Values Program
1996	Search Committee, Director of Career Planning and Placement Center
1996 – 1998	Stanford University Board of Trustees Committee on Development
1996 – 1999	Writing Advisory Board
1996 – 2010	University Undergraduate Advisory Council
1996 – 1997	Residences Task Force
1996 – 1999	Executive Committee, Center for Materials Research
1997	Search Committee, Dean of Admissions & Financial Aid

1997 – 1999	Residence Programs Implementation Group
1998 – 2001	Stanford University Board of Trustees Committee on Nominations
1998 – 1999	Director, Center for Materials Research
1999	Elected Chair, Faculty Senate of the Academic Council
1999 – 2010	Dean of the Freshman/Sophomore College
1999 – 2010	Freeman–Thornton Vice Provost for Undergraduate Education
2000	Co–Chair, Search Committee, Dean of Admissions & Financial Aid
2000 – 2001	Provost’s Task Force, University Needs Assessment
2000 – 2003	Strategic Planning Board, Cantor Center for the Visual Arts
2001	Search Committee, Dean of Humanities and Sciences
2001	Chair, Search Committee, Vice Provost for Student Affairs
2001	Interim Vice Provost for Student Affairs
2004 – 2005	Co–Chair, Search Committee, Dean of Admissions & Financial Aid
2004 – 2007	Dean of Potter College
2005 – 2006	Search Committee, Director of Athletics
2006	Co–Chair, Stanford Task Force Evaluating Education in the Residences
2006 – 2007	Search Committee, University Registrar
2006 – 2010	Co–Chair, WASC Reaccreditation Steering Committee

Stanford School of Engineering Service

1986 – 1990	EE–ICL Faculty Search Committees (Silicon Processing; Sensors)
1986 – 1992	Radiation Safety Committee
1987 – 1992	School of Engineering Committee on Computers
1991 – 2001	Member of Undergraduate Council
1991 – 2006	Faculty Advisor, Tau Beta Pi Engineering Honor Society
1992 – 1993	MSE–EE Faculty Search Committee (Magnetic Materials & Devices)
1992 – 1994	MSE–Chem. E. Faculty Search Committee (Comput. Materials Science)
1992 – 1993	Associate Dean for Undergraduate Affairs
1992 – 2009	Faculty Advisor, Stanford Solar Car Project
1993 – 2001	Senior Associate Dean for Student Affairs

Stanford Departmental Service

1986 – 1987	Student Faculty Liaison Committee, Dept. of Materials Science
1986 – 1987	Industrial Affiliates Program Committee, Dept. of Materials Science
1986 – 1990	Advanced Degree Committee, Dept. of Materials Science
1986 – 1990	Curriculum Committee, Dept. of Materials Science
1986 – 1991	Project Director, X–ray Diffraction Lab., Dept. of Materials Science
1986 – 1991	Chair, Computer Committee, Dept. of Materials Science
1987 – 1991	Chair, Space and Building Committee, Dept. of Materials Science
1990 – 1995	Chair, Financial Aids Committee, Dept. of Materials Science
1991 – 1995	Associate Chairman, Dept. of Materials Science and Engineering
1996 – 1999	Chairman, Department of Materials Science & Engineering

Referee

Journals	Journal of Applied Physics, Applied Physics Letters, Journal of Materials Science, Journal of the Electrochemical Society, Physical Review B, IEEE Electron Device Letters, Journal of Materials Research, Physical Review Letters, Journal of Engineering Education
Agencies	National Science Foundation, Electric Power Research Institute, Air Force Office of Scientific Research, Department of Energy, Army Research Labs, Science and Engineering Research Council (Singapore)

Professional Society Memberships

American Physical Society	American Society of Metals
Materials Research Society	The Metallurgical Society of AIME
Microscopy Society of America	Institute of Electrical & Electronic Engineers

Patents

- “Drug Delivery Systems using Mesoporous Oxide Films,” Dimitrios Pantelidis and John C. Bravman, US Patent 7,981,441, issued July 19, 2011.
- “Bioactive Material Delivery Systems Comprising Sol-Gel Compositions,” Dimitrios Pantelidis, John C. Bravman, Jonathan Rothbard, and Richard Klein, US Patent 8,097,269, issued January 17, 2012.

Consulting Activities – Technical Advisory

1985 – 1987	Fairchild Semiconductor, Palo Alto, CA. (technical consulting)
1986 – 1992	National Semiconductor, Santa Clara, CA. (technical consulting)
1987 – 1990	Intel Corporation, Santa Clara, CA. (technical consulting)
1992 – 2000	Mayfield Fund, Menlo Park, CA. (technical consulting)
1994 – 2002	Chair, National Advisory Board, Materials Science & Technology Programs, Sandia National Laboratories
1998 – 2003	Mohr, Davidow, Menlo Park, CA. (technical consulting)
2001 – 2005	Reflectivity, Inc., Sunnyvale, CA. (technical consulting)
2003 – 2007	Applied Plasma Lab, Inc., Santa Clara, CA. (Board of Directors)
2004 – 2010	Medlogics, Santa Rosa, CA. (technical consulting)
2005 – 2007	Sharefare, Los Altos, CA. (Technical Advisory Board)
2006 – 2008	HSH Nordbank, New York, NY. (technical due diligence)
2007	Virgin Fuels, London, U.K. (technical due diligence)

Consulting Activities – Intellectual Property, Trade Secret, Product Liability

1999 – 2000

Robins, Kaplan, Miller and Ciresi

- for Raychem; patent litigation, consultant, 1999 – 2000. [*completed*]

2001 – 2024

Perkins Coie

- for Semitool in Semitool *v.* Applied Materials, Novellus, and Ebara; patent litigation, testifying expert, 2001 – 2004. [Case CV 01-1066-BR (Applied Materials), [Case CV 01-874-BR (Novellus), Case CV 01-873-BR (Ebara); US District Court, District of Oregon.] [*completed*]
- for Semitool in Semitool *v.* DMS; patent litigation, testifying expert, 2004 – 2005. [Case 01-1391 WHA; US District Court, Northern District of California, San Francisco Division; filed April 9, 2001] [*completed*]
- for Micron in Ohmi *v.* Intel, IBM, Samsung and Micron; patent litigation, testifying expert, 2006 – 2007. [Case 2-05-CV-209; US District Court, Eastern District of Texas, Marshall Division; filed June 2, 2005.] [*completed*]
- for On Semiconductor in On Semiconductor *v.* Hynix Semiconductor, Elpida Memory, Nanya Technology, and Hittite Microwave; patent litigation, testifying expert, 2010. [Case 6-09-CV-390; United States District Court for the Eastern District of Texas.] [*completed*]
- for Intel in Semcon Tech LLC *v.* Intel Corporation; patent litigation, testifying expert, 2012 – 2013. [Case 12-531-RGA; United States District Court for Delaware.] [*completed*]
- for Intel, Global Foundries, and Micron in Intel Corporation, Global Foundries U.S., Inc., and Micron Technology, Inc., Petitioners, *v.* Daniel L. Flamm; IPR Petitions on U.S Patent RE40,264; 2016-2018. [*completed*]
- for STMicroelectronics, in Ocean Semiconductor LLC *v.* STMicroelectronics, Inc., patent litigation, testifying expert, 2021 – 2023. [Case 6:20-cv-01215-ADA, W.D. Texas.] [*completed*]
- for Intel, in Request for Ruling Relating to the Limited Exclusion Order Issued by the International Trade Commission in Certain Chemical Mechanical Planarization Slurries and Components Thereof, Inv. No. 337-TA-1204 [*completed*]
- for Mohith Verghese and Carl White, in (1) ASM America, Inc. (Plaintiff) *v.* Mohith Verghese and Carl White (Defendants) (Superior Court of the State of Arizona, County of Maricopa and United States District Court, District of Arizona), and (2) Mohith Verghese and Carl White (Plaintiffs) *v.* ASM America, Inc. (Defendant) (United States District Court, District of Arizona), testifying expert, 2024. [*completed*]

2003 – present

Kirkland & Ellis

- for Honeywell Electronic Materials; patent litigation; consultant, 2003. [*completed*]
- for Samsung in Ohmi *v.* Intel, IBM, Samsung and Micron; patent litigation; testifying expert, 2006 – 2007. [Case 2-05-CV-209; US District Court, Eastern District of Texas, Marshall Division; filed June 2, 2005.] [*completed*]

- for Samsung in Samsung *v.* Matsushita; patent litigation; testifying expert, 2006 – 2007. [Case 6:06-CV-154 LED; US District Court, Eastern District of Texas, Tyler Division; filed September 15, 2005.] *[completed]*
- for Agere Systems in Agere *v.* Atmel; patent litigation; testifying expert, 2006. [Case 02–CV–864; US District Court, Eastern District of Pennsylvania] *[completed]*
- for Samsung in On Semiconductor *v.* Samsung; patent litigation; testifying expert, 2008 – 2009 [Case 07–449 (JJF); US District Court, District of Delaware]; and in Samsung *v.* On Semiconductor; patent litigation; testifying expert, 2008 – 2009 [Case 06–720 (JJF); US District Court, District of Delaware]. *[completed]*
- for Samsung in Spansion *v.* Samsung; patent litigation; testifying expert, 2009 [Investigation 337–TA–664; US International Trade Commission, Washington D.C.] *[completed]*
- for Samsung in Spansion *v.* Samsung; patent litigation; testifying expert, 2010 – 2011 [Case 08–855 (SLR); US District Court, District of Delaware]. *[completed]*
- for Samsung in Spansion *v.* Samsung; patent litigation; testifying expert, 2011 – 2012 [Case No. 3:10-CV-685 (WMC); US District Court, Western District of Wisconsin]. *[completed]*
- for Intel in IP Bridge *v.* Intel, Petitioners; IPR Petitions on U.S Patents 6346736, 6424041, 6709950, 6967409, 7417289, 7893501; 2017–2019. *[completed]*
- for Intel, Acer, ASUSTeK, Lenovo, and Micro-Star International, in Tela Innovations, Inc. *v.* Intel, Acer, ASUSTeK, Lenovo, and Micro-Star International; patent litigation; testifying expert, 2019–2020 [Case 337-TA-1148; US International Trade Commission, Washington D.C.] *[completed]*
- for Applied Materials, in Demaray LLC *v.* Intel Corporation [Case No. 6:20-cv-00634-ADA (W.D. Tex.)] and [Demaray LLC *v.* Samsung Electronics Co. Ltd., Case No. 6:20-cv-00636-ADA (W.D. Tex.)] 2022 – 2023. *[completed]*
- for Applied Materials, in Applied Materials, Inc. *v.* Demaray LLC [Case 5:20-CV-9341-EJD; US District Court, Northern District of California, filed December 24, 2020].

2003**McDermott, Will & Emery**

- for Motorola/Freescale in Motorola *v.* Analog Devices; patent litigation, testifying expert, 2003. [Case 1:03-CV-0131; US District Court, Eastern District of Texas, Beaumont Division; filed March 4, 2003.] *[completed]*

2003 – 2004**Jones Day**

- for Motorola/Freescale in Motorola *v.* Analog Devices; patent litigation, testifying expert, 2003 – 2004. [Case 1:03-CV-0131; US District Court, Eastern District of Texas, Beaumont Division; filed March 4, 2003.] *[completed]*

2004 – 2005**Heller Erhman White & McAuliffe**

- for Fujitsu in *Fujitsu v. Sumitomo, Amkor, and Cirrus Logic*; product liability, testifying expert, 2004 – 2005. [Case 1-03-CV-009885; Superior Court of the State of California, County of Santa Clara; filed November 25, 2003.] [*completed*]
- for Western Digital in *Reiber et al. v. Western Digital et al.*, patent litigation, testifying expert, 2008. [Investigation No. 337-TA-616.] [*completed*]

2004 – present Paul Hastings, Janofsky & Walker

- for SMIC in *TSMC v. SMIC*; trade secrets, consultant, 2004 – 2005. [Case C-03-5761 MMC; US District Court, Northern District of California.] [*completed*]
- for Samsung Electronics in *Elm 3DS Innovations v. Samsung Electronics*; patent litigation, testifying expert, 2022 – 2023. [Case 1-14-cv-01430-LPS-CJB; US District Court, Delaware.] [*completed*]
- for GlobalFoundries in *Katana v. GlobalFoundries*, testifying expert, 2022 – 2023 [Case No. 6-22-cv-00191]

2005 – 2006 Townsend and Townsend and Crew

- for Hynix in *Toshiba v. Hynix*; patent litigation, testifying expert, 2005 – 2006. [Case 3:04-CV-04708 VRW; US District Court, Northern District of California; filed September 1, 2006.] [*completed*]

2005 – 2007 Kecker and Van Nest

- for Intel in *Ohmi v. Intel, Samsung and Micron*; patent litigation, testifying expert, 2005 – 2007. [Case 2–05CV–209; US District Court, Eastern District of Texas, Marshall Division; filed June 2, 2005.] [*completed*]
- for Sputtered Films Inc. in *SFI v. Advanced Modular Sputtering*; trade secret litigation, consultant, 2006. [Case 0113-2336; Superior Court of the State of California, Anacapa Division; filed December 23, 2003.] [*completed*]

2005 – present Fish and Richardson

- for Hynix in *Toshiba v. Hynix*; patent litigation, testifying expert, 2005 – 2006. [Investigation 337–TA–552; US International Trade Commission, Washington D.C.] [*completed*]
- for Metconnex in *JDS Uniphase v. Metconnex*; patent litigation, consultant, 2005 – 2006. [Case 05-5371-AG(SS); US District Court, Central District of California, Southern Division; filed July 25, 2005.] [*completed*]
- for Hynix in *Toshiba v. Hynix*; patent litigation, testifying expert, 2007 – 2008. [Investigation 337–TA–592; US International Trade Commission, Washington D.C.] [*completed*]
- for Samsung in *Renesas v. Samsung*; patent litigation, testifying expert, 2007 – 2008. [Investigation 337–TA–595; US International Trade Commission, Washington D.C.] [*completed*]
- for Applied Materials, in *Applied Materials v. SSC*, patent litigation, testifying expert, 2007 – 2008 [*completed*]

- for Samsung in Spansion *v.* Samsung; patent litigation, testifying expert, 2010 – 2011. [Investigation 337-TA-735; US International Trade Commission, Washington D.C.] *[completed]*
- for Invensense in STMicroelectronics *v.* Invensense; patent litigation, testifying expert, 2013 – 2014. [Case CV 12-02475 (JSW), US District Court, Northern District of California, San Francisco Division.] *[completed]*
- for Macronix in Spansion *v.* Macronix; patent litigation, testifying expert, 2014 – 2015. [Investigation 337-TA-916; US International Trade Commission, Washington D.C.] *[completed]*
- for Samsung in KIPB LLC *v.* Samsung; patent litigation, testifying expert, 2020 – present. [Case No. 2:19-cv-00056-JRG-RSP; US District Court, Eastern District of Texas, Marshall Division.] *[completed]*
- for General Motors LLC, Volkswagen AG and Volkswagen Group of America, Inc., and Bayerische Motoren Werke AG and BMW of North America, LLC, in Arigna *v.* General Motors LLC, Volkswagen AG and Volkswagen Group of America, Inc., and Bayerische Motoren Werke AG and BMW of North America, LLC; testifying expert, 2021 – 2023, [International Trade Commission, Investigation No. 337-TA-1267]. *[completed]*
- for Samsung in Daedalus *v.* Samsung; patent litigation, testifying expert, 2022 – 2023. [Case No: 2:22-cv-353; Eastern District of Texas, Marshall Division.] *[completed]*
- for Micron in the Matter of Certain Memory Devices and Electronic Devices Containing the Same, Investigation No. 337-TA-1406; MimirIP LLC *v.* Micron Technology, Inc. et. al., Civil Action No.:2:24-cv-405 (E.D. Tex.); MimirIP LLC *v.* Micron Technology, Inc. et. al., Civil Action No.:2:24-cv-407 (E.D. Tex.) 2024 *[completed]*

2006 – 2014

Irell & Manella

- for Tessera in Tessera, Inc. *v.* Amkor; patent litigation, testifying expert, 2006 – 2008. [International Court of Arbitration of The International Chamber of Commerce] *[completed]*
- for Tessera in Tessera, Inc. *v.* Advanced Micro Devices, Inc., Spansion Inc., Spansion Technology Inc., Spansion LLC, Advanced Semiconductor Engineering, Inc., ASE (US) Inc., ChipMOS Technologies Inc., ChipMOS USA Inc., Siliconware Precision Industries Co. Ltd, Siliconware USA Inc., STMicroelectronics N.V. STMicroelectronics, Inc., STATSChipPac Ltd., STATSChipPac, Inc., STATSChipPac (BVI) Limited; patent litigation, testifying expert, 2006 – 2008. [Case 4:05-cv-04063 CW); US District Court, Northern District of California; filed January 31, 2006 (Second Amended Complaint).] *[completed]*
- for Chi Mei Optoelectronics in LG Display *v.* Chi Mei Optoelectronics; patent litigation, testifying expert, 2009 – 2011. [Case 06-726, 07-357, 08-355 (D. Dell); US District Court, Delaware.] *[completed]*
- for Tessera in Tessera, Inc. *v.* Amkor; patent litigation, testifying expert, 2010 – 2011. [International Court of Arbitration of The International Chamber of Commerce] *[completed]*

- for Tessera in Tessera, Inc. *v.* Sony Corporation, 2012 – 2013. [Case 5:11-CV-04399-EJD, US District Court, Northern District of California.] *[completed]*
- for Tessera in Power Technology Inc. *v.* Tessera, Inc. 2012 – 2014. [Case 4:10-CV-00945-CW, US District Court, Northern District of California.] *[completed]*
- for Tessera in Tessera *v.* UTAC, 2015 – 16, patent litigation, testifying expert. *[completed]*

2006 – present **Sheppard, Mullin, Richter, & Hampton**

- for Northrup Grumman; contract dispute, consultant, 2006 – 2010. *[completed]*
- for Qorvo, in Qorvo *v.* Akoustis, Inc.; patent litigation, testifying expert, 2022 – present. [Case 21-cv-01417, District of Delaware.] *[completed]*
- for Qorvo, in Akoustis Technologies, Inc. & Akoustis, Inc. *v.* Qorvo, Inc.; patent litigation, testifying expert, 2024 – present. [Case 2:23-cv-00180-JRG-RSP, Eastern District of Texas, Marshall Division.]

2006 – 2007 **Foley & Lardner**

- for Schlumberger Technology Corporation in Memory Corporation *v.* Kentucky Oil Technology N.V., Peter Besselink, Memory Metals Holland, B.V., and Kentucky Oil Technology N.V. *v.* Memory Corporation and Schlumberger Technology Corporation; trade secrets, consultant, 2006 – 2008. [Case 04–CV–03843 RMW (HRL); US District Court, Northern District of California.] *[completed]*

2007 – 2009 **Baker Botts**

- for Agere in Agere *v.* Sony; patent litigation, testifying expert, 2007 – 2009. [Case 2:06–CV–00079 (TJW); US District Court, Eastern District of Texas, Marshall Division.] *[completed]*
- for Samsung in Samsung *v.* Pioneer Corporation; patent litigation, testifying expert, 2007 – 2008. [Case 2:06–CV–384 (DF); US District Court, Eastern District of Texas, Marshall Division.] *[completed]*

2007 – 2008 **Wolf, Greenfield & Sacks**

- for Nanya Technology Corporation in Fujitsu *v.* Nanya; patent litigation, testifying expert, 2007 – 2008. [Case 4:06-CV-06-06613 (CW); US District Court, Northern District of California, Oakland Division.] *[completed]*

2008 – 2009 **Quinn Emanuel Urquhart Oliver & Hedges**

- for Samsung in Samsung *v.* SanDisk; patent litigation, testifying expert, 2008 – 2009. *[completed]*

2008 – 2021 **Winston & Strawn**

- for S.O.I.TEC Silicon on Insulator Technologies in S.O.I.TEC Silicon on Insulator Technologies *v.* MEMC Electronic Materials; patent litigation, testifying expert, 2008 – present. [Case 1:08-cv-292-SLR; US District Court, District of Delaware.] *[completed]*
- for Corning, testifying expert, 2019–2021. *[completed]*

2009 – 2011	Lerner, David, Littenberg, Krumholz & Mentlik for Tessera in Tessera <i>v.</i> ST Microelectronics; patent re-examination, consulting expert, 2009 – 2011. [<i>completed</i>]
2009 – 2013	Morrison & Foerster - for Advanced Microfabrication Equipment in Applied Materials <i>v.</i> Advanced Microfabrication Equipment; patent litigation, testifying expert, 2009. [Case C07–05248 JW (PVT), US District Court, North. District of California, San Jose Division.] [<i>completed</i>] - for Invensas in STMicroelectronics <i>v.</i> Invensas; patent litigation, testifying expert, 2013 – 2014. [Case CV 12-02475 (JSW), US District Court, Northern District of California, San Francisco Division.] [<i>completed</i>]
2009	Sidley Austin for Samsung in Samsung <i>v.</i> Pioneer Corporation; patent litigation, testifying expert, 2009. [Case 2:06–CV–384 (DF); US District Court, Eastern District of Texas, Marshall Division.] [<i>completed</i>]
2009	Thompson & Knight for LSI in Qimonda <i>v.</i> LSI; patent litigation, testifying expert, 2009. [Investigation 337–TA–665; US International Trade Commission, Washington D.C.] [<i>completed</i>]
2009	O’Melveny & Myers - for Samsung in Semiconductor Energy Laboratory <i>v.</i> Samsung; patent litigation, testifying expert, 2009 – 2010. [Case 3:09–CV–0001; US District Court, Western District of Wisconsin.] [<i>completed</i>] - for Micron in Elm3DS <i>v.</i> Micron <i>v.</i> Samsung; patent litigation, testifying expert, 2021. [<i>completed</i>]
2010 – 2011	Farella, Braun & Martell for Volterra Semiconductor in Volterra <i>v.</i> Infineon Technologies; patent litigation, testifying expert, 2010 – 2011. [Case No. CV-08-5129 (JCS); US District Court, Northern District of California, San Francisco Division.] [<i>completed</i>]
2011	Faegre & Benson for CoorsTek in CoorsTek <i>v.</i> Reiber & Reiber; patent litigation, testifying expert, 2011. [Case 08-CV-1133-KMT-CBS; US District Court, District of Colorado.] [<i>completed</i>]
2012 – 2021	WilmerHale for Apple, Intel, and Hewlett-Packard, in X2Y Attenuators LLC <i>v.</i> Apple, Intel and Hewlett-Packard, patent litigation, testifying expert, 2012. [ITC Investigation No. 337-781.] [<i>completed</i>]

- for Intel, in Zond LLC v. Intel, patent litigation, testifying expert, 2013 – 2014. [Case No. CV-1:13-11570-JLT; US District Court.] *[completed]*
- for Gillette, in Gillette v. Zond, LLC, testifying expert, 2015. [Patent 8,125,155; Case Nos. IPR 2014–00477 and 00479.] *[completed]*
- for Gillette, in Gillette v. Zond, LLC, testifying expert, 2015. [Patent 6,896,775; Case Nos. IPR 2014-00578 and 00604.] *[completed]*
- for Gillette, in Gillette v. Zond, LLC, testifying expert, 2015. [Patent 6,896,773; Case Nos. IPR 2014-00580 and 00726.] *[completed]*
- for Gillette and TSMC, in Gillette, Fujitsu Semiconductor Limited, Advanced Micro Devices, Renesas Electronics Corporation, Renesas Electronics America, Global Foundries U.S., Global Foundries Dresden Module One LLC & Co. KG, Global Foundries Dresden Module Two LLC & Co. KG, Toshiba America Electronic Components, Toshiba America, Toshiba America Information Systems, and Toshiba Corporation v. Zond, LLC, testifying expert, 2015. [Patent 7,808,184; Case Nos. IPR 2014-00799 and 00803.] *[completed]*
- for Apple in Apple v. Ericsson; patent litigation, 2015. [ITC Investigation No. 337-953.] *[completed]*
- for Intel in DSS v. Intel Corporation; patent litigation, 2015. [Case No. CV-6:15-130; US District Court.] *[completed]*
- for Intel in DSS v. Intel Corporation; patent litigation, 2015. [Patent 5,965,924; Case Nos. IPR 2015-TBD and Patent 6,784,552; Case Nos. IPR 2015-TBD.] *[completed]*
- For Intel in VLSI Technology v. Intel, Petitioners; IPR Petitions on U.S Patents 7709303, 8268672; 2017 – 2019. *[completed]*

2015**Latham & Watkins**

- for NVIDIA, in Samsung v. NVIDIA, patent litigation, testifying expert, 2015. [Case No. CV-3:14-757; US District Court.] *[completed]*

2015 – 2016**Troutman Sanders**

- for Soitec in Silicon Genesis v. Soitec; patent litigation, testifying expert, 2015-16. [Investigation 337–TA–966; US International Trade Com., Washington D.C.] *[completed]*

2016 – present**Covington & Burling**

- for Tessera, in Tessera v. Broadcom; patent litigation, testifying expert, 2016-17 [Investigation 337–TA–1010; US International Trade Com., Washington D.C.] *[completed]*
- for Hanwha Q CELLS in Hanwha Q Cells v. JinkoSolar Holding Co., Ltd., JinkoSolar (U.S.) Inc., Jinko Solar (U.S.) Industries Inc., Jinko Solar Co., Ltd., Zhejiang Jinko Solar Co., Ltd., Jinko Solar Technology Sdn. Bhd., LONGi Green Energy Technology Co., Ltd., Longi Solar Technology Co., Ltd., LONGi (H.K.) Trading Ltd., LONGi (Kuching) Sdn. Bhd., Taizhou LONGi Solar Technology Ltd., Zhejiang LONGi Solar Technology Ltd., Hefei LONGi Solar Technology Ltd., LONGi Solar Technology (U.S.) Inc., REC Solar Holdings AS, REC Solar Pte. Ltd. and REC Americas, LLC; patent

litigation, testifying expert, 2019 [Investigation 337-TA-1151; US International Trade Com., Washington D.C.] [*completed*]

- for Invensas Corporation and Tessera Advanced Technologies, Inc., in Invensas Corporation and Tessera Advanced Technologies v. NVIDIA Corporation; patent litigation, testifying expert, 2019-2020 [Case No. 1:19-cv-00861; US District Court] [*completed*]
- for Samsung Display Co., IP Review of Certain AUO Patents. [*completed*]
- for Hynix, in SK Hynix and SK Hynix America v. Longhorn IP LLC, Trenchant Blade Technologies LLC, and Hamilcar Barca IP LLC; patent litigation, testifying expert, 2022 – 2023. [Case No. 3:22-cv-3915 (N.D. Cal.)]. [*completed*]
- for Samsung Electronics, in Redstone Logics LLC v. Samsung Electronics et al., No. 2:22-cv-0233-JRG (E.D. Tex.) 2024 [*completed*]
- for Samsung, in SiOnyx, LLC v. Samsung Electronics, Co., Ltd., In the Matter of Certain Sensors with Pixels and Products Containing the Same, Investigation No. 337-TA-1403 at the U.S. International Trade Commission; SiOnyx, LLC v. Samsung Elecs. Co., Ltd. et al., Case No. 2:24-cv-291 (E.D. Tex.); SiOnyx, LLC v. Samsung Elecs., Co. Ltd. et al., Case No. 2:24-cv-408 (E.D. Tex.); 2024 – 2025 [*completed*]

2018 – 2021

Mintz Levin

- for Innovative Foundry Technologies, consulting expert. [ITC investigation 337-TA-1149] [*completed*]

2019 – 2021

Orrick

- for Micron Technology, Inc., in North Star Innovations, Inc. v. Micron Technology, Inc., patent litigation, testifying expert, 2019–present. [Case No. 17-cv-506-LPS-CJB; US District Court.] [*completed*]
- for Micron Technology, Inc., in Lone Star Innovations LLC, Inc. v. Micron Technology, Inc., patent litigation, testifying expert, 2019–present. [Case No. 3:17-cv-05458-WHA; US District Court.] [*completed*]

2019 – 2021

Step toe & Johnson

- for Chilis Electronics Corp., in Cynotec Company, Ltd. v. Chilis Electronics Corp, Inc., patent litigation, testifying expert, 2019–2021. [Case No.: 4:18-cv-00939-PJH; US District Court.] [*completed*]

2014 – 2021

Desmarais

- for Round Rock Research LLC, in Round Rock Research LLC v. SanDisk Corporation; consulting expert. 2014–15. [*completed*]
- for Apple, in nCAP Licensing, LLC, nCAP Telecommunications LLC, nCAP Medical, LLC v. Apple Inc, consulting expert. [No.: 2:17-cv-00905-HCN-CMR]. [*completed*]
- for Samsung Electronics Ltd. et al., in Samsung Electronics Ltd. et al. v. Trenchant Blade Technologies LLC, Inter Parties Review; testifying expert, 2020–2021. [*completed*]

- for Samsung in Demaray LLC v. Samsung Electronics Co., Ltd. et al., 6:20-cv-00636; testifying expert, 2020–2021[*completed*]

2021 – present **Finnegan**

- for General Motors LLC, Volkswagen AG and Volkswagen Group of America, Inc., and Bayerische Motoren Werke AG and BMW of North America, LLC, in Arigna v. General Motors LLC, Volkswagen AG and Volkswagen Group of America, Inc., and Bayerische Motoren Werke AG and BMW of North America, LLC; testifying expert, 2021 – 2023, [International Trade Commission, Investigation No. 337-TA-1267]. [*completed*]
- for Dupont, in CMC Materials v. DuPont de Nemours, Inc., Rohm and Haas Electronic Materials CMP LLC, Rohm and Haas Electronic Materials CMP Asia Inc. (d/b/a Rohm and Haas Electronic Materials CMP Asia Inc., Taiwan Branch (U.S.A.)), Rohm and Haas Electronic Materials Asia-Pacific Co., Ltd., Rohm and Haas Electronic Materials K.K., and Rohm and Haas Electronic Materials LLC; testifying expert, 2023 – 2024 [US District Court, Delaware, C.A. No. 20-738-MN] [*completed*]

2022 – 2023 **King & Spaulding**

- for Applied Materials, in Applied Materials v. Mattson Technology; Canfeng Lai; and Does 1-20; testifying expert, 2022 – 2023. [*completed*]

2022 – present **Weil Gotshall & Manges**

- for GlobalFoundries, in International Business Machines Corp. v. GlobalFoundries U.S. Inc.; testifying expert, 2022 – present. [Index No. 653625/2021, Supreme Court of the State of New York, Commercial Division]. **

2024 – present **Kellogg Hansen**

- for Corning Eagle XG Glass, potential ITC Investigation.; testifying expert, 2024 – 2026. [*completed*]

2024 – present **White & Case**

- for Chilisin in Cynotec Company, Ltd. v. Chilisin Electronics Corp. et al., N.D. Cal. Case No. 4:18-cv-00939, testifying expert, 2024 – present. **

Note: The designation “[*completed*]” at the end of an entry indicates that Dr. Bravman’s involvement in the matter has ended no later than the publication date of this document. Matters indicated with a ** remain active as of that date.

Doctoral Students of Prof. John C. Bravman

	<u>Name</u>	<u>Dissertation Title</u>	<u>Completion Date</u>
1.	David C. Paine	A Microstructural Study of LPCVD Tungsten Thin Films for Use as a Contact Material to Silicon.	September 1988
2.	Ming Bing Chang	Deep Level Related Studies in GaAs with the Constant Capacitance Voltage Transient Technique.	October 1989
3.	Soonil Hong	Free Standing Multilayer Thin Film Microstructures for Electronic Systems.	December 1990
4.	Bruce M. Lairson	Critical Currents and Flux Creep in YBa ₂ Cu ₃ O _{7-δ} Superconducting Thin Films.	August 1991
5.	Paul L. Meissner	Barrier Height Reduction and Interface Chemistry in Pd–Ge Based Contacts to GaAs.	August 1991
6.	Ramnath Venkatraman	Plasticity and Flow Stresses in Al Thin Films on Silicon.	May 1992
7.	Stephen K. Streiffer	Nucleation and Growth of YBa ₂ Cu ₃ O _{7-δ} Films on Lattice–Matched and Mismatched Substrates.	November 1992
8.	Paul R. Besser	X–ray Determination of Thermal Strains, Stresses, and Relaxation in Passivated Al Lines During Thermal Cycling	August 1993
9.	Thomas N. Marieb	In-Situ Observations of Void Nucleation and Growth in Passivated Metal Lines	January 1994
10.	Richard Vinci	Thermal Stresses and Strains in Copper Films and Interconnect Structures	August 1994
11.	Cynthia Lee	Diffusion of Implanted Dopants in GaAs/AlGaAs.	December 1994
12.	Eden Zielinski	The Influence of Strain Energy on Abnormal Grain Growth in Copper Thin Films	August 1995
13.	Yaser M. Haddara	Transient Diffusion of Dopants in Gallium Arsenide	March 1997
14.	Charlie Yu	High Temperature Mechanical Properties of Thin Films used in VLSI Fabrication	August 1997
15.	Samantha Lee	The Effects of Passivation on Electromigration Behavior in Aluminum Interconnects	August 1998
16.	Jonathan Doan	In–Situ Studies of Electromigration in Thin Films	December 1998
17.	Guido Cornella	Monotonic and Cyclic Testing of Thin Film Materials	December 1998
18.	Seok Hee Lee	Electromigration in Cu Thin Films	December 2000
19.	Nicole Meier	Electromigration in Cu Thin Films	March 2002
20.	Dimitrios Pantelidis	Mechanisms of Adhesion of Organic/Inorganic Interfaces	June 2003

	Reinforced by Silane Coupling Agents and Self-Assembling Mesoporous Oxide Thin Films	
21. Ping Zhang	Thin Film Mechanical Behavior	May 2003
22. Bryan Valek	Thin Film Mechanical Behavior	March 2003
23. Maura Jenkins	Selection and Use of Silane Adhesion Promoters in Microelectronic Packaging	June 2003
24. Gaurav Verma	Laser Silicidation for Deep Submicron MOSFETs	August 2003

Invited Presentations at Conferences, Workshops, etc.

1. March 1985 "High Resolution Microstructural Studies of VLSI Interfaces," Electronic Materials Symposium, Northern California Section, AIME, Santa Clara, CA.
2. October 1985 "Microstructural Characterization of LPCVD Tungsten Interfaces," Second Annual Workshop on Tungsten & Other Refractory Metals, Albuquerque, NM.
3. January 1986 "High Resolution Microstructural Studies of Semiconductor Materials," 1986 Annual Meeting of the Society of Photo-Optical Instrumentation Engineers, Los Angeles, CA.
4. February 1987 "High Resolution Microstructural Studies of Semiconductor Materials," Golden Gate Materials Technology Conference, San Francisco, CA.
5. June, 1987 "Structural Characterization of Silicon – LPCVD Tungsten Interfaces," 13th Western Regional Meeting for Electron Microscopy and Microbeam Analysis, Concord, CA.
6. July 1987 "TEM Studies of Semiconductor Materials," Annual Meeting of the International Metallographic Society, Monterey, CA.
7. September 1987 "Synthesis and Properties of Superconducting Parasites," Regional Meeting of the Materials Research Society, Univ. of Washington, Seattle, WA.
8. January 1988 "TEM Studies of VLSI Materials," Northern California American Vacuum Society Thin Film Microanalysis Seminar, San Jose, CA.
9. January 1988 "Synthesis and Properties of Superconducting Perovskites," 1988 Annual Meeting of The Metallurgical Society, Phoenix, AZ.
10. April 1988 "Morphology and Microstructure of LPCVD Tungsten Films," 1988 Spring Meeting of the Materials Research Society, Reno, NV.
11. September 1989 "High Resolution Electron Microscopy of Semiconductor Interfaces," 12th Brazilian Congress on Electron Microscopy, Rio de Janeiro, Brazil.
12. April 1990 "Beam Deflection Techniques for Studying Thin Film Mechanical Behavior," 1990 Spring Meeting of the Materials Research Society, San Francisco, CA.

13. June, 1991 “Mechanical Testing of Thin Films,” 6th International Conference on Solid State Sensors and Actuators,” San Francisco, CA.
14. November 1991 “Mechanical Properties of Thin Film Aluminum,” American Vacuum Society Annual Meeting, Seattle, WA.
15. March 1992 “Mechanical Testing of Thin Film Media,” The Metallurgical Society Annual Meeting, San Diego, CA.
16. April 1992 “Mechanical Testing of Thin Film Materials,” 1992 Spring Meeting of the Materials Research Society, San Francisco, CA.
17. March 1993 “X-ray Determination of Strains, Stresses and Relaxation in Interconnection Metallizations,” Second International Workshop on Stress Induced Phenomena in Metallizations, Austin, TX.
18. June 1994 “Microstructure and Stresses in Sputtered Copper Films,” Symposium on Micromechanisms in Stressed Conductor Lines and Thin Films, Ringberg, Germany.
19. December 1994 “In-situ Electromigration Testing of Aluminum Metallizations,” 1994 Fall Meeting of the Materials Research Society, Boston, MA.
20. April 1995 “Development of High Voltage SEM for In-situ Electromigration Testing,” 1995 Spring Meeting of the Materials Research Society, San Francisco, CA.
21. May 1995 “Development of High Voltage SEM for In-situ Electromigration Testing,” 1995 Meeting, European Materials Research Society, Strasbourg, France.
22. September 1998 “In-Situ High Voltage Scanning Electron Microscopy for Electromigration Studies,” 14th Quadrennial International Conference on Electron Microscopy, Cancun, Mexico.
23. June 2001 “Micromechanical Testing of Free-Standing Thin Films for MEMS Applications,” 2001 Mechanics and Materials Conference, San Diego. (Keynote)
24. June 2001 “In-Situ Dynamic Studies of Electromigration in Copper Metallizations,” 2001 Mechanics and Materials Conference, San Diego. (Keynote)
25. December 2001 “Micromechanical Testing of Free-Standing Thin Films for MEMS Applications,” 2001 Fall Meeting of the Materials Research Society, Boston, MA.

Invited Presentations at Universities, Corporations, etc.

26. December 1984 “Transmission Electron Microscopy of Semiconductor Materials,” Department of Materials Science, Oregon Graduate Center.
27. January 1985 “Transmission Electron Microscopy of VLSI Interfaces,” Rockwell International, Semiconductor Products Division, Los Angeles, CA.

28. January 1985 “Transmission Electron Microscopy: A Tool for VLSI Research,” Hewlett Packard Research Laboratories, Palo Alto, CA.
29. March 1985 “Morphological Aspects of Silicon – Silicon Dioxide VLSI Interfaces,” Texas Instruments Central Research Laboratories, Dallas, TX.
30. October 1987 “Transmission Electron Microscopy of Thin Films and Interfaces,” Department of Materials Science and Mineral Engineering, Univ. of California at Berkeley.
31. May 1988 “Progress and Prospects in High Temperature Superconductors,” Raychem Corporation, Redwood City, CA.
32. June 1988 “An Assessment of LPCVD Tungsten for VLSI Metallizations,” Xerox Corp., Palo Alto Research Center, Palo Alto, CA.
33. April 1989 “Transmission Electron Microscopy: A Tool for VLSI Research,” National Semiconductor Corporation, Santa Clara, CA.
34. September 1989 “Mechanical Properties of Thin Film Materials,” 12th Brazilian Congress on Electron Microscopy, Rio de Janeiro, Brazil.
35. October 1989 “Mechanical Properties of Thin Film Materials,” Motorola Corporation, Phoenix, AZ.
36. February 1991 “Microstructural Studies of Y–Ba–Cu–O High T_c Thin Films,” 1991 Stanford Symposium on Applications of Contemporary Electron Microscopy, Stanford, CA.
37. May 1991 “Mechanical Testing Techniques for Thin Film Structures,” Microelectronics Center of North Carolina, Research Triangle Park, NC.
38. September 1991 “Processing and Structure of Y–Ba–Cu–O High T_c Thin Films,” Physical Sciences Colloquium, IBM Almaden Research Center, San Jose, CA.
39. November 1991 “Mechanical Properties of Thin Films,” Department Colloquium, Materials Science and Engineering, U. C. Berkeley.
40. March 1992 “Mechanical Testing of Thin Film Media,” Department Colloquium, Materials Science and Engineering, University of Southern California, Los Angeles, CA.
41. November 1992 “Mechanical Testing Techniques for Thin Film Structures,” Opening Dedication Ceremony, Max Planck Institut, Stuttgart, Germany.
42. November 1992 “Micron–Scale Mechanical Testing of Films and Layered Media,” Opening Dedication Ceremony, Max Planck Institut für Metallforschung; Stuttgart, Germany.
43. September 1993 “Modern Techniques for Surface and Interface Analysis,” National Defense Academy, Japan.
44. October 1993 “X-ray Methods for Determining Strains and Stresses in Thin Films and Interconnection Lines,” AFOSR Workshop on Stress and Reliability in Thin Film and Optical Materials, The Aerospace Corporation, Los Angeles, CA.

45. October 1995 “Development of High Voltage SEM for In-situ Electromigration Testing,” University of California at Los Angeles.
46. November 1996 “Development of High Voltage SEM for In-situ Electromigration Testing,” University of Michigan at Ann Arbor.
47. August, 1997 “How to be an Effective Engineering Educator,” Linkoping University, Sweden
48. August 1997 “How to be an Effective Engineering Educator,” Upsala University, Sweden
49. August 1997 High Voltage Scanning Electron Microscopy for In-situ Electromigration Testing,” Royal Institute of Technology, Sweden
50. November 1997 High Voltage Scanning Electron Microscopy for In-situ Electromigration Testing,” Georgia Institute of Technology.
51. February 1998 “The Future of Engineering Education at Major Research Universities: Course Correction or New Paradigm?” Symposium on Engineering Education for the 21st Century, Nagoya, Japan.
52. March 1998 “Quantitative Assessments of Electromigration Lifetime,” Lehigh University
53. November 1998 “A New Method for Studying Electromigration in IC Interconnects,” Lawrence Berkeley Laboratory, Advanced Light Source Colloquium Series.

Publications (approximately 4500 citations; h-index ~ 40)

Archival Refereed Journals

1. “TEM Studies of the Polycrystalline Silicon – Silicon Dioxide Interface,” J.C. Bravman and R. Sinclair, *Thin Solid Films*, 104, 153–61 (1983).
2. “Observation of Rapid Field Aided Diffusion of Silver in Metal Oxide Semiconductor Structures,” J.D. McBrayer, R.M. Swanson, T.W. Sigmon and J.C. Bravman, *Applied Physics Letters*, 43, 653–54 (1983).
3. “The Preparation of Cross-section Specimens for Transmission Electron Microscopy,” J.C. Bravman and R. Sinclair, *Journal of Electron Microscopy Technique*, 1, 52–61 (1984).
4. “Structure and Morphology of Polycrystalline Silicon – Single Crystal Silicon Interfaces,” J.C. Bravman, G.L. Patton and J.D. Plummer, *Journal of Applied Physics*, 57, 2279–82 (1985).
5. “Physics, Technology and Modeling of Polysilicon Emitter Contacts for VLSI Bipolar Transistors,” G.L. Patton, J.C. Bravman and J.D. Plummer, *IEEE Transactions on Electron Devices*, ED-33, 1754–68 (1986).
6. “Damage Calculation and Measurement for GaAs Amorphized by Si-Implantation,” W.G. Opyd, J.F. Gibbons, J.C. Bravman and M.A. Parker, *Applied Physics Letters*, 49, 974–976 (1986).
7. “Observations of Beta-tungsten deposited by Low Pressure Chemical Vapor Deposition,” D. C. Paine, J.C. Bravman and C. Y. Yang, *Applied Physics Letters*, 50, 498–500 (1987).
8. “The Mechanical Deflection of Cantilever Microbeams: A New Technique for Testing the Mechanical Properties of Thin Films,” T. P. Weihs, S. Hong, J.C. Bravman and W.D. Nix, *Journal of Materials Research*, 3, 931–942 (1988).
9. “Molecular Beam Epitaxy of Layered Dy–Ba–Cu–O Compounds,” D. G. Schlom, J. N. Eckstein, E. S. Hellman, S.K. Streiffer, J. S. Harris, M. R. Beasley, J.C. Bravman, T. H. Geballe, C. Webb, K. von Dessenneck and F. Turner, *Applied Physics Letters*, 53, 1660–1662. (1988).
10. “Microstructural Characterization of YBa₂Cu₃O_{7-d} Thin Films on SrTiO₃ Using Four Axis X-ray Diffraction,” J. Sizemore, R. Barton, A. Marshall and J.C. Bravman, *IEEE Transactions on Magnetics*, 25 (2), 2245-2249 (1989).
11. “Determination of High Field Enhanced Emission Rates with the Constant Capacitance–Voltage Transient Technique,” M.B. Chang, H. Tomokage, J.J. Shiao, R. H. Bube, and J.C. Bravman, *Journal of Applied Physics*, 65, 2734–2738 (1989).
12. “Phase Characterization of Dysprosium Barium Copper Oxide Thin Films Grown on Strontium Titanate by Molecular Beam Epitaxy,” E. S. Hellman, D. G. Schlom, A. F. Marshall, S.K. Streiffer, J. S. Harris, M. R. Beasley, J.C. Bravman and T. H. Geballe, *Journal of Materials Research*, 4, 476–95 (1989).
13. “Elimination of Current Dissipation in High Transition Temperature Superconductors,” J.Z. Sun, B. Lairson, C.B. Eom, J.C. Bravman and T.H. Geballe, *Science*, 247, 307–309 (1990).
14. “Microwave Properties of Highly Oriented YBa₂Cu₃O_{7- δ} Thin films,” A. Inam, X.D. Wu, L. Nazar, M.S. Hegde, C.T. Rogers, T. Venkatesan, R.W. Simon, K. Daly, H. Padamsee, J.

- Kirchgessner, D. Moffat, D. Rubin, Q.S. Shu, D. Kalolitis, A. Fathy, V. Pendrick, R. Brown, B. Brycki, E. Belohoubek, L. Drabek, G. Gruner, R. Hammond, F. Gamble, B.M. Lairson and J.C. Bravman, *Applied Physics Letters*, 56, 1178–80 (1990).
15. “Reduction of Magnetization Decay Rate in High T_c Superconductors,” B. Lairson, J.Z. Sun, J.C. Bravman and T.H. Geballe, *Physical Review B*, 42, 1008–1011 (1990).
 16. “Measuring Stiffnesses and Residual Stresses of Silicon Nitride Thin Films,” S. Hong, T.P. Weihs, J.C. Bravman and W.D. Nix, *Journal of Electronic Materials*, 19, 903–909 (1990).
 17. “Molecular Beam Epitaxial Growth of Layered Bi–Sr–Ca–Cu–O Compounds,” D. G. Schlom, A. F. Marshall, J.T. Sizemore, Z.J. Chen, J. N. Eckstein, I. Bozovic, K.E. von Dessionneck, J.S. Harris and J.C. Bravman, *Journal of Crystal Growth*, 102, 361–375 (1990).
 18. “Mechanical Properties and Microstructural Characterization of Al–0.5% Cu Thin Films,” R. Venkatraman, J.C. Bravman, P.W. Davies, P.A. Flinn, D.B. Fraser and W.D. Nix, *Journal of Electronic Materials*, 19, 1231–7 (1990).
 19. “The Microstructure of Fe and Ag Thin Films Grown by Molecular Beam Epitaxy on GaAs (001),” C.J. Chien, R.F.C. Farrow and J.C. Bravman, *Journal of Applied Physics*, 68, 4343–5 (1990).
 20. “Vortex Pinning by Twin Boundaries in $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Thin Films,” B.M. Lairson, S.K. Streiffer and J.C. Bravman, *Physical Review B*, 42, 10067–74 (1990).
 21. “Magnetic ordering in strain-controlled epitaxial rare-earth films,” R. F. C. Farrow, S. S. P. Parkin, W. Bennett, A. Segmuller, J. Chien, and J. C. Bravman, *Journal of Applied Physics*, 67, 5733 (1990).
 22. “Growth of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Thin Films on Vicinal MgO Surfaces,” S.K. Streiffer, B.M. Lairson and J.C. Bravman, *Applied Physics Letters*, 57, 2501–3 (1990).
 23. “Synthesis and Properties of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Thin Films Grown In-Situ by 90° Off-Axis Single Magnetron Sputtering,” C.B. Eom, J.Z. Sun, S.K. Streiffer, A.F. Marshall, B.M. Lairson, K. Yamamoto, S.M. Anlage, J.C. Bravman, T. H. Geballe, S.S. Laderman and R. C. Taber, *Physica C*, 171, 354–82 (1990).
 24. “Role of Atomic Oxygen produced by ECR Plasma in the Oxidation of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Thin Films studied by In-situ Resistivity Measurements,” K. Yamamoto, B.M. Lairson, C.B. Eom, R. H. Hammond, J.C. Bravman and T.H. Geballe, *Applied Physics Letters*, 57, 1936–8 (1990).
 25. “Thermal Activation of Vortex Motion in $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Films at Low Temperatures,” B.M. Lairson, J.Z. Sun, T.H. Geballe, M. R. Beasley and J.C. Bravman, *Physical Review B*, 43, 10405–12 (1991).
 26. “Magnetic Relaxation, J–E Characteristics and Possible Dissipation Mechanisms for High T_c Superconducting Films of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$,” J.Z. Sun, C.B. Eom, B.M. Lairson, J.C. Bravman and T. H. Geballe, *Physical Review B*, 43, 3002–8 (1991).
 27. “Oxidation Kinetics of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ Thin Films in the Presence of Atomic Oxygen and Molecular Oxygen by In-situ Resistivity Measurements,” K. Yamamoto, B.M. Lairson, J.C. Bravman and T.H. Geballe, *Journal of Applied Physics*, 69, 7189–7201 (1991).
 28. “Microstructure of Ultrathin Films of $\text{YBa}_2\text{Cu}_3\text{O}_{7-d}$ on MgO,” S.K. Streiffer, B.M. Lairson, C.B. Eom, J.C. Bravman and T. H. Geballe, *Physical Review B*, 43, 13007–13018 (1991).

29. "Thickness Dependence of the Twin Density in YBa₂Cu₃O_{7-d} Thin Films Sputtered onto MgO Substrates," S.K. Streiffer, E.M. Zielinski, B.M. Lairson, and J.C. Bravman, *Applied Physics Letters*, 58, 2171–3 (1991).
30. "The Effect of Laser Reflowing on the Variation of Stress with Thermal Cycling in Aluminum Thin Films," R. Venkatraman, S. Chen and J.C. Bravman, *Journal of Vacuum Science and Technology A*, 9, 2536–42 (1991).
31. "Separation of Film Thickness and Grain Boundary Strengthening Effects in Al Thin Films on Si," R. Venkatraman and J.C. Bravman, *Journal of Materials Research*, 7, 2040–2048 (1992).
32. "Limits on the use of tunneling to describe the Pd–Ge ohmic contact to GaAs," A. Herrera-Gómez, P. L. Meissner, J. C. Bravman and W. E. Spicer, *Journal of Vacuum Science and Technology A*, 10, 1029 (1992).
33. "Nucleation and Growth Mechanisms of a,b–Axis Oriented YBa₂Cu₃O_{7-d} Films on LaAlO₃," S.K. Streiffer, B.M. Lairson, E.M. Zielinski, and J.C. Bravman, *Physical Review B*, 47, 11431–38 (1993).
34. "Finite-Element Modeling and X-ray Determination of Stress in Passivated Al Lines During Thermal Cycling," P.R. Besser, A. Sauter-Mack, D. Fraser, and J.C. Bravman, *Journal of the Electrochemical Society*, 140, 1769–73 (1993).
35. "An X-ray Method for Direct Determination of the Stress State and Strain Relaxation in Micro-scale Passivated Metallization Lines During Thermal Cycling," P.R. Besser, S. Brennan, and J.C. Bravman, *Journal of Materials Research*, 9, 13–24 (1994).
36. "Elastic Strain Gradients and X-ray Line Broadening Effects as a Function of Temperature in Aluminum Thin Films on Silicon," R. Venkatraman, P.R. Besser, S. Brennan and J.C. Bravman, *Journal of Materials Research*, 9, 328–35 (1994).
37. "Understanding and Electrochemical Control of YBa₂Cu₃O_{7-x} Thin Film Epitaxy on Yttrium Stabilized Zirconia," J.L. MacManus-Driscoll, T.H. Geballe, and J.C. Bravman, *Journal of Applied Physics*, 75, 412–422 (1994).
38. "Effect of Ion Energy on the Diffusion of Si Implanted into GaAs," C.C. Lee, M.D. Deal, K.S. Jones, H.G. Robinson, and J.C. Bravman, *Journal of the Electrochemical Society*, 141, 2245–2249 (1994).
39. "Eliminating Dopant Diffusion after Ion Implantation by Surface Etching," C.C. Lee, M. E. Deal and J.C. Bravman, *Applied Physics Letters*, 64, 3302–3304 (1994).
40. "Observation of Voids Induced by Mechanical Stress and Electromigration in Passivated Al Lines Deposited at Different Purity Levels," T. Marieb, J.C. Bravman, P. Flinn, D. S. Gardner, and M. Madden, *Applied Physics Letters*, 64, 2424–6 (1994).
41. "The Effects of Silver and Lead on the Phase Stability of the Bi–2212 and Bi–2223 Phases Above and Below the Solidus Temperature," J.L. MacManus-Driscoll, J.C. Bravman, R. J. Savoy, G. Gorman and R.B. Beyers, *Journal of the American Ceramic Society*, 77, 2305–13 (1994).
42. "Effects of Barrier Layer and Annealing on Abnormal Grain Growth in Copper Thin Films," E.M. Zielinski, R.P. Vinci, and J.C. Bravman, *Journal of Applied Physics*, 76, 4516–23 (1994).

43. "Studies of Structural Disorder in $\text{ReBa}_2\text{Cu}_3\text{O}_{7-x}$ Thin Films as a Function of Rare Earth Ionic Radius and Film Deposition Conditions," J.L. MacManus-Driscoll, J.A. Alonso, P. Wang, T.H. Geballe and J.C. Bravman, *Physica C*, 232, 288-308 (1994).
44. "Phase Equilibria and Melt Processing of $\text{Bi}_2\text{Sr}_2\text{Ca}_1\text{Cu}_2\text{O}_{8+x}$ Tapes at Reduced Oxygen Partial Pressures, J.L. MacManus-Driscoll, P. Wang, J.C. Bravman and R.B. Beyers, *Applied Physics Letters*, 65, 2872-4 (1994).
45. "The Influence of Strain Energy on Abnormal Grain Growth in Copper Thin Films," E.M. Zielinski, R.P. Vinci, and J.C. Bravman, *Applied Physics Letters*, 67, 1078 - 1080 (1995).
46. "Transport Characterization of Calcium-Doped YBCO Thin Films," J.T. Kucera and J.C. Bravman, *Physical Review B*, 51, 8582-90 (1995).
47. "Thermal Strain and Stress in Copper Thin Films," R.P. Vinci, E.M. Zielinski, J.C. Bravman, *Thin Solid Films*, 262, 142-153 (1995). (Invited Review)
48. "Effects of Barrier Layer and Processing Conditions on Thin Film Cu Microstructure." E.M. Zielinski, R.P. Vinci, J.C. Bravman, *Journal of Electronic Materials*, 24, 1485-1492 (1995).
49. "Observations of Electromigration Induced Void Nucleation and Growth in Polycrystalline and Near-bamboo Passivated Al Lines," T. Marieb, P. Flinn, J.C. Bravman, D. Gardner, M. Madden, *Journal of Applied Physics*, 78, 1026-32 (1995).
50. "Diffusion of Implanted Be in $\text{Al}_x\text{Ga}_{1-x}\text{As}$ as a Function of Al Concentration and Anneal Temperature," C.C. Lee, M. Deal and J.C. Bravman, *Applied Physics Letters*, 66, 355-7 (1995).
51. "Phase Equilibria in the Y-Ba-Cu-O System and Melt Processing of Ag Clad $\text{Y}_1\text{Ba}_2\text{Cu}_3\text{O}_{7-x}$ Tapes at Reduced Oxygen Partial Pressures," J.L. MacManus-Driscoll, J.C. Bravman and R.B. Beyers, *Physica C*, 241, 401-13 (1995).
52. "Pseudo-Quaternary Phase Relations Near $\text{Bi}_2\text{Sr}_2\text{Ca}_1\text{Cu}_2\text{O}_{8+x}$ in Reduced Oxygen Pressures," J.L. MacManus-Driscoll, J.C. Bravman and R.B. Beyers, *Physica C*, 251, 71-88 (1995).
53. "Modeling Diffusion in Gallium Arsenide: Recent Work," Y.M. Haddara, C.C. Lee, J.C. Hu, M.D. Deal & J.C. Bravman, *MRS Bulletin*, 20, #4, 41-50 (1995).
54. "Measurement and Interpretation of Strain Relaxation in Passivated Al-0.5%Cu Lines," P.R. Besser, T.N. Marieb, P.A. Flinn, and J.C. Bravman, *Journal of Materials Research*, 11, 184-193 (1996).
55. "The Effect of Encapsulant Material on the Diffusion of Beryllium in Molecular Beam Epitaxially Grown Gallium Arsenide," Y.M. Haddara, M.D. Deal and J.C. Bravman, *Applied Physics Letters*, 68, 1939-41 (1996).
56. "An Analysis Technique for Extraction of Thin Film Stresses from X-ray Data," G. Cornella, S.H. Lee, W.D. Nix, and J.C. Bravman, *Applied Physics Letters*, 71, 2949-52 (1997).
57. "Transient Diffusion of Beryllium and Silicon in Gallium Arsenide," Y. Haddara and J.C. Bravman, *Annual Reviews of Materials Science*, 28, 185-214 (1998)
58. "Void nucleation on intentionally added defects in Al interconnects," J. C. Doan, S.-H. Lee, J. C. Bravman, P. A. Flinn and T. N. Marieb, *Applied Physics Letters*, 75, 633-35 (1999).

59. "The Evolution of the Resistance of Aluminum Interconnects during Electromigration," J. C. Doan, J. C. Bravman, P. A. Flinn, and T. N. Marieb, *Microelectronics Reliability*, 40, 981-990 (2000).
60. A High-Voltage Scanning Electron Microscopy System for In-Situ Electromigration Testing," J. C. Doan, S.-H. Lee, N. E. Meier, J. C. Bravman, P. A. Flinn, T. N. Marieb and M. C. Madden, *Review of Scientific Instruments*, 71, 2848-2854 (2000).
61. "Stress Relaxation of Free Standing Aluminum Beams for Microelectromechanical Systems Applications," H.-J. Lee, G. Cornella and J. C. Bravman, *Applied Physics Letters*, 76, 3415-17 (2000).
62. "Differential Thermal Budget in Laser Processing: Application to Formation of Titanium Silicide," J. C. Bravman, G. Verma, and S. Talwar, *IEEE Electron Device Letters*, 21, #10, 482-84, (2000).
63. "Effects of Dielectric Materials on Electro-migration Failure," J. C. Doan, S. Lee, S.-H. Lee, P. A. Flinn, J. C. Bravman and T. N. Marieb, *Journal of Applied Physics*, 89, 7797-7808 (2001).
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65. "Creating Process Margin in Laser Thermal Processing: Application to Formation of Titanium Silicide," G. Verma, S. Talwar and J. C. Bravman, *Applied Physics Letters*, 78, 925-27 (2001).
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67. "Miniature Nernstian Oxygen Sensors for Deposition and Growth Environments," E. van Setten, T. M. Gur, D. Blank, J. C. Bravman and M. R. Beasley, *Review of Scientific Instruments*, 73, 156-161 (2002).
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EXHIBIT B

Jack C. Lee
Professor Emeritus
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Current Status

Professor Emeritus
Department of Electrical and Computer Engineering
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Research / teaching experience and expertise:

- Integrated circuits manufacturing
- Photovoltaic/solar cell technologies
- Alternative channel devices
- Dielectric and passivation processes
- Display technologies
- Semiconductor memory devices

Education

9/84 – 8/88 Ph.D., Electrical Engineering, University of California at Berkeley
9/80 – 12/81 M.S., Electrical Engineering, University of California at Los Angeles
9/76 – 6/80 B.S., Electrical Engineering (with highest honors),
University of California at Los Angeles

Professional Career

9/23 – present Professor Emeritus
Department of Electrical and Computer Engineering
The University of Texas at Austin
9/00 – 8/23 Cullen Trust for Higher Education Endowed Professorship in Engineering #4
Department of Electrical and Computer Engineering
The University of Texas at Austin
9/96 – 8/23 Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
9/92 – 8/96 Associate Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
9/88 – 8/92 Assistant Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
1/88 – 6/88 Lecturer, Electrical Engineering and Computer Science Department
University of California at Berkeley
6/79 – 8/84 Member of Technical Staff, High-Speed Bipolar Device Program
TRW Microelectronics Center, Redondo Beach, CA

Honors and Awards

IEEE Electron Devices Society Distinguished Lecturer, 2004 - 2016.
Fellow, The Institute of Electrical and Electronic Engineers (IEEE), 2002 “For
contributions to the understanding and development of ultra-thin dielectrics and their
application to silicon devices”
Gordon Lepley IV Endowed Memorial Teaching Award, ECE Department, The
University of Texas at Austin, 2004

Cullen Trust For Higher Education Endowed Professorship in Engineering, 2000-present
 Dean's Fellow, College of Engineering, The University of Texas at Austin, 1999, 2003
 Lockheed Fort-Worth Division Award for Excellence in Engineering Teaching,
 College of Engineering, The University of Texas at Austin, 1996
 Award of Excellence, Halliburton Foundation, 1993
 Departmental Teaching Award, College of Engineering, The University of Texas at
 Austin, 1993
 SRC Inventor Recognition Award, Semiconductor Research Corporation, 1991
 Hughes Aircraft Company Endowed Faculty Fellowship in Engineering,
 The University of Texas at Austin, 1991- 2000
 Outstanding Engineering Teaching by an Assistant Professor, College of Engineering,
 The University of Texas at Austin, 1991
 Best Paper Award, SEMATECH Centers of Excellence Coordination Meeting, 1990.
 Dow Outstanding Young Faculty Award, American Society for Engineering Education,
 1990
 Engineering Research Initiation Award, Engineering Foundation of the United
 Engineering Trustees, 1989
 Best Paper Award, IEEE International Reliability Physics Symposium, 1988

Litigation Experience as Testifying Experts

The following is a list of cases in which I have provided expert testimony, including through a declaration, report or testimony at deposition or trial since 2016

- HD Silicon Solutions LLC v. NXP USA
 July 2026 –
 (Representing defendant, NXP)
- Marlin Semiconductor Limited v. TSMC
 December 2025 –
 (Representing defendant, TSMC)
- Longitude Flash Memory Solutions and IPValue Management v. SanDisk
 April 2025 – February 2026
 (Representing defendant, SanDisk)
- Marlin Semiconductor Limited v. TSMC
 April 2025 –
 (Representing defendant, TSMC)
- Advanced Integrated Circuit Process LLC v. Taiwan Semiconductor Mfg. Co. Ltd
 October 2024 –
 (Representing defendant, TSMC)
- Yangtze Memory Technologies v. Micron
 January 2024 –
 (Representing defendant, Micron)

- BeSang v. Micron
March 2023 –
(Representing defendant, Micron)
- The Trustees of Purdue University v. Wolfspeed, Inc.
February 2022 – March 2025
(Representing defendant, Wolfspeed)
- GlobalFoundries - IPR
March 2021 –
(Representing petitioner and filed an IPR declaration)
- HD Silicon Solutions LLC v. Microchip Technology Inc.
January 2021 – February 2025
(Representing defendant, Microchip Technology Inc.)
- Longitude Licensing Limited vs Western Digital
March 2023 – January 2024
(Representing defendant, Western Digital)
- Impinj, Inc. v. NXP USA, Inc.
September 2019 – July 2023
(Representing defendant, NXP USA, Inc.)
- Monterey Research, LLC v. Qualcomm Inc.
February 2020 – May 2022
(Representing defendant, Qualcomm)
- Bell Semiconductor, LLC v. Texas Instruments Incorporated
July 2020 – March 2021
(Representing defendant, Texas Instruments)
- Greenthread LLC v. Samsung
February 2020 – July 2020
(Representing plaintiff, Greenthread LLC)
- MLC Intellectual Property LLC vs. Micron Technology, Inc.,
June 2016 – January 2021
(Representing plaintiff, MLC Intellectual Property LLC)
- Hanwha v. LONGi Solar Technology
April 2019 – June 2020
(Representing defendant LONGi Solar Technology)
- Vista Peak Ventures, LLC v. Innolux Corp.

January 2019 – March 2019
(Representing IPR Petitioner, Innolux Corp.)

- Home Semiconductor Corp. v. Samsung
August 2018 – June 2020
(Representing plaintiff, Home Semiconductor Corp.)
- Mariner IC Inc. v. Asustek Computer Inc.
May 2018 – August 2019
(Representing plaintiff, Mariner IC Inc.)
- Godo Kaisha IP Bridge 1 v. OmniVision Technologies, Inc.
January 2018 – December 2018
(Representing defendant, OmniVision Technologies, Inc.)
- Macronix v. Toshiba
May 2017 – February 2018
(Representing defendant, Toshiba)
- University of Illinois v. Micron Technology
September 2009 – November 2017
(Representing plaintiff, Univ. of Illinois)
- Silicon Genesis v. Soitec
October 2016 – March 2017
(Representing plaintiff, Silicon Genesis)

Non-Litigation Consulting

- Applied Materials, Inc.
Teaching short courses on semiconductor physics, devices, design/layout, and fabrication technologies (2-3 courses per year).

Jack C. Lee is a Professor Emeritus of the Electrical and Computer Engineering Department. He held the Cullen Trust for Higher Education Endowed Professorship in Engineering # 4 at The University of Texas at Austin from 2000-2023. He received the B.S. and M.S. degrees in electrical engineering from University of California, Los Angeles, in 1980 and 1981, respectively; and the Ph.D. degree in electrical engineering from University of California, Berkeley, in 1988. From 1979 to 1984, he was a Member of Technical Staff at the TRW Microelectronics Center, CA, in the High-Speed Bipolar Device Program. He worked on bipolar device and circuit design, fabrication and testing. In 1988, he joined the faculty of The University of Texas at Austin. His current research

interests include semiconductor devices including 3D structures such as FinFETs, fabrication processes, characterization and modeling, dielectric process, characterization and reliability, high-K gate dielectrics and electrode, semiconductor memory applications, alternative channel materials, display technologies and photovoltaic/solar cell technologies. Dr. Lee has over 40 years of experience in semiconductor technology and dielectric processing. He has published over 600 journal publications and conference proceedings and 9 issued US patents; and co-authored 8 book and book chapters. He has also been recognized with many teaching and research awards. Dr. Lee is a Fellow of IEEE and had been a Distinguished Lecturer for the IEEE Electron Devices Society.

Publications

Jack C. Lee

Refereed Journal Publications

1. J. Lee, K. Mayaram and C. Hu, "A Theoretical Study of Gate/Drain Offset in LDD MOSFET's," IEEE Electron Device Letters, vol. EDL-7, no. 3, p. 152 - 154, March 1986.
2. J. Lee, I-C Chen and C. Hu, "Comparison Between CVD and Thermal Oxide Dielectric Integrity," IEEE Electron Device Letters, vol. EDL-7, no. 9, p. 506 - 509, September 1986.
3. K. Mayaram, J. Lee and C. Hu, "A Model for the Electric Field in Lightly Doped Drain Structures," IEEE Transactions on Electron Devices, vol. ED-34, no.7, p. 1509 - 1518, July 1987.
4. J. Lee and C. Hu, "Polarity Asymmetry of Oxides Grown on Polycrystalline Silicon," IEEE Transactions on Electron Devices, vol. ED-35, no. 7, p. 1063 - 1070, July 1988.
5. J. Lee, C. Hegarty and C. Hu, "Electrical Characteristics of MOSFET's Using Low-Pressure Chemical Vapor Deposited Oxide," IEEE Electron Device Letters, vol. EDL-9, no. 7, p. 324 - 327, July 1988.
6. J. Lee, I-C Chen and C. Hu, "Modeling and Characterization of Gate Oxide Reliability," Special Issue of IEEE Transactions on Electron Devices on Reliability, Vol. ED-35, no. 12, p. 2268 - 2278, December 1988.
7. H. Hwang, W. Ting, D. L. Kwong, J. Lee, L. Buhrow and R. Bowling, "Electrical Characteristics of Reoxidized-nitrided CVD Oxide," Applied Physics Letters, 55(8), p. 755 - 756, 21 August, 1989.
8. R. Moazzami, J. Lee and C. Hu, "Temperature Acceleration of Time-Dependent Dielectric Breakdown," Special issue of IEEE Transactions on Electron Devices on Vacuum Microelectronic Devices, vol. ED-36, no. 11, p. 2462 - 2465, November 1989.
9. H. Hwang, W. Ting, D.L. Kwong, J. Lee, L. Buhrow and R.A. Bowling, "Effects of Dynamic Stressing on Nitrided and Reoxidized-Nitrided Chemical Vapor Deposited Gate Oxides," IEEE Electron Device Letters, vol. EDL-10, no. 12, p. 568 - 570, December 1989.
10. J. Lin, S. Banerjee, J. Lee, and C. Teng, "Soft Breakdown in Titanium-Silicided Shallow Source/Drain Junction," IEEE Electron Device Letters, vol. EDL-11, no. 5, p. 191 - 193, May 1990.
11. K. Park, S. Batra, J. Lin, S. Yoganathan, J. Lee, S. Banerjee, S. Sun, J. Yeargain, and G. Lux, "Anomalous Capacitance-Voltage Behavior Due to Dopant Segregation and

Carrier Trapping in Arsenic-Implanted Polysilicon and Polycide Gates," *Applied Physics Letters*, vol. 56, no. 23, p.2325 - 2327, June 4, 1990.

12. J. Lin, S. Banerjee, J. Lee, and C. Teng, "Anomalous Current-Voltage Behavior in Titanium-Silicided Source/Drain Junctions," *Journal of Applied Physics*, vol. 68, no. 3, p. 1082 - 1087, August 1, 1990.
13. H. Hwang, W. Ting, B. Maiti, D.L. Kwong and J. Lee, "Electrical Characteristics of Ultrathin Gate Dielectrics Prepared by Rapid Thermal Oxidation of Si in N₂O," *Applied Physics Letters*, vol. 57, no. 10, p. 1010 - 1011, Sept. 3, 1990.
14. S. Batra, K. Park, J. Lin, S. Yoganathan, J. Lee, S. Banerjee, S. Sun, J. Yeargain and G. Lux "Effects of Dopant Redistribution, Segregation and Carrier Trapping in As-Implanted MOS Gates," *IEEE Transactions on Electron Devices*, vol. 37, no. 11, p. 2322 - 2330, October, 1990.
15. S. Bhattacharya, S. Banerjee, J. Lee, A. Tasch and A. Chatterjee, "The Impact of Trench Isolation on Latch-up Immunity in Bulk, Non-epitaxial CMOS," *IEEE Electron Device Letters*, vol. EDL-12, no. 2, p. 77 - 79, February, 1991.
16. W. Ting, H. Hwang, J. Lee and D. L. Kwong, "Composition and Growth Kinetics of Ultrathin SiO₂ Films Formed by Oxidizing Si Substrate in N₂O," *Applied Physics Letters*, vol. 57, p. 2808 - 2810, 1990.
17. W. Ting, P.C. Li, G. Q. Lo, J. Lee and D.L. Kwong, "Metal-Oxide Semiconductor Characteristics of Rapid Thermal Processed Chemical Vapor Deposited SiO₂ Gate Dielectrics," *Solid State Electronics*, vol. 34, no. 4, p 385 - 388, 1991.
18. W. Ting, H. Hwang, J. Lee and D.L. Kwong, "Growth Kinetics of Ultrathin SiO₂ Films Prepared by Rapid Thermal Oxidation of Si Substrates in N₂O," *Journal of Applied Physics*, vol. 70, no. 2, p. 1072 - 1074, July 15, 1991.
19. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee and G. Lux, "Enhancement of Boron Diffusion Through Gate Oxides in MOS Devices During Rapid Thermal Silicidation," *Applied Physics Letters*, vol. 58, no. 19, p. 2123 - 2125, May 1991.
20. H. Hwang, W. Ting, D. L. Kwong and J. Lee, "A Physical Model for Boron Penetration Through Oxynitride Gate Dielectric Prepared by Rapid Thermal Processing in N₂O," *Applied Physics Letters*, vol. 59, no. 13, p. 1581 - 1582, September 23, 1991.
21. H. Hwang, W. Ting, D. L. Kwong and J. Lee, "Improved Reliability Characteristics of Submicron nMOSFET's with Oxynitride Gate Dielectrics Prepared by Rapid Thermal Oxidation in N₂O," *IEEE Electron Device Letters*, vol. 12, no.9, p.495-497, September 1991.

22. J. Carrano, C. Sudhama, V. Chikarmane, J. Lee, A. Tasch, W. Shepherd and N. Abt, "Electrical and Reliability Properties of PZT Thin Films for ULSI DRAM Applications," IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control, vol. 38, no. 6, p. 690 - 703, November 1991.
23. V. Chikarmane, C. Sudhama, J. Kim, J. Lee, A. Tasch and S. Novak, "Comparative Study of The Perovskite Phase Microstructure Evolution and Electrical Properties of PZT Thin Film Capacitors Annealed in Oxygen and Nitrogen Ambients," Applied Physics Letters, vol. 59, no. 22, p. 2850 - 2852, Nov. 25, 1991.
24. M. Hao and J. Lee, "Electrical Characteristics of Oxynitrides Grown on Textured Single-Crystal Silicon," Applied Physics Letters, vol. 60, no. 4, p. 445 - 447, January 27, 1992 .
25. S. Bhattacharya, S. Banerjee, J. Lee, A. Tasch and A. Chatterjee, "Parametric Study of Latchup-Immunity of Deep Trench-Isolated, Bulk, Non-epitaxial CMOS," IEEE Trans. Elec. Dev., vol. 39, no. 4, p. 921 - 931, April 1992.
26. V. Chikarmane, J. Kim, C. Sudhama, J. Lee and A. Tasch, "Annealing of Lead Zirconate Titanate (65/35) Thin Films for Storage Dielectric Applications: Phase Transformations and Electrical Characteristics," Journal of Electronic Materials, vol. 21, no. 5, p. 503 - 512, May 1992.
27. V. Chikarmane, C. Sudhama, J. Kim, J. Lee and A. Tasch, "Effects of Post-Deposition Annealing Ambient on the Electrical Characteristics and Phase Transformation Kinetics of Sputtered Lead Zirconate titanate (65/35) Thin Film Capacitors," Journal of Vacuum Science and Technology, vol. 10, no. 4, p. 1562 - 1568, July 1992.
28. V.K. Mathews, R. L. Maddox, P.C. Fazan, J. Rosato, H. Hwang and J. Lee, "Degradation of Junction Leakage in Devices Subjected to Gate Oxidation in Nitrous Oxide," IEEE Electron Device Letters, vol. 13, no. 12, p. 648 - 650, Dec. 1992.
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32. H. Hwang, J. Lee, P. Fazan and C. Dennison, "Hot-Carrier Reliability Characteristics of Narrow Width MOSFETs," *Solid-State Electronics*, Vol. 36, No. 4, p. 665 - 666, 1993.
33. B. Maiti and J. Lee, "A New Low-Thermal Budget Process for Ultrathin Oxynitride Dielectrics," *Journal of Electronic Materials*, 1992.
34. J. Lin, W. Chen, S. Banerjee and J. Lee, "Cobalt Disilicide as a Dopant Diffusion Source for Polysilicon Gates in MOS Devices," *Journal of Electronic Materials*, vol. 22, p. 667 - 673, 1993.
35. W. Chen, J. Lin, S. Banerjee and J. Lee, "Thermal Stability and Dopant Drive-Out Characteristics of CoSi₂ Polycide Gates," *Journal of Applied Physics*, vol. 73, p. 4712 - 4714, May 1993.
36. J. Lee, V. Chikarmane, C. Sudhama, and J. Kim, "Sputtered PZT Thin Films for Memory Applications," *Journal of Integrated Ferroelectrics*, vol. 3, 1993.
37. C. Sudhama, J. Kim, J. Lee, W. Shepherd and E. Meyer, "The Effects of Lanthanum Doping on the Electrical Properties of Sol-Gel Derived Ferroelectric Lead Zirconate Titanate (PZT) for ULSI DRAM Applications," *Journal of Vacuum Science and Technology*, p. 1302 - 1309, July / Aug 1993.
38. M. Y. Hao, H. Hwang and J. Lee, "Silicon-Implanted SiO₂ for Non-volatile Memory Applications," *Solid-State Electronics*, vol. 36, p. 1321 - 1324, 1993.
39. M. Y. Hao, H. Hwang and J. Lee, "Memory Effects of Silicon-Implanted Oxides for EEPROM Applications," *Applied Physics Letters*, vol. 62, No. 13, p. 1530 - 1532, March 1993.
40. S. Batra, K. Picone, K. Park, S. Bhattacharya, S. Banerjee, J. Lee, M. Manning and C. Dennison, "Study of Lateral Non-Uniformity as a Function of Junction Depth in Ultra-Shallow Junctions and Its Effects on Leakage Behavior in As-Deposited Polycrystalline Si and Amorphous Si Diodes," *Solid-State Electronics* vol. 36, no. 7, p. 955 - 960, 1993.
41. J. Lee, C. Sudhama, J. Kim and R. Khamankar, (Invited Paper) "High Dielectric Constant Ferroelectric Thin Films for DRAM Applications", *Extended Abstracts of International Conf. on Solid State Devices and Materials*, p. 850 - 852, 1993.
42. W. M. Chen, S. K. Banerjee, and J. C. Lee, "Degradation mechanism and improvement of thermal stability of CoSi₂/polycrystalline Si layers," *Applied Physics Letters*, vol. 64, no. 12, p. 1505 - 1507, 1994.
43. W. M. Chen, J. Lin, S. K. Banerjee, and J. C. Lee, "Simultaneous Shallow-Junction Formation and Gate Doping p-channel Metal-Semiconductor-Oxide Field-Effect

Transistor Process Using Cobalt Silicide as a Diffusion/Doping Source," *Applied Physics Letters*, vol. 64, no. 3, p. 345 - 347, 1994.

44. M. Y. Hao, B. Maiti, and J. Lee, "Novel Process for Reliable Ultrathin Tunnel Dielectrics," *Appl. Phys. Lett.*, vol. 64, p. 2102 - 2104, April 1994.
45. B. Jiang, C. Sudhama, R. Khamankar, J. Kim and J. Lee, "Effects of Nonlinear Storage Capacitor on DRAM READ/WRITE," *IEEE Electron Device Letters*, vol. 15, no. 4, p.126 - 128, April 1994.
46. J. Kim, C. Sudhama, R. Khamankar, B. Jiang, J. Lee, P. Maniar, R. Moazzami, R. Jones and C. J. Mogab, "La Doped PZT Thin Films for Gigabit DRAM Technology," 1994 Symposium on VLSI Tech. Digest, p. 151 - 152, 1994.
47. M. Y. Hao, K. Lai, W. M. Chen, and J. Lee, " Surface Cleaning Effect on Dielectric Integrity for Ultrathin Oxynitrides Grown in N₂O," *Applied Physics Letters*, vol. 65, no. 9, p. 1133 - 1135, August 1994.
48. K. Lai, M. Y. Hao, C. Y. Hu, W. M. Chen, and J. Lee, " Effects of Surface Preparation on the Electrical and Reliability Properties of Ultrathin Thermal Oxide," *IEEE Electron Device Letters*, vol. ED-15, no. 11, p. 446 - 448, November, 1994.
49. M. Y. Hao, W. M. Chen, K. Lai, M. Gardner, J. Fulford, and J. Lee, "Correlation of Dielectric Breakdown with Hole Transport for Ultrathin Thermal Oxides and N₂O Oxynitrides," *Applied Physics Letters*, vol. 66, no. 9, p. 1126 - 1128, February 27, 1995.
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295. Yao-Feng Chang, Burt Fowler, Fei Zhou, Kwangsub Byun, Jack C. Lee, "Study of SiO_x-Based Resistive Switching Memory (ReRAM) in Integrated One Diode – One Resistor (1D-1R) Architecture" IEEE Semiconductor Interface Specialists Conference (SISC), December 10-13 (2014).
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EXHIBIT C

DAVID KUAN-YU LIU, Ph.D.

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PROFESSIONAL PROFILE

- Technical expert with over 20 years of industrial experience in the development of advanced CMOS devices and process technology, with special emphasis on non-volatile semiconductor memory devices and process technology, and high-voltage semiconductor devices, with additional 18 years of experience serving as expert consultant.
 - Named inventor on 99 US patents.
- Consultant and expert witness for past 18 years in semiconductor intellectual property evaluation and patent litigation, with deposition experience and successful ITC trial testifying experience.
- Successful track record in transitioning products from development phase to high-volume manufacturing phase, both as an individual contributor and as manager/director.
- Co-founder of an LLC in the development and licensing of intellectual property with emphasis on embedded flash memory technology.
- Co-founder of a semiconductor memory start-up company (Progressant Technologies, Inc.); participated in its successful sale to Synopsys, Inc.
- Fluent in Mandarin Chinese (native language) and English; over 10 years experience as real-time translator of church sermons and proficient in written translation.

EDUCATION

Stanford University, Palo Alto, California

Ph.D., Electrical Engineering (1989)

Thesis: Physics and Technology of Novel Conductivity-Modulated Power MOSFET's

M.S., Electrical Engineering (1985)

Coursework in integrated-circuit technology, semiconductor device physics, computer modeling of semiconductor devices and fabrication processes, and solid-state physics

University of California, Berkeley, California

B.S., Electrical Engineering (1983)

Certificate of Distinction (awarded annually to the top EECS graduate)

EXPERT WITNESS WORK - CASE EXPERIENCE:

Case: **Advanced Memory Technologies, Inc. v. Micron Technology, Inc.**
Jurisdiction: Civil Action No. 1:25-cv-01036 (W.D. Tex.)
Work Product: Expert consulting work relating to Booster Circuit. Patent analysis and prior art review toward preparation of EPR petition support and draft of declaration.
Nature of Case: EPR.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 11/25 – Present

Case: **Apple v. WeCrevention, Inc.**
Jurisdiction: Post-Grant Patent Matters Adverse to WeCrevention, Inc.
Work Product: Expert consulting work relating to High Speed Chip Memory Module. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: IPR.
Counsel: Retained by Fish & Richardson (Contact: Dr. Hyun Jin In)
Date: 11/25 – Present

Case: **Monterey Research LLC v. Renesas Electronics Corp.**
Jurisdiction: Civil Action No. 2:24-cv-00238-JRG (E.D. Tex.)
Work Product: Expert consulting work relating to Charge Trapping Nonvolatile Memory. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: Civil action and IPR.
Counsel: Retained by Morrison Foerster (Contact: Yuka Teraguchi)
Date: 03/25 – Present

Case: **Palisade Techs., LLP v. Micron Tech., Inc.**
Jurisdiction: Civil Action N.D. Cal, No. 7:24-cv-00262 (W.D. Tex.)
Work Product: Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents.
Nature of Case: Civil action and IPR.
Counsel: Retained by O'Melveny & Myers LLP (Contact: John Kappos)
Date: 12/24 – 01/26

Case:	Yangtze Memory Technologies Company, Ltd. vs. Micron Technology, Inc.
Jurisdiction:	Civil Action N.D. Cal, No. 3:223-cv-05792-RFL Easter District of Texas
Work Product:	Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 4 patents. <i>Deposed on 3 petitions that have been instituted.</i>
Nature of Case:	Civil action and IPR.
Counsel:	Retained by Orrick, Herrington & Sutcliffe (Contact: Jason Lang)
Date:	03/24 – Present
Case:	Force MOS Technology Co., Ltd. vs. Inergy Technology, Inc.
Jurisdiction:	Civil Action E.D. TX, No. 2:22-cv-00460-JRG Easter District of Texas
Work Product:	Expert consulting work relating to Vertical Trench DMOS process technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents. Declaration for claim construction on 3 patents. <i>Deposed on Claim Construction Declaration, and deposed on declarations for IPR petitions, and deposed on reply declaration.</i> Continuing work on claim construction and technical consulting.
Nature of Case:	Civil action and IPR.
Counsel:	Retained by Baker McKenzie (Contact: McKenzie Martin)
Date:	10/23 – 10/25
Case:	GlobalFoundries U.S. vs. International Business Machines
Jurisdiction:	03348-KMK-AEK (S.D.N.Y. 2023)
Work Product:	Expert consulting work relating to trade secrets involving advanced CMOS process and device technology (Nanosheet technology). Reviewed evidence and summarized findings.
Nature of Case:	Civil action.
Counsel:	Retained by Latham & Watkins (Contact: Sami Al-Marzoog)
Date:	10/23 – 2/25
Case:	Element Capital. vs. Beijing BOE Display Technology Co., Ltd.
Jurisdiction:	Civil Action No. 2:22-cv-00118-JRG Eastern District of Texas
Work Product:	Expert consulting work relating to OLED technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 3 patents.
Nature of Case:	Civil action.
Counsel:	Retained by Fish & Richardson (Contact: Linhong Zhang)
Date:	4/23 – 12/23

Case: **Sonrai Memory Ltd. vs. Micron Technology, Inc.,**
Jurisdiction: Civil Action No. 6:22-cv-000855-FB
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration resulting in institution of
the petition and ***deposed on supplemental declaration.***
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 12/22 – 7/24

Case: **Daedalus Prime LLC vs. Samsung Electronics Co.,**
Jurisdiction: Civil Action E.D. TX, Marshall Div., No. 2:22-cv-353
Eastern District of Texas, Marshall Div., and
Investigation No. 337-TA-1336
Work Product: Expert consulting work relating to FINFET process technology.
Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration on 2 patents. Source code
database viewing (GDS layout and process flow documents) for
supporting of non-infringement position.
Nature of Case: Civil action and ITC Investigation on Certain Semiconductor Devices,
Mobile Devices Containing the Same and Components Thereof..
Counsel: Retained by Fish & Richardson (Contact: Kevin Su)
Date: 10/22 – 4/23

Case: **Netlist, Inc. vs. Samsung Electronics Co. Ltd. et al.,**
Jurisdiction: Civil Action No. 2:21-cv-00463
Eastern District of Texas
Work Product: Expert consulting work relating to memory modules.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tony Nguyen)
Date: 8/22 – 12/22

Case: **Vervain, LLC vs. Micron Technology & related IPR Petition**
Jurisdiction: Action No. 6:21-cv-00487-ADA
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration on 4 patents, all 4 petitions
instituted, and ***deposed on both initial declaration and reply
declaration on all 4 patents.***
Counsel: Retained by Orrick, Herrington & Sutcliffe LLP (Contact: Jared
Bobrow)
Date: 8/21 – 2/23

Case:	Monterey Research vs. Qualcomm PTAB Matter
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Jones Day (Contact: David Cochran)
Date:	12/21 – 2/22
Case:	<i>Semiconductor Energy Laboratory Co., OLED Patent Matters</i>
Jurisdiction:	Action No. 8:21-cv-00554-JAK-ADS Central District of California
Work Product:	Expert for infringement on 3 patents relating to OLED driver circuit and layout, responsibilities include source code viewing, GDS layers viewing, and claim charts review.
Nature of Case:	Civil action.
Counsel:	Retained by Fish & Richardson (Contact: Bryan Basso)
Date:	9/21 – 4/22
Case:	<i>Apple Inc. IPR Petition for U.S. Patents 6,622,108 & 6,807,505</i>
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Erise IP (Contact: Paul Hart)
Date:	8/21 – 9/21
Case:	<i>Texas Instruments IPR Petition</i>
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Fish & Richardson (Contact: Josh Griswold)
Date:	9/20 – 4/21
Case:	<i>Micron Technology vs. UMC/JinHua</i>
Jurisdiction:	Case No. 4:17-CV-06932-MMC United States District Court for the Northern District of California
Work Product:	Testifying expert, trade secret documents review, and declaration submission.
Nature of Case:	Trade secret contention of Micron confidential documents.
Counsel:	Retained by Jones Day (Contact: Randy Kay)
Date:	3/17 – 12/23
Case:	<i>SK Hynix vs. BitMicro</i>
Jurisdiction:	Investigation No. 337-TA-1097 United States International Trade Commission
Work Product:	Testifying expert for non-infringement on 2 patents, issued expert rebuttal report, <i>deposition taken</i> , issued witness statement.
Nature of Case:	ITC infringement case concerning module stacking of SDRAM.
Counsel:	Retained by Fish Richardson (Contact: Joseph Colaianni)
Date:	11/18 – 4/19

Case: ***Rohm Company IPR Petition***
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Fish & Richardson (Contact: Matt Colvin)
Date: 4/18 – 8/18

Case: ***Apple Inc., vs. Qualcomm Incorporated***
Jurisdiction: Case No. 3:17-CV-02402
United States District Court for the Southern District of California
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration, ***2 depositions taken.***
Counsel: Fish & Richardson (Contact: Tony Nguyen)
Date: 1/18 – 5/19

Case: ***Macronix vs. Toshiba***
Jurisdiction: Investigation No. 337-TA-1046
United States International Trade Commission
Work Product: Testifying expert for infringement and non-invalidity on 2 patents, claim construction analysis, technical tutorial, database viewing for evidence gathering, issuing expert report and expert rebuttal report, ***deposition taken for 2 patents,*** issued witness statement and rebuttal witness statement, ***testified in ITC court on both infringement and invalidity.***
Nature of Case: ITC infringement case concerning advanced NAND Flash memory.
Counsel: Retained by Fish Richardson (Contact: David Barken)
Date: 4/17 – 10/18

Case: ***Novatek Microelectronics Corp. IPR Petition***
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Kroub, Silbersher & Kolmykov PLLC (Contact: Sergey Kolmykov)
Date: 2/17 – 4/17

Case: ***Taiwan Semiconductor Manufacturing Company IPR Petition***
Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel: Retained by Fish & Richardson (Contact: John Goetz)
Date: 7/15 – 4/17

Case: **Triune Systems, LLC vs. Wayne Chen, et al.,**
Jurisdiction: Case No. 296-03209-2013,
District Court of Collin County, TX
Work Product: Testifying expert, database viewing, expert report, ***video-taped testimony***
Nature of Case: Defending against trade secret contention.
Counsel: Retained by Gordon & Rees LLP (Contact: Robert Bragalone)
Date: 5/15 – 9/15

Case: **Macronix vs. Spansion**
Jurisdiction: Investigation No. 337-TA-922
United States International Trade Commission
Work Product: Testifying expert, claim construction analysis, technical tutorial, database viewing for evidence gathering, expert report, expert rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Fish Richardson (Contact: David Barken)
Date: 9/14 – 1/15

Case: **Macronix vs. Spansion**
Jurisdiction: Investigation No. 337-TA-909
United States International Trade Commission
Work Product: Database viewing for evidence gathering, support technical content of expert report, support technical content against rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Winston & Strawn (Contact Michael Murray)
Date: 5/14 – 1/15

Case: **Keranos, LLC v. Analog Devices et. al.**
Jurisdiction: Civil Case No. 2:10-CV-00207
United States District Court for the Eastern District of Texas
Work Product: Claim Chart Generation and Claim Construction
Nature of Case: Patent infringement case concerning split gate Flash memory cell
Counsel: Retained by Agility IP (Contact Michelle Breit)
Date: 2/09 – 2/14

Case: **Silicon Storage Technology, Inc. v. Xicor, LLC**
Jurisdiction: Civil Case No. CV 10-01515
United States District Court for the Northern District of California
Work Product: Expert report and declarations
Nature of Case: Matter concerning United States Reissue Patent RE38,370 on the process of flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 6/11 – 10/11

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3:10CV-481
United States District Court for the Northern District of Texas
Work Product: Declarations
Nature of Case: Defending Motion for Attorney Fees.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/10 – 6/10

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3-08cv0977M
United States District Court for the Northern District of Texas
Work Product: Testifying expert: issuing expert report, declarations, and ***deposition taken.***
Nature of Case: Patent infringement case concerning minimizing source current during erase of Flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/09 – 5/10

PROFESSIONAL EXPERIENCE:

10/07 to present: **Technical Consultant/Expert Witness**

Serving as a technical consultant and advisor in the area of semiconductor device and process technologies. Expert witness in ITC cases and district court cases, with successful testifying experience in trial hearing. Depositions taken at least 12 times for both litigation cases and IPR petitions. Experienced in source code viewings of circuit schematics and GDS layouts; seasoned in reverse engineering of integrated circuits, as well as techniques for de-processing and analyzing semiconductor chips. Responsibilities including gathering and reviewing of evidence, issuing declaration on claim construction, giving technical tutorial, issuing expert report and expert rebuttal report. Technical consultant for evaluating numerous flash memory and CMOS image sensor patent portfolios for licensing and acquisition consideration. Assessed qualitative merits of each patent portfolio and identified potential infringers and potential licensees. Generated numerous claim charts against potential infringers based on published patents, technical papers and datasheets, and reverse engineering reports.

10/07 to 4/11:

Jonker, LLC

Inventor and Partner. Developed an intellectual property (IP) portfolio (7 U.S. patents granted and 3 U.S. patent applications pending) on zero-cost CMOS-logic-compatible embedded flash memory. Successfully negotiated the acquisition of this IP portfolio by a major semiconductor company.

8/04 to 10/07:

Maxim Integrated Products, San Jose, California

Senior Scientist. Responsible for developing embedded non-volatile memory process technology for battery management products. Duties included supervising the development of integrated process flows at the 0.35 μ m and 0.18 μ m CMOS technology nodes. Analyzed and solved process-related reliability issues.

5/00 to 8/04:

Xilinx, Incorporated, San Jose, California

Senior Manager. Responsible for developing non-volatile memory process technology for flash and CPLD products, as well as advanced CMOS process technology (75nm node). Duties included supervising the development of integrated non-volatile memory process flows at multiple CMOS technology nodes to meet product application requirements. Analyzed and solved process-related reliability issues.

2/00 to 5/00

Progressant Technologies, Fremont, California

Co-Founder. Facilitated IP development and sale of the company to Synopsys, Inc.

9/98 to 2/00

Programmable Silicon Solutions, Sunnyvale, California

Director of Process Engineering. Responsible for developing a fully logic compatible embedded flash memory technology for WSMC (a semiconductor foundry). Developed a new integrated process flow to implement a proprietary flash cell in a high-performance logic process, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement. Applied interpersonal skills to align the goals of the foundry technology development group to that of PSS. Participated in the strategic planning of a technology roadmap and foundry strategy for the company.

9/97 to 9/98

AMIC Technology, Santa Clara, California

Director of Flash Technology. Responsible for developing 0.35μm ETOX-based flash memory technology. Prepared the business plan and technology roadmap. Developed a new integrated process flow, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement.

5/96 to 9/97

Altera Corp., San Jose, California

Device Engineering Manager. Responsible for technology development, process integration, and foundry interface for 0.35μm generation of logic and EEPROM programmable logic device technology. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, and product yield enhancement.

7/95 to 5/96

Information Storage Devices, San Jose, California

Technology Development Manager/Program Manager. Led a project team of 10 in technology development, process integration, and first product introduction, with a new foundry. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, resolution of sorting issues and yield enhancement.

5/92 to 7/95

Advanced Micro Devices, Sunnyvale, California

Member of Technical Staff. Led the development of low-energy/DINOR-type flash EPROM technology. Key individual contributor in optimizing the process and design of flash cell and periphery devices in AMD's 0.5μm and 0.35μm flash EPROM technologies. Duties included process integration, device modeling, and development of triple-well technology and high-voltage transistors for negative gate erase operation.

1/89 to 4/92

Texas Instruments, Dallas, Texas

Member of Technical Staff. Research and development of 16Mb generation of flash EPROM technology, as well as a new generation of antifuse-based FPGAs. Duties included process integration, device modeling, high-voltage CMOS process integration, and investigation of a novel source-side injection mechanism for EPROM channel hot-electron programming.

AWARDS/HONORS

U.C. Berkeley Certificate of Distinction Award, 1983

AEA Faculty Development Fellowship, 1983-1985

AMD Spotlight Award, 1995

PATENTS

1. U.S. Patent 5,106,773, "Programmable Gate Array and Methods for Its Fabrication" (with K.-L. Chen and H. Tigelaar), issued April 21, 1992.
2. U.S. Patent 5,166,557, "Gate Array with Built-in Programmable Circuitry" (with K.-L. Chen), issued November 24, 1992.
3. U.S. Patent 5,202,576, "Asymmetrical Non-Volatile Memory Cell, Arrays, and Methods for Fabricating the Same" (with M. Wong), issued April 13, 1993.
4. U.S. Patent 5,219,782, "Sublithographic Antifuse and method for Manufacturing" (with K.-L. Chen), issued June 15, 1993.
5. U.S. Patent 5,250,464, "Method of Making a Low Capacitance, Low Resistance Sidewall Antifuse Structure" (with M. Wong), issued October 5, 1993.
6. U.S. Patent 5,264,384, "Method of Making a Non-Volatile Memory Cell" (with C. Kaya), issued November 23, 1993.
7. U.S. Patent 5,300,803, "Source Side Injection Non-volatile Memory Cell," issued April 5, 1994.
8. U.S. Patent 5,365,105, "Sidewall Antifuse Structure and Method for Making," issued November 15, 1994.
9. U.S. Patent 5,371,402, "Low Capacitance, Low Resistance Sidewall Antifuse Structure and Process" (with M. Wong), issued December, 1994
10. U.S. Patent 5,395,797, "Antifuse Structure and Method of Fabrication," issued March 7, 1995.
11. U.S. Patent 5,470,773, "Method of Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch" (with Y. Sun and C. Chang), issued November 28, 1995.
12. U.S. Patent 5,482,880, "Non-Volatile Memory and Fabrication Method," (with C. Kaya), issued January 9 1996.
13. U.S. Patent 5,517,443, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued May 14, 1996.
14. U.S. Patent 5,521,867, "Adjustable Threshold Voltage Conversion Circuit" (with Jian Chen), issued May 28, 1996.
15. U.S. Patent 5,534,455, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued July 9, 1996.

16. U.S. Patent 5,541,875, "High Energy Buried Layer Implant To Provide A Low Resistance P-Well in A Flash EPROM Array" (with Jian Chen), issued May 28, 1996.
17. U.S. Patent 5,590,076, "Channel Hot-Carrier Page Write," issued December 31, 1996.
18. U.S. Patent 5,596,531, "Method for Decreasing the Discharge Time of A Flash Memory Cell," issued January 21, 1997.
19. U.S. Patent 5,612,914, "Asymmetrical Non-volatile Memory Cell, Arrays and Methods for Fabricating Same" (with Man Wong), issued May 18, 1997.
20. U.S. Patent 5,624,859, "Method for Providing Device Isolation and Off-State Leakage Current for a Semiconductor Device," issued April 29, 1997.
21. U.S. Patent 5,625,220, "Sublithographic Antifuse," (with K.L. Chen), issued April 29, 1997.
22. U.S. Patent 5,646,430, "Non-volatile Memory Cell Having Lightly-Doped Source Region" (with C. Kaya), Issued July 8, 1997.
23. U.S. Patent 5,650,964, "Method of Inhibiting Degradation of Ultra Short Channel Charge-carrying Devices during Discharge," (with Jian Chen), Issued July 22, 1997.
24. U.S. Patent 5,652,155, "Method for Making Semiconductor circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued July 27, 1997.
25. U.S. Patent 5,656,509, "Method and Test Structure for Determining Gouging in a Flash EPROM Cell During SAS Etch," Issued July 29, 1997
26. U.S. Patent 5,661,059, "Boron Penetration to Suppress Short Channel Effect in P-channel Device," Issued August 26, 1997.
27. U.S. Patent 5,674,764, "Method of Making Asymmetrical Non-volatile Memory Cell," Issued October 7, 1997.
28. U.S. Patent 5,693,972, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self-Aligned Source (SAS) Etch in a Semiconductor Device," Issued December 2, 1997.
29. U.S. Patent 5,751,631, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued May 12, 1998
30. U.S. Patent 5,789,295, "Method of Eliminating or Reducing Poly1 Oxidation at Stacked Gate Edge in Flash EPROM Process," Issued August 4, 1998.
31. U.S. Patent 5,814,854, "Highly scalable FLASH EEPROM cell," Issued September 29, 1998.
32. U.S. Patent 5,814,864, "Semiconductor Circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued September 29, 1998.

33. U.S. Patent 5,912,836, "Circuit for Detecting Both Charge Gain and Charge Loss Properties in a Non-Volatile Memory Array," Issued June 15, 1999.
34. U.S. Patent 5,930,174, "Circuit and Method for Erasing Flash Memory Array," Issued July 27, 1999.
35. U.S. Patent 5,981,994, "Method and Semiconductor circuit for Maintaining Integrity of Field Threshold Voltage Requirements," Issued November 9, 1999.
36. U.S. Patent 5,995,418, "Circuit and Method for Erasing Flash Memory Array," Issued November 30, 1999.
37. U.S. Patent 6,026,017, "Compact nonvolatile memory," Issued February 15, 2000.
38. U.S. Patent 6,027,974, "Nonvolatile memory," Issued February 22, 2000.
39. U.S. Patent 6,088,263, "Non-volatile memory using substrate electrons," Issued July 11, 2000.
40. U.S. Patent 6,091,636, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued July 18, 2000.
41. U.S. Patent 6,127,225, "Memory cell having implanted region formed between select and sense transistors," Issued October 3, 2000.
42. U.S. Patent 6,159,800, "Method of forming memory cell," Issued December 12, 2000.
43. U.S. Patent 6,188,604, "Flash memory cell & array with improved pre-program and erase characteristics," Issued February 13, 2001.
44. U.S. Patent 6,185,133, "Flash EPROM using junction hot hole injection for erase," Issued February 6, 2001.
45. U.S. Patent 6,252,799, "Device with embedded Flash and EERPOM memories," Issued June 26, 2001.
46. U.S. Patent 6,326,265, "Device with Embedded Flash and EPROM Memories," Issued December 4, 2001.
47. U.S. Patent 6,417,550, "High voltage MOS devices with high gated-diode breakdown voltage and punch-through voltage," Issued July 9, 2002.
48. U.S. Patent 6,479,862, "Charge trapping device and method for implementing a transistor having a negative differential resistance mode," Issued November 12, 2002.
49. U.S. Patent 6,512,274, "CMOS-process compatible, tunable NDR (negative differential resistance) device and method of operating same," Issued January 28, 2003.
50. U.S. Patent 6,596,617, "CMOS compatible process for making a tunable negative differential resistance (NDR) device," Issued July 22, 2003.

51. U.S. Patent 6,624,026, "Nonvolatile Memory," Issued September 23, 2003.
52. U.S. Patent 6,680,245, "Method for making both a negative differential resistance (NDR) device and a non-NDR device using a common MOS process," Issued January 20, 2004.
53. U.S. Patent 6,686,631, "Negative differential resistance (NDR) device and method of operating same," Issued February 3, 2004.
54. U.S. Patent 6,693,027, "Method for configuring a device to include a negative differential resistance (NDR) characteristic," Issued February 17, 2004.
55. U.S. Patent 6,700,155, "Charge trapping device and method for implementing a transistor having a configurable threshold," Issued March 2, 2004.
56. U.S. Patent 6,711,063, "EEPROM memory cell array architecture for substantially eliminating leakage," Issued March 23, 2004.
57. U.S. Patent 6,835,979, "Nonvolatile Memory," Issued December 28, 2004.
58. U.S. Patent 6,969,894, "Variable threshold semiconductor device and method of operating same," Issued November 29, 2005.
59. U.S. Patent 6,972,234, "High Voltage MOS Devices with High Gated-Diode Breakdown Voltage and Punch-through Voltage," Issued December 6, 2005.
60. U.S. Patent 6,972,465, "CMOS process compatible, tunable negative differential resistance (NDR) device and method of operating same," Issued December 6, 2005.
61. U.S. Patent 7,067,873, "Charge Trapping Device," Issued June 27, 2006.
62. U.S. Patent 7,091,077, "Method of directionally trimming polysilicon width," Issued August 15, 2006.
63. U.S. Patent 7,109,078, "CMOS compatible process for making a charge trapping device," Issued September 19, 2006.
64. U.S. Patent 7,301,194, "Shrinkable and highly coupled double poly EEPROM with Inverter," Issued November 27, 2007.
65. U.S. Patent 7,436,710, "EEPROM memory device with cell having NMOS in a P pocket as a control gate, PMOS program/erase transistor, and PMOS access transistor in a common well," Issued October 14, 2008.
66. U.S. Patent 7,535,758, "One or Multiple-Times Programming Device," Issued May 19, 2009.
67. U.S. Patent 7,782,668, "Integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 24, 2010.

68. U.S. Patent 7,787,295, "Integrated circuit embedded with Non-volatile Multiple-Time Programmable memory having variable coupling," Issued August 31, 2010.
69. U.S. Patent 7,787,304, "Method of making integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 31, 2010
70. U.S. Patent 7,787,309, "Method of operating integrated circuit embedded with Non-volatile One-Time Programmable and Multiple-Time Programmable memory," Issued August 31, 2010.
71. U.S. Patent 7,791,955, "Method of erasing a block of memory cells," Issued September 7, 2010.
72. U.S. Patent 7,835,184, "EEPROM memory cell with first-dopant-type control gate transistor, and second-dopant type program/erase and access transistors formed in common well," Issued November 16, 2010
73. U.S. Patent 7,835,186, "Method of programming a selected memory cell," Issued November 16, 2010.
74. U.S. Patent 7,852,672, "Integrated Circuit embedded with Non-Volatile programmable memory having variable coupling," Issued December 14, 2010
75. U.S. Patent 7,876,615, "Method of operating integrated circuit embedded with Non-Volatile programmable memory having variable coupling related application data," Issued January 25, 2011
76. U.S. Patent 7,920,426, "Non-volatile memory programmable through areal capacitive coupling," Issued April 5, 2011
77. U.S. Patent 7,944,750, "Multi-programmable Non-Volatile Memory Cell," Issued May 17, 2011
78. U.S. Patent 8,203,861, "Non-Volatile One-Time Programmable and Multiple-Time Programmable Device," Issued June 19, 2012
79. U.S. Patent 8,208,299, "Integrated Circuit Embedded with Non-Volatile Programmable Memory Having Variable Coupling and Separate Read/Write Paths," Issued June 26, 2012
80. U.S. Patent 8,300,470, "Two Terminal Programmable Hot Channel Electron Non-Volatile Memory," Issued October 30, 2012
81. U.S. Patent 8,305,805, "Common Drain Non-Volatile Multiple-Time Programmable Memory," Issued November 6, 2012
82. U.S. Patent 8,325,519, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Ratio," Issued December 4, 2012.

83. U.S. Patent 8,580,622, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued November 12, 2013.
84. U.S. Patent 8,582,342, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued November 12, 2013.
85. U.S. Patent 8,599,612, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Related Application data," Issued December 3, 2013.
86. U.S. Patent 8,705,263, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued April 22, 2014.
87. U.S. Patent 8,988,103, "Capacitively Coupled Logic Gate," Issued March 24, 2015.
88. U.S. Patent 9,064,192, "Automatic Valid Vote Count Storage using Secure Embedded Non-Volatile Memory," Issued June 23, 2015.
89. U.S. Patent 9,224,739, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued December 29, 2015.
90. U.S. Patent 9,305,931, "Zero Cost NVM Cell Using High Voltage Devices in Analog Process," Issued April 5, 2016.
91. U.S. Patent 9,324,438, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued April 26, 2016.
92. U.S. Patent 9,558,813, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued January 31, 2017.
93. U.S. Patent 9,570,161, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
94. U.S. Patent 9,570,175, "Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
95. U.S. Patent 10,061,738, "Ephemeral Peripheral Device," Issued August 28, 2018.
96. U.S. Patent 10,115,467, "One time Accessible (OTA) Non-Volatile Memory," Issued October 30, 2018.
97. U.S. Patent 10,198,889, "Automatic Valid Vote Count Storage Using Secure Embedded Non-Volatile Memory," Issued February 5, 2019
98. U.S. Patent 10,559,700, "Zero Cost ANVM Cell Using High Voltage Devices in Analog Process," Issued February 11, 2020.
99. U.S. Patent 10,629,607, "Semiconductor CMOS Non-Volatile Memory Device," Issued April 21, 2020

PUBLICATIONS

1. S.D. Leeke, D. K. Y. Liu and J. P. McVittie, "Plasma mode trench etching with direct hydrocarbon injection," *Materials Research Society Symposium Proceedings*, Vol. 68, pp. 21-27, 1986.
2. D. K. Y. Liu and J. D. Plummer, "A novel trench-injector power device with low ON-resistance and high switching speed," *IEEE Electron Device Letters*, Vol. 9, No. 7, pp. 321-323, 1988.
3. D. M. Boisvert, D. K. Y. Liu and J. D. Plummer, "Circuit approaches to increasing IGBT switching speed," *IEEE Journal of Solid-State Circuits*, Vol. 23, No. 5, pp. 1276-1279, 1988.
4. D. K. Y. Liu and J. D. Plummer, "Device physics and optimization of conductivity-modulated power MOSFET's," *IEEE Transactions on Electron Devices*, Vol. 35, No. 12, pp. 2457-2458, 1988.
5. D. K. Y. Liu, K.-L. Chen, H. Tigelaar, J. Paterson and S. O. Chen, "Scaled dielectric antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 12, No. 4, pp. 151-153, 1991.
6. M. Wong, D. K. Y. Liu, M. M. Moslehi and D. W. Reed, "Preoxidation treatment using HCl/HF vapor," *IEEE Electron Device Letters*, Vol. 12, No. 8, pp. 425-426, 1991.
7. D. K. Y. Liu, C. Kaya, M. Wong, J. Paterson and P. Shah, "Optimization of a source-side injection FAMOS cell for flash EPROM applications," *International Electron Devices Meeting Technical Digest*, pp. 315-318, 1991.
8. K.-L. Chen, D. K. Y. Liu, G. Misium, W. M. Gosney, S.-J. Wang, J. Camp and H. Tigelaar, "A sublithographic antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 13, No. 1, pp. 53-55, 1992.
9. K. T. San, C. Kaya, D. K. Y. Liu, T. P. Ma, and P. Shah, "A new technique for determining capacitive coupling coefficients in flash EPROMs," *IEEE Electron Device Letters*, Vol. 13, No. 6, pp. 328-331, 1992.
10. C. Kaya, D. K. Y. Liu, J. Paterson and P. Shah, "Buried source-side injection (BSSI) for flash EPROM programming," *IEEE Electron Device Letters*, Vol. 13, No. 9, pp. 465-467, 1992.
11. M. Wong, D. K. Y. Liu and S. S.-W. Huang, "Analysis of the subthreshold slope and the linear transconductance techniques for the extraction of the capacitance coupling coefficients of floating-gate devices," *IEEE Electron Device Letters*, Vol. 13, No. 11, pp. 566-568, 1992.
12. D. K. Y. Liu and J. D. Plummer, "Design and analysis of a new conductivity-modulated power MOSFET," *IEEE Transactions on Electron Devices*, Vol. 40, No. 2, pp. 428-438, 1993.
13. J. Z. Peng, S. Haddad, H. Fang, C. Chang, S. Longcor, B. Ho, Y. Sun, D. Liu, Y. Tang, J. Hsu, S. Luan, J. Lien, "Flash EPROM endurance simulation using physics-based models," In *International Electron Devices Meeting Technical Digest*, pp. 295-298, 1994.

14. J. Chen, J. Hsu, S. Luan, Y. Tang, D. Liu, S. Haddad, C. Chang, S. Longcor, J. Lien, "Short Channel enhanced degradation during discharge of Flash EEPROM memory Cell," *International Electron Devices Meeting Technical Digest*, pp. 331-334, 1995.
15. V. H. Chan and D. Liu, "An Enhanced Erase Mechanism During Channel Fowler-Nordheim Tunneling in Flash EPROM Memory Devices," *IEEE Electron Devices Letters*, Vol. 20, No. 3, pp. 140-142, 1999.

EXHIBIT D

PROFESSIONAL SUMMARY

Currently a Registered Professional Engineer (#79454) and recognized as an inventor on 31 US and a number of foreign patents, I am President of McAlexander Sound, Inc., and the Managing Director of McAlexander Sound Pte Ltd. I have focused my expertise to support a number of clients in product, process, and operations analysis and investigation. Fifty-two years of experience in microcircuit and semiconductor technologies has developed my skills in areas of circuit design and analysis, device fabrication and assembly, testing, marketing, control system design and analysis, manufacturing operations and respective areas of quality, reliability, and defect / failure analysis. I am also the President and CEO of MDFHoldings, Inc., an IP holding company currently engaged in the field of GPS Tracking, and was formerly, among others, a Manager with QM Partners, LP, supporting clients in IP management and licensing. I have:

- designed and managed development, testing, and evaluation of devices including Dynamic Random Access Memories (DRAMs), Static Random Access Memories (SRAMs), Charge Coupled Devices (CCDs), Shift Registers (SRs), and functional circuits including I/O buffers for address and data, decoders, clocks, sense amplifiers, fault tolerant, parallel-to-serial data paths for video applications, level shifters, converters, pumps, and logic, as well as wireless communication systems and MEMs applications;
- designed, built, and managed 4 physical / chemical analysis laboratories to assess cause of defect and reliability of products, including nanostructure integrated circuits, systems, and product packaging;
- managed operations including engineering, training, and quality assurance for device fabrication, assembly, test, analysis, and reliability assessment, as well as manufacturing control, each of which involved both volatile and non-volatile memory; testing, analysis, and control involving use of mechanical calibration and measuring equipment, including optical, scanning e-beam, IR, capacitive, and laser using phase contrast and Fast Fourier Transform (FFT) for High Aspect Ratio Inspection (HARI) applications; audio and video system design and installation;
- taught courses in solid state device physics, integrated circuit design, integrated circuit fabrication, and statistical control;
- provided expert services, investigating both process and design technologies of various devices (microprocessor and controller, memory, programmable logic, card, tag, module, mixed signal, custom, and other), systems (PC and peripheral, computer, control, laser measurement, switch, architecture, software, and other), and consumer products (medical, TV, telephone, VCR, facsimile, copier, lighting, game, and other);
- provided nuclear radiation hardness testing services for military and space clients; and

PROFESSIONAL SUMMARY

- managed the design and installation of audio sound and video systems for private and commercial enterprises.

From 1986-1990, I was Executive Vice President of EPI Technologies, Inc., prior to joining the staff at Cochran Consulting, Inc. where I served as senior managing consultant from 1991-2002. From 1972 to 1986, I was employed by Texas Instruments Incorporated.

EXPERIENCE PROFILE

1988-present

McAlexander Sound, Inc. (McASI) - Richardson, TX
President

- o System, Product, and Process investigation, expert witness services for protection of intellectual property;
- o Patent portfolio development and valuation;
- o Product liability and insurance claim investigation, expert witness services for matters involving such claims;
- o Quality Systems consulting and engineering;
- o Radiation Hardness Testing Technical Representative;
- o Acoustic system design/installation
- o Technical Advisor in High Aspect Ratio and Surface Contour Measurement using Direct-to-Digital Holography.

2005-present

McAlexander Sound Pte Ltd - Singapore
Managing Director

- o System, Product, and Process investigation, expert witness services for protection of intellectual property;
- o Patent portfolio development and valuation;
- o Contract consultation;
- o Not assigned any patents and is not engaged in any IP related licensing or litigation activities.
- o Consumer product marketing

2002-present

MDFHoldings, Inc. – Las Vegas, NV
CEO

- o IP holding and licensing company;
- o Assigned U.S. Patent Nos. 7,340,260 and 7,657,265 and EP 1 557 058 B1, each related to an object locating system;
- o Has not been and is not engaged in any IP related litigation or licensing activities.

2006-2020

QM Partners, L.P. – Texas
Manager

- o Management of development, licensing, prosecution and exploitation of assigned intellectual property;
- o Not assigned any patents and is not engaged in any IP related licensing or litigation activities.

EXPERIENCE PROFILE (continued)

- 2006-2020 **Guardian Technologies, LLC – Texas**
Manager
- o IP holding and licensing company;
 - o Litigated U.S. Patent No. 5,657,076 related to security surveillance 2009-2011 (patent expired 2015);
 - o Assigned U.S. Patent No. 8,334,775 related to RFID tracking (reassigned 2020 to personal);
 - o Not engaged in any IP related licensing or litigation activities.
- 2006-2020 **Appropriate Holdings, LLC – Delaware**
Manager
- o IP holding company;
 - o Not assigned any patents, has not been and is not engaged in any IP related licensing or litigation activities.
- 1996-2010 **RMC Management, LLP - Plano, TX**
Partner
- o Asset management.
- 1991-2002 **Cochran Consulting, Inc. (CCI) - Richardson, TX**
Managing Consultant
- o System, Product, and Process investigation, expert witness services for protection of intellectual property;
 - o Design, process, and product reliability;
 - o Defect and failure analysis.
- 1986-Nov'90 **EPI Technologies, Inc. - Richardson, TX**
Executive Vice President and Company Officer
- o Managed Advanced Technology Div., QA, and Engineering, including software program development;
 - o Developed strategic, space/energy market growth plans;
 - o Negotiated the acquisition of and managed a radiation company;
 - o Designed and managed physical analysis, radiation effects, and environmental stress laboratories, including optical and e-beam measurement;

EXPERIENCE PROFILE (continued)

- o Achieved > 30% annual revenue growth and profitability for each laboratory the first 12 months;
- o Product and Process investigation services for protection of intellectual property.

1972 – 1986 **Texas Instruments, Inc.** - Dallas/Houston, TX; Singapore

'84 - '86 **Quality/Reliability Assurance Manager, TI Dallas** Advanced DRAM semiconductor wafer fabrication facility

- o Developed/implemented on-line, computerized SPC software tools for dimensioning analysis and control and pattern recognition;
- o Coordinated people development, design-of-experiments;
- o Managed chemical and physical analysis laboratories;
- o Implemented control systems to assure product, process, material, equipment, and facility compliance, including Cost of Quality analysis.

'82 - '84 **Quality/Reliability Assurance and Engineering Manager, TI Singapore** assembly/test facility

- o Developed, implemented, and operated an effective Quality/Reliability Assurance program for assembly processing including optical pattern recognition for equipment registration;
- o Supervised 225 people for 7 day/week operation, including QRA, Computer Systems software development, and Training;
- o Trained engineers in Solid State Physics, device fabrication, and statistical process control.

'81 - '82 **Engineering Operations Manager, TI Houston**

- o Managed DRAM memory product cost center;
- o Responsible for division test software generation, product assembly and test quality / yield, cost reduction and quality improvement;
- o Provided technical customer interface for marketing;
- o Coordinated TI Singapore engineering test/assembly.

EXPERIENCE PROFILE (continued)

'79 - '81

Product Engineering Manager, TI Houston

- o Responsible for yield improvement, technical customer interface, quality improvement, design evaluation, and device characterization for DRAM and CCD products;
- o Developed device specifications and test software.

'72 - '79

Design Section Manager / Engineer, TI Houston

- o Responsible for design and development, process compatibility, production introduction of Dynamic Ram products;
- o Activities included electrical and physical layout, SPICE model simulation, test program generation, and product implementation for MOS Dynamic Ram products.

1969 - 1972

U. S. Army - Coventry, Rhode Island; Seoul, Korea
Captain, Air Defense Artillery

- o Served one year as Communications Officer in Korea;
- o Served two years as Tactical Officer, New England Defense.

ORGANIZATIONS, PUBLICATIONS, EDUCATION**PROFESSIONAL ORGANIZATIONS AND AWARDS**

- 1 - Institute of Electrical and Electronics Engineers, Inc. (IEEE), Senior Member. Societies:
Computer, Electron Devices, Solid State Circuits
- 2 - Licensing Executives Society (LES)
- 3 - National Society of Professional Engineers
- 4 - Texas Board of Professional Engineers, Registered License #79454
- 5 - American Society of Civil Engineers
- 6 - Architectural Engineering Institute
- 7 - 2000/2001 Nationwide Register's Who's Who in Executives and Businesses
- 8 - 1996/1997 Strathmore's Who's Who Registry of Business Leaders

PUBLICATIONS

- 1- NUS Proceedings of Engineering Convention '83, Aug '83, pgs. 139-142, The Memory Challenge.
- 2- Archives of Biochemistry and Biophysics, Dec'81, Vol. 212, No. 2, Equilibrium Constants under Physiological Conditions for the Reactions of D-3-Phosphoglycerate Dehydrogenase and L-Phosphoserine Aminotransferase.
- 3- International Electron Devices Meeting, Dec '79, pgs. 355-357, Sub 100ns 16K x 1 MOS Dynamic RAM Using a Grounded Substrate.

EDUCATION PROFILE

1980 - 1985	Taught Solid State Device Physics, Semiconductor Processing, Circuit Design Techniques, and Statistical Quality Control Methods Effectiveness Training and Japanese Manufacturing Techniques, Participative Problem Solving courses
1975 - 1976	1.5 years Graduate Study in Neural Science, the University of Texas Graduate School of Biomedical Science
1965 - 1969	BSEE, North Carolina State University

PATENTS (US-31, Foreign-8)

4,239,993	(1980) High Performance Dynamic Sense Amplifier with Active Loads
4,280,070	(1981) Balanced Input Buffer Circuit for Semiconductor Memory
4,288,706	(1981) Noise Immunity in Input Buffer Circuit for Semiconductor Memory
4,370,575	(1983) High Performance Dynamic Sense Amplifier with Active Loads
4,418,293	(1983) High Performance Dynamic Sense Amplifier with Multiple Column Outputs
4,533,843	(1985) High Performance Dynamic Sense Amplifier with Voltage Boost for Row Address Lines
4,543,500	(1985) High Performance Dynamic Sense Amplifier Voltage Boost for Row Address Lines
4,543,501	(1985) High Performance Dynamic Sense Amplifier with Dual Channel Grounding Transistor
4,748,349	(1988) High Performance Dynamic Sense Amplifier with Voltage Boost for Row Address Lines
6,172,640 B1	(2001) Pet Locator
6,236,358 B1	(2001) Mobile Object Locator
6,421,001 B1	(2002) Object Locator
6,441,778 B1	(2002) Pet Locator
6,480,147 B2	(2002) Portable Position Determining Device
6,518,919 B1	(2003) Mobile Object Locator
6,771,213 B2	(2004) Object Locator
6,859,171 B2	(2005) Mobile Object Locator

PATENTS (continued)

7,113,126 B2	(2006) Portable Positioning Determining Device
7,179,674 B2	(2007) Bi-Directional Released-Beam Sensor
7,209,075 B2	(2007) Mobile Object Locator
7,324,044 B2	(2008) Object Locator
7,336,227 B2	(2008) Portable Position Determining Device
7,340,260 B2	(2008) System and Method for Tracking the Location of Multiple Mobile Radio Transceiver Units (assigned to MDFHoldings, Inc.)
7,353,706 B2	(2008) Weighted Released-Beam Sensor
7,397,097 B2	(2008) Integrated Released Beam Layer Structure Fabricated in Trenches and Manufacturing Method Thereof
7,564,405 B2	(2009) Object Locator
7,657,265 B2	(2010) System and Method for Tracking the Location of Multiple Mobile Radio Transceiver Units (assigned to MDFHoldings, Inc.)
7,760,137 B2	(2010) Portable Positioning Determining Device
7,764,228 B2	(2010) Portable Positioning Determining Device
7,989,906 B2	(2011) Bi-Directional released-Beam Sensor
8,334,775 B2	(2012) RFID-Based Asset Security and Tracking System, Apparatus and Method (assigned to personal)
JP 55-053640 B4	(1980) Defect Resistant Semiconductor Memory Cell
JP 59-044720 B4	(1984) Semiconductor High Speed Read/Write Memory Unit
DE2935121 C2	(1980) Clock Voltage Generator for Semiconductor Memory with Reduced Power Dissipation
DE3043651 A1	(1981) Clock Voltage Generator for Semiconductor Memory with Reduced Power Dissipation

PATENTS (continued)

- | | |
|-----------------|---|
| GB2032211 B2 | (1980) High Performance Dynamic MOS Read/Write Memory |
| EP 1 557 058 B1 | (2011) System and method for tracking the location of multiple mobile radio transceiver units (assigned to MDFHoldings, Inc.)
[States: AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HU IE IT LI LU MC NL PT RO SE SI SK TR] |
| EP 1 676 809 B1 | (2010) Weighted released-beam sensor
[States: DE FR GB IT] |
| EP 1 676 810 B1 | (2010) Bi-directional released-beam sensor
[States: DE FR GB IT] |

CASES

Cases over at least the past 7 years, either active or closed, in which I have signed a Protective Order, have testified as an expert either at a trial, hearing, or deposition, or have submitted statements / opinions by declaration, affidavit, or report, are:

CASE¹	CASE NUMBER	LOCATION	YEAR	TYPE²
IMS v. Micron* (*Weil Gotchal/ Orrick)	1:14-cv-01480- RGA	DE	2015 - 2022	P,t
(Patents related to flash memory)				
3rd Eye Surveillance* v United States (*firm: Kennedy Law)	1:15-cv-00501- CFL	Court of Federal Claims, DC	2015 -	P,t,TH
(Patents related to real-time image processing to response agency)				
Vivint* v Alarm.com (*firm: Weil)	2:15-cv-00392- CW-BCW	UT	2017 - 2023	P,t
(Patents related to Electronic Message Delivery)				
Blitzsafe* v Bosch (*firm: Brown Rudnick)	2:17-cv-00105- JRG	EDTX, Marshall	2017 - 2019	P,t
(Patents related to audio device integration systems)				
Blitzsafe* v Daimler & Mercedes (*firm: Brown Rudnick)	2:17-cv-00422- JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				
Blitzsafe* v BMW (*firm: Brown Rudnick)	2:17-cv-00418- JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				
Blitzsafe* v Mazda (*firm: Brown Rudnick)	2:17-cv-00423- JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				

¹ * = Client

² P=Patent, C=Contract, ©=Copyright, TS=Trade Secret, AT=Antitrust; CA=Class Action; t=testified, TH=Trial/Hearing

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Blitzsafe* v Mitsubishi (*firm: Brown Rudnick)	2:17-cv-00430-JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				
Blitzsafe* v Subaru (*firm: Brown Rudnick)	2:17-cv-00421-JRG	EDTX, Marshall	2017 - 2019	P,t
(Patents related to audio device integration systems)				
Blitzsafe* v Tata & Jaguar (*firm: Brown Rudnick)	2:17-cv-00424-JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				
Blitzsafe* v Volvo (*firm: Brown Rudnick)	2:17-cv-00420-JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to audio device integration systems)				
AGIS* v HTC (*firm: Brown Rudnick)	2:17-00514-JRG	EDTX, Marshall	2017 - 2019	P,t
(Patents related to Patents related to interactive remote communications)				
AGIS* v LG (*firm: Brown Rudnick)	2:17-00515-JRG	EDTX, Marshall	2017 - 2019	P,t
(Patents related to Patents related to interactive remote communications)				
AGIS* v Apple (*firm: Brown Rudnick)	2:17-00516-JRG	EDTX, Marshall	2017 - 2019	P,t
(Patents related to Patents related to interactive remote communications)				
AGIS* v ZTE (*firm: Brown Rudnick)	2:17-00517-JRG	EDTX, Marshall	2017 - 2019	P
(Patents related to Patents related to interactive remote communications)				
ZF Micro Solutions* v. Tat Investment (*firm: Berger)	1-09-CV-134970	Santa Clara County	2018 - 2026	C,t,TH
(Business tort, unfair practice)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Univ of S FL* v BRIT (*firm: Dickinson Wright)	3:18-CV-0250-K	NDTX, Dallas	2018 - 2019	P,t
(Patent related to Workstation Interface – Digital Mammography)				
Cypress Insurance v Hynix* (*firm: Bird Marella)	C17-467 RAJ	WDWA, Seattle	2018 - 2019	C,t,TH
(Contract dispute related to product delivery)				
MPS* v Intersil (*firm: Perkins)	16-1125-LPS	DE	2018 - 2021	TS,t
(Trade Secret dispute)				
GoGo v Squire Patton Boggs* (*firm: Cozen O'Connor)	2016-L-007789	Cook County, IL	2018 - 2019	P
(Patent Malpractice claim)				
X2Y* v Intel (*firm: Dovel)	3:18-cv-1394-HZ	DOR, Portland	2018 - 2020	P
(Patents related to package layer metalization)				
Power Integration* v ON Semi (*firm: Fish & Richardson)	5:16-cv-06371- BLF	NDCA, San Jose	2019	P,t
(Patents related to power converter regulation)				
ON Semi v Power Integration* (*firm: Fish & Richardson)	1:17-cv-00247- LPS-CJB	DE	2019	P,t
(Patents related to power converter regulation)				
Semcon* v Amazon (*firm: Brown Rudnick)	2:18-CV-00192- JRG	EDTX, Marshall	2019	P
(Patent related to Adaptive Power Control)				
Semcon* v Asustek (*firm: Brown Rudnick)	2:18-CV-00193- JRG	EDTX, Marshall	2019	P
(Patent related to adaptive power control)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Semcon* v Kyocera (*firm: Brown Rudnick)	2:18-CV-00197-JRG	EDTX, Marshall	2019	P
(Patent related to adaptive power control)				
Innovation Sciences* v Amazon (*firm: Davidson Berquist)	4:18-cv-00474-ALM	EDTX, Sherman	2019 - 2020	P,t,TH
(Patent related to Internet communication)				
Innovation Sciences* v HTC (*firm: Davidson Berquist)	4:18-cv-00476-ALM	EDTX, Sherman	2019 - 2020	P,t
(Patent related to Internet communication)				
Innovation Sciences* v Resideo (*firm: Davidson Berquist)	4:18-cv-00475-ALM	EDTX, Sherman	2019 - 2020	P,t
(Patent related to Internet communication)				
Innovation Sciences* v Vector (*firm: Davidson Berquist)	4:18-cv-00477-ALM	EDTX, Sherman	2019	P
(Patent related to Internet communication)				
CXT* v JC Penny (*firm: Brown Rudnick)	2:18-cv-00171-RWS-RSP (lead) 2:18-CV-233-RWS-RSP	EDTX, Marshall	2019 - 2020	P,t
(Patents related to storage management and information distribution)				
Photonics* v Lenovo (*firm: Brown Rudnick)	1:18-cv-00489-MN	DE	2019	P
(Patents related to GDS-II Layout)				
Quest NetTech Corp* v Apple (*firm: Brown Rudnick)	2:19-CV-00118-JRG	EDTX, Marshall	2019 - 2020	P
(Patent related to programmable credit card system)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Pilot, Inc.* v Schumacher (*firm: Bradford)	1:19-cv-05982-EEC	NDIL, Eastern Div	2019 - 2021	P
(Patents related to battery chargers)				
Vocalife* v Amazon (*firm: Brown Rudnick)	2:19-cv-00123-JRG	EDTX, Marshall	2019 - 2020	P,t,TH
(Patent related to microphone array)				
Semcon* v Vuitton (*firm: Brown Rudnick)	2:19-CV-00122-JRG	EDTX, Marshall	2019	P,t
(Patent related to adaptive power control)				
Semcon* v Shenzhen (*firm: Brown Rudnick)	2:18-CV-00196-JRG	EDTX, Marshall	2019	P,t
(Patent related to adaptive power control)				
Ultravision* v GoVision (*firm: Brown Rudnick)	2:18-cv-00100-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v Shenzhen (*firm: Brown Rudnick)	2:18-cv-00103-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v Aoto (*firm: Brown Rudnick)	2:18-cv-00113-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v Leyard (*firm: Brown Rudnick)	2:18-cv-00102-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v NEC (*firm: Brown Rudnick)	2:18-cv-00150-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to Modular Display Panels and LED lighting)				
Ultravision* v Prismaflex (*firm: Brown Rudnick)	2:18-cv-00108-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Ultravision* v Unilumin (*firm: Brown Rudnick)	2:18-cv-00116-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v Yaham (*firm: Brown Rudnick)	2:18-cv-00118-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Ultravision* v Absen (*firm: Brown Rudnick)	2:18-cv-00112-JRG-RSP	EDTX, Marshall	2019	P
(Patents related to modular display panels and LED lighting)				
Birchett* v City of Fort Worth (*firm: Kennedy Law)	DC-19-06941	Dallas County	2019	C,t
(Whistleblower)				
Innovation Sciences* v HTC (*firm: Davidson Berquist)	337-TA-1180	ITC	2020 - 2021	P,t
(Patent related to wireless communication)				
Innovation Sciences* v Resideo (*firm: Davidson Berquist)	337-TA-1180	ITC	2020	P,t
(Patent related to Internet communication)				
CXT* v Neiman Marcus (*firm: Brown Rudnick)	2:19-cv-00270-RWS-RSP	EDTX, Marshall	2020	P
(Patents related to storage management and information distribution)				
CXT* v General Nutrition (*firm: Brown Rudnick)	2:19-cv-00271-RWS-RSP	EDTX, Marshall	2020	P
(Patents related to storage management and information distribution)				
CXT* v Steven Madden (*firm: Brown Rudnick)	2:19-cv-00272-RWS-RSP	EDTX, Marshall	2020	P
(Patents related to storage management and information distribution)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Consolidated* v Amazon (*firm: Devlin Law)	1:19-cv-01715- RGA	DE	2020 – 2021	P,t
(Patents related to on-line shopping)				
Consolidated* v Ford (*firm: Devlin Law)	1:19-cv-01916- RGA	DE	2020 - 2021	P,t
(Patents related to on-line shopping)				
Consolidated* v JC Penney (*firm: Devlin Law)	1:19-cv-01918- RGA	DE	2020 - 2021	P,t
(Patents related to on-line shopping)				
Consolidated* v Ebay (*firm: Devlin Law)	1:19-cv-01580- RGA	DE	2020 - 2021	P,t
(Patents related to on-line shopping)				
Consolidated* v Walmart (*firm: Devlin Law)	1:19-cv-01581- RGA	DE	2020 - 2021	P,t
(Patents related to on-line shopping)				
Semcon* v TCT Mobile (*firm: Brown Rudnick)	2:18-CV-00194- JRG	EDTX, Marshall	2020	P
(Patent related to adaptive power control)				
BNR v Samsung* (*firm: Fish & Richardson)	2:19-cv-00286- JRG	EDTX, Marshall	2020	P
(Patents related to portable communication devices)				
AGIS* v Google (*firm: BRudnick, Fabricant)	2:19-cv-00361- JRG	EDTX, Marshall	2020 - 2022	P,t
(Patents related to interactive remote communications)				
AGIS* v Waze (*firm: BRudnick, Fabricant)	2:19-cv-00359- JRG	EDTX, Marshall	2020 - 2022	P,t
(Patents related to interactive remote communications)				
Burke* v City of Fort Worth (*firm: Kennedy Law)	DC-19-07239	Dallas County	2020	C,t
(Whistleblower)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Infinera v Oyster Optics* (*firm: Davidson Berquist)	IPR2020-00325	PTAB	2020	P,t
(Patent related to fiber optic telecommunications)				
Blitzsafe* v GM (*firm: Fabricant)	2:19-cv-00377-JRG	EDTX, Marshall	2020	P,t
(Patents related to audio device integration systems)				
Blitzsafe* v FCA (*firm: Fabricant)	2:19-cv-00378-JRG	EDTX, Marshall	2020	P,t
(Patents related to audio device integration systems)				
Kuster v Western Digital* (*firm: Shearman & Sterling)	6:20-cv-00563	WDTX, Waco	2020	P,t
(Patent related to memory stack storage)				
TruSun* v Eaton (*firm: Duane Morris)	6:19-cv-00656-ADA	WDTX, Waco	2020 - 2022	P,t
(Patent related to a light-generating system)				
RJR v PMI* (*firm: Latham Watkins)	1:20-cv-00393-LO-TCB	EDVA, Alexandria	2020 - 2022	P,t,TH
(Patents related to e-cigarette controls)				
Koss* v Apple et al (*firm: K L Gates)	6:20-cv-00665	WDTX, Waco	2020 - 2022	P,t
(Patents related to Wireless Headphones)				
Koss* v Bose (*firm: K L Gates)	6:20-cv-00661	WDTX, Waco	2020	P
(Patents related to Wireless Headphones)				
Ipcom v Sprint et al (Nokia*) (*Alston Bird)	2:20-cv-321	EDTX, Marshall	2020 - 2022	P
(Patents related to CDMA communication)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Ipcom v Verizon et al (Nokia*) (*Alston Bird)	2:20-cv-323	EDTX, Marshall	2020 - 2022	P
(Patents related to CDMA communication)				
Pilot* v CarKu (*firm: Bradford / Sheridan)	7131799991	Amazon Arbitration	2020	P,t
(Patent related to jump starters)				
MPS* v Meraki (*firm: Perkins Coie)	6:2020cv00876- ADA	WDTX, Waco	2020 - 2022	P,TS,C
(Patents related to synchronous rectifier control)				
AGIS* v Uber (*firm: Fabricant)	2:21-cv-00026- JRG-RSP)	EDTX, Marshall	2021	P
(Patents related to interactive remote communications)				
AGIS* v WhatsApp (*firm: Fabricant)	2:21-cv-00029- JRG-RSP)	EDTX, Marshall	2021	P
(Patents related to interactive remote communications)				
One E-Way* v Apple (*firm: Knobbe Martin)	2:20-cv-06339- JSK-PD	CDCA	2021 - 2023	P,t
(Patent related to Bluetooth Headphones)				
BMW v Omega Patents* (*firm: Allen Dyer)	IPR2021-00181	PTAB	2021	P,t
(Patent related to remote start)				
Pilot* v The NOCO Co (*firm: Bradford / Sheridan)	2:20-cv-01452- SRB	AZ	2021 - 2023	P,t
(Patents related to auto chargers)				
Express Mobile* v Atlassian (*firm: Mololamken)	6:20-cv-00805- ADA	WDTX, Austin	2021	P
(Patents related to browser-based web site generation)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
AGIS* v Lyft (*firm: Fabricant)	2:21-cv-00024-JRG-RSP	EDTX, Marshall NDCA	2021 - 2022	P
(Patents related to interactive remote communications)				
AGIS* v T-Mobile (*firm: Fabricant)	2:21-cv-00072-JRG-RSP	EDTX, Marshall	2021	P
(Patents related to interactive remote communications)				
Altria* v RJR (*firm: Weil Gotshal)	1:20-cv-00472	MDNC	2021 - 2022	P,t,TH
(Patents related to e-cig)				
Apple v One E-Way* (*firm: Knobbe Martin)	IPR2021-00283 IPR2021-00286	PTAB	2021 - 2022	P,t
(Patent related to Bluetooth Headphones)				
Altria* v RJR (*firm: Weil Gotshal)	IPR2021-00650 IPR2021-00652 IPR2021-00725 IPR2021-00744 IPR2021-00745 IPR2021-00746 IPR2021-00747 IPR2021-00793	PTAB	2021 - 2022	P,t
(Patents related to e-cig control)				
Kiwi Connection* v Anker (*firm: Bradford)	8519300821	Amazon Arbitration	2021	P,t
(Patent related to bidirectional connector)				
Kiwi Connection* v JBL (*firm: Bradford)	8674855301	Amazon Arbitration	2021	P,t
(Patent related to bidirectional connector)				
Platform Science v Omnitracs* (*firm: Kirkland & Ellis)	IPR2020-0187 IPR2020-1517 IPR2020-1518	PTAB	2021 – 2022	P,t
(Patent related to critical event reporting)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Netlist v Samsung* (*firm: Bird Marella)	8:20-cv-993-MCS	CDCA, Southern	2021	C,t
(Contract dispute)				
Qualcomm v Monterey Research* (*firm: Desmarais)	IPR2021-00119 IPR2021-00120	PTAB	2021 - 2022	P,t
(Patents related to semiconductor processing)				
Nanya v Monterey Research* (*firm: Desmarais)	IPR2021-00172	PTAB	2021 - 2022	P,t
(Patent related to semiconductor processing)				
Apple / Bose v Koss* (*firm: K L Gates)	IPR2021-00255 IPR2021-00297 IPR2021-00305 IPR2021-00381 IPR2021-00592	PTAB	2021 - 2022	P,t
(Patents related to Wireless Headphones)				
Vocalife* v Amazon2 (*firm: Fabricant)	2 :20-cv-00401- JRG-RSP	EDTX, Marshall	2021 - 2023	P,t
(Patent related to microphone array)				
Vocalife* v Google (*firm: Fabricant)	2 :21-cv-00124- JRG-RSP	EDTX, Marshall	2021 - 2023	P,t
(Patent related to microphone array)				
Luxshare* v Amphenol (*firm: K L Gates)	IPR2022-00132	PTAB	2021 - 2022	P,t
(Patent related to high performance connector)				
Vervain v Micron* (*firm: Orrick)	6:20-cv-487-ADA	WDTX, Waco	2021 - 2023	P,t
(Patents related to MLC & SLC endurance)				
Express Mobile* v Dropbox (*firm: Mololamken)	3:21-cv-01145	NDCA	2021	P
(Patents related to browser-based web site generation)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Lexmark v Universal Imaging* (*firm: Allen Dyer)	8:18-cv-1047-T-17AEP	MDFL, Tampa	2021 - 2023	P
(Patents related to printer cartridge authentication)				
The NOCO Co v Pilot* (*firm: Bradford / Sheridan)	IPR2021-00777 IPR2021-01232 IPR2023-00167 IPR2023-00810	PTAB	2021	P,t
(Patents related to auto chargers)				
Delta T* v MacroAir (*firm: Latham/VedderPrice)	5:20-cv-00728-GW-GJS	CDCA, LA	2022 - 2023	P
(Patents related to fan sensors and control)				
MPS* v Dong & Sheng (*firm: Perkins Coie)	4 :20-cv-6752-JSW	NCCA, Oakland	2022 - 2023	P, TS
(Patents related to synchronous rectifier control)				
Vocalife* v Sonos (*firm: Fabricant)	2 :21-cv-00129-JRG	EDTX, Marshall	2022	P,t
(Patent related to microphone array)				
Vocalife* v Bose (*firm: Fabricant)	2 :21-cv-00128-JRG	EDTX, Marshall	2022	P,t
(Patent related to microphone array)				
Enovsys* v T-Mobile (*firm: Eichmann)	2:21-cv-00368-JRG	EDTX, Marshall	2022	P,t
(Patents related to precise location determination and secure disclosure)				
Enovsys* v Verizon (*firm: Eichmann)	2:21-cv-00315-JRG-RSP	EDTX, Marshall	2022	P,t
(Patents related to precise location determination and secure disclosure)				
Lyft v AGIS* (*firm: Fabricant)	5 :21-cv-04653-BLF	NDCA, San Jose	2022 - 2023	P
(Patents related to interactive remote communications)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Amazon v Vocalife* (*firm: Fabricant)	IPR2021-01331 IPR2022-00004 IPR2022-00005	PTAB	2022	P,t
(Patent related to microphone array)				
MPS* v Meraki (*firm: Perkins Coie)	4:22-cv-01986 JSW	NDCA, Oakland	2022 - 2023	P,TS,C
(Patents related to synchronous rectifier control)				
Signode v Polychem* (*firm: Latham)	1 :22-cv-00519- UNA	DE	2022	P
(Patents related to a strapping apparatus)				
Pilot* v Hulkman (*firm: BRADFORD)	10427111251	Amazon Arbitration	2022	P,t
(Patent related to jump starters)				
Netlist v Samsung* (*firm: Fish & Richardson)	2:21-cv-463	EDTX, Marshall	2022 - 2024	P,t,TH
(Patents related to DIMM DDR3 signal timing, power management, and rank multiplication)				
Orange* v Autel (*firm: Sughrue Mion)	2 :21-cv-240	EDTX, Marshall	2022 - 2023	P,t,TH
(Patent related to Tire Pressure Monitoring Systems)				
Bell Power v MPS* (*firm: Perkins Coie)	6 :21-cv-00655- ADA	WDTX, Waco	2022 -	P
(Patents related to power converter/regulator control)				
VPR v Jupiter* (*firm: Noroozi PC)	2:20-CV-02185- DJH	AZ	2023	P,t
(Patent related to electronic vape device with air flow sensor)				
Multimodal* v Guangdong (*firm: Fabricant)	2:21-cv-00436- JRG-RSP	EDTX, Marshall	2023	P
(Patents related to communication over a network, call completion)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Multimodal v ZTE (*firm: Fabricant)	2:21-cv-00437-JRG-RSP	EDTX, Marshall	2023	P
(Patents related to communication over a network, call completion)				
Entropic* v Charter Communications et al. (*firm: K L Gates)	2 :22-cv-00125-JRG	EDTX, Marshall	2022	P
(Patents related to modems and broadband content distribution)				
Oasis Tooling* v Global Foundries U.S. (*firm: Kramer Levin)	22-00312-CJB	DE	2022 - 2024	P,t
(Patents related to chip design process workflow)				
Oasis Tooling* v Siemens Industry Software (*firm: Kramer Levin)	22-00151-CJB	DE	2022 - 2024	P,t
(Patents related to chip design process workflow)				
AGIS* ITC (*firm: Fabricant)	337-TA-1347	ITC	2023	P
(Patents related to Location-Sharing Systems)				
Monterey* v Nanya (*firm: Desmarais)	19-2090-NIQA-LAS	DE	2023 – 2024	P
(Patents directed to semiconductor technology)				
Geotab v Omega* (*firm: Allen Dyer)	IPR2023-00504	PTAB	2023	P,t
(Patent related to multi-vehicle communication)				
Adv Lighting Concepts* v mate LLC (*firm: Munsch)	IPR2023-01257 IPR2023-01259 IPR2023-01264 IPR2023-01265	PTAB	2023	P
(Patent related to LED drivers)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
SitePro* v Waterbridge (*firm: Perkins Coie)	6:23-cv-00115- ADA-DTG	WDTX, Waco	2023	P,TS,C
(Patents and Trade Secrets related to remote delivery control)				
Nanya v Monterey* (*firm: Botos Churchill)	EPR 6,825,526C1 EPR 6,680,516	USPTO	2023	P,t
(Patents directed to semiconductor technology)				
ICPillar* v ARM (*firm: Eichmann)	6_23-cv-00115- ADA-DTG	WDTX, Austin	2023 - 2024	P
(Patents related to computer method for IC design)				
GoTV Streaming v Netflix* (*firm: Wilkie Farr)	EPR 8478245 EPR 8989715	USPTO	2023	P,t
(Patent related to rendering content on a wireless device)				
Greenthread v MPS* (*firm: Perkins Coie)	23-579-RGA	DE	2024	P
(Patents related to semiconductor structure with graded doping)				
MPS* v REED (*firm: Perkins Coie)	1:23-cv-1155-JFM 1:24-cv-165-JFM 1:24-cv-166-JFM	DE	2023 -	P,t
(Patent related to bootstrap refresh control)				
Verizon v Omega* (*firm: Allen Dyer)	IPR2023--01162	PTAB	2023	P,t
(Patent related to multi-vehicle communication)				
MOSAID v MediaTek* (*firm: Fish & Richardson)	2:23-cv-00129- JRG-RSP	EDTX, Marshall	2023 - 2024	P
(Patents related to IC Power Management)				
GoTV Streaming v Netflix* (*firm: Wilkie Farr)	2 :22-cv-07556- RGK-SHK	CDCA	2023	P
(Patent related to rendering content on a wireless device)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Lifetime* v Qima (*firm: Fabricant)	2:23-cv-00216-JRG	EDTX, Marshall	2024	P,t
(Patents related to Quality Management)				
Stellar* v Motorola Solutions (*firm: Devlin)	4:23-cv-00750-SDJ	EDTX, Sherman	2024 -	P,t
(Patents related to write protect recording)				
Enovsys* v Lyft (*firm: Seth Law)	23-cv-05157-EJD	NDCA, San Jose	2024	P
(Patents related to mobile unit tracking)				
ARM v ICPillar* (*firm: Lowenstein)	IPR2024-00476 IPR2024-00566	PTAB	2024	P,t
(Patents related to computer method for IC design)				
Maxell v TCL* (*firm: King & Spalding / Merchant & Gould)	5:23-CV-00108-RWS-JBB	EDTX, Texarkana	2024 - 2025	P,t
(Patents related to image display apparatus)				
Enovsys* v Uber (*firm: Seth Law)	23-cv-04549-EJD	NDCA, San Jose	2024	P
(Patents related to mobile unit tracking)				
Uber v Enovsys* (*firm: Seth Law)	IPR2024-00825 IPR2024-00826 IPR2024-00827	PTAB	2024	P,t
(Patents related to mobile unit tracking)				
JUUL v NJOY* (*firm: Weil Gotshal)	337-TA-1368 IPR2024-00231	ITC PTAB	2024	P,t,TH
(Patent related to vaping devices)				
RJR v Guangdong et al.* (*firm: Mei & Mark)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
RJR v Kangvape* (*firm: Fish & Richardson)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v Breeze et al.* (*firm: Goodwin Procter)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v Pastel et al.* (*firm: Keller Heckman)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v Shenzhen Yanyang et al.* (*firm: Rimon)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v Maduro* (*firm: Thompson Hine)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v TheSy* (*firm: Thompson Hine)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
RJR v SV3* (*firm: Sterne Kessler)	337-TA-1410	ITC	2024 - 2025	P,t
(Patent related to vaping devices)				
REED v MPS* (*firm: Perkins Coie)	IPR2024-00871	PTAB	2024 -	P,t
(Patent related to bootstrap refresh control)				
SoundClear* v Amazon (*firm: Daignault Iyer)	1 :24-cv-00728	EDVA, Alexandria	2024 -	P,t
(Patents related to audio control and noise reduction)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
SoundClear* v Google (*firm: Daignault Iyer)	1 :24-cv-00729	EDVA, Alexandria	2024 -	P
(Patents related to audio control and noise reduction)				
SoundClear* v Google (*firm: Daignault Iyer)	IPR2025-00344 IPR2025-00345	PTAB	2025 - 2026	P
(Patents related to audio control and noise reduction)				
SoundClear* v Amazon (*firm: Daignault Iyer)	IPR2025-00565 IPR2025-00673	PTAB	2025 - 2026	P
(Patents related to audio control and noise reduction)				
Innolux v Phenix* (*firm: Womble Bond)	IPR2025-00043 IPR2025-00044	PTAB	2025	P,t
(Patents related to Gamma reference generation)				
Hisense v Phenix* (*firm: Womble Bond)	IPR2025-01004	PTAB	2025	P,t
(Patents related to Gamma reference generation)				
Phenix* v Innolux (*firm: Womble Bond)	2:23-cv-00478- RWS-RSP	EDTX, Marshall	2024 - 2025	P
(Patents related to Gamma reference generation)				
Phenix* v AU Optronics) (*firm: Womble Bond)	2_23-cv-00477- RWS-RSP	EDTX, Marshall	2024 - 2026	P,t,TH
(Patents related to Gamma reference generation)				
8KHZ* v Meta (*firm: Brown Rudnick)	622-cv-00575- ADA-DTG	WDTX	2025 - 2026	P
(Patents related to sound localization)				
MyChoice* v TAIIV (*firm: Perkins Coie)	2:23-CV-0507- JRG-RSP	EDTX, Marshall	2025	P,t,TH
(Patents related to sound localization)				
Longhorn AG* v Hyundai (*firm: Fabricant)	2 :24-cv-00554- JRG-RSP	EDTX, Marshall	2025	P,t
(Patents related to vehicle control systems)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Longhorn AG* v Nissan (*firm: Fabricant)	2 :24-cv-00397-JRG-RSP	EDTX, Marshall	2025	P
(Patents related to vehicle control systems)				
Longhorn AG* v VW (*firm: Fabricant)	2 :24-cv-00933-JRG-RSP	EDTX, Marshall	2025 -	P,t
(Patents related to vehicle control systems)				
Longhorn AG* v Mazda (*firm: Fabricant)	2 :24-cv-00686-JRG-RSP	EDTX, Marshall	2025 -	P
(Patents related to vehicle control systems)				
Longhorn AG* v Volvo (*firm: Fabricant)	2 :24-cv-00603-JRG-RSP	EDTX, Marshall	2025 -	P
(Patents related to vehicle control systems)				
Longitude* v TSMC et al (*firm: Mintz ...)	337-TA-1443	ITC	2025 -	P
(Patents related to semiconductor/IC processing)				
Longitude* v TSMC et al (*firm: Mintz ...)	IPR2025-00847 IPR2025-00848 IPR2025-00864 IPR2025-00865 IPR2025-00879 IPR2025-01054 IPR2025-01082 IPR2025-01265 IPR2025-01444 IPR2025-01484 IPR2025-01527 IPR2026-00058 IPR2026-00061 IPR2026-00130 IPR2026-00220 IPR2026-00261	PTAB	2025 -	P,t
(Patents related to semiconductor/IC processing)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Semisilicon v Changzhou JE* (*firm: Rimon)	1:25-cv-07388	NDIL	2025 -	P
(Patents related to LED lamps and systems)				
Zepp* v Amazwear et al. (*firm: Rimon)	2:24-cv-07182- JLS-AGR	CDCA	2026 -	P
(Patents related to wearable device sensors)				
Smart Speaker* v Amazon (*firm: Fabricant)	2:25-cv-00707- JRG	EDTX, Marshall	2026 -	P
(Patents related to server based control systems)				
Cellspin Soft v SZ DJI Tech*: (*firm: EviLaw / Cleary)	2 :25-cv-948	EDTX, Marshall	2026 -	P
Patents related to image transmission)				
MR Licensing* v Seagate (*firm: Fabricant)	2:25-CV-00596- JRG	EDTX, Marshall	2026 -	P
(Patents related to memory authentication and data relocation)				
MPS* v REED (*firm: Cleary)	EPR	PTAB	2026	P,t
(Patent related to switching power supplies)				
UMBRA* v Juniper (*firm: Devlin)	1:24-cv-01288- JLH-CJB	DE	2026	P
(Patents related to network architecture, information flow and routing)				
UMBRA* v PAN (*firm: Devlin)	2:25-CV-00635- JRG-RSP	EDTX, Marshall	2026	P
(Patents related to network architecture, information flow and routing)				
UMBRA* v Zscaler (*firm: Devlin)	2:25-CV-01063- JRG-RSP	EDTX, Marshall	2026	P
(Patents related to network architecture, information flow and routing)				
Longhorn AG* v Mitsubishi (*firm: Fabricant)	2_24-cv-00685- JRG-RSP	EDTX, Marshall	2026 -	P
(Patents related to vehicle control systems)				

Cases (continued)

CASE ¹	CASE NUMBER	LOCATION	YEAR	TYPE ²
Skybell* v Alarm.com (*firm: Cleary)	1:25-cv-01105- RDA-WBP	EDVA, Alexandria	2026 -	TS
(Investigation related to Trade Secrets)				

Cases (continued)

I have worked with other clients in various areas of my expertise, including “system, product, and process investigation,” “patent valuation,” “product liability and insurance claim investigation,” “quality systems consulting and engineering,” and “IP licensing.” This work generally relates to patents and/or trade secrets and may involve analysis of products either defensively or offensively. In no case does any of the work involve design of circuits, processes, packaging, software, or systems.

Non-confidentially, I have represented a radiation effects testing company, ICS Radiation Technology. Further, I have worked with investment companies, such as Hatcreek Partners, reviewing potential investment opportunities, and have participated as a technical advisor to nLine Corporation, a company that developed a product for semiconductor wafer inspection using holographic High Aspect Ratio Inspection (HARI) technology. Work associated with each of these matters occurred more than 10 years ago.

These and other non-confidentially related companies are:

The Aspire International Foundation (2019-present)

- Partner;
- Rediscovering and redefining identity and purpose through experiential equine learning programs;
- Not assigned any patents and not engaged in any IP related activities.

Casualty Consulting Group of America (2013-present)

- Partner;
- Assessment of structural damage caused by extreme weather events;
- Not assigned any patents and not engaged in any IP related activities.

VCSY (2013-2020)

- Advisory board Member;
- Technology related to encrypted communication.

Bethel Cannon Group / Bethel Cannon Holdings (2011-present)

- Partner;
- Counseling / Event / Retreat Center / Hunting Lodge;
- Not assigned any patents and not engaged in any IP related activities.

Spirit Song Youth Equestrian Academy / Spirit Song Holdings (2011-present)

- CEO;
- Equine Assisted Learning Program for abused / traumatized youth, families, corporations, and churches;
- Not assigned any patents and not engaged in any IP related activities.

Cases (continued)E³A (2016-present)

- President (2018-2023);
- Certification training of Equine Practitioners;
- Not assigned any patents and not engaged in any IP related activities.

Novo Tellus Capital Partners (2016-2017)

- Novo Tellus is an investment firm. I served as a technical advisor in reviewing the technology and contracts of investment opportunities.

UTAC (2008-2016)

- Technical consultant in matters related to packaging test and assembly, patents, and contracts.

ICS Radiation Technology (1988-2009)

- Consulting work related to nuclear radiation effects testing.

Creative Management Consultants (CMC) (2003-2004)

- Consulting work related to Internet services, such as access service to clients and web site hosting services; providing business co-op services and internet product purchasing sites.

Hatcreek Partners (1999-2003)

- Hatcreek Partners is an investment firm. I served as a technical advisor in reviewing the technology of investment opportunities.

nLine Corporation (1999-2003)

- Member of Technical Advisory Board;
- nLine Corp.'s business related to semiconductor holographic High Aspect Ratio Inspection (HARI) technology.

Texas Instruments (pre 2002)

- Patent evaluation and application consulting.

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EXHIBIT E

CARLA S. MULHERN

Senior Advisor

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Ms. Mulhern specializes in the application of microeconomic principles to issues arising in complex business litigation. She has served as an expert witness on damages issues in commercial litigation matters, including intellectual property and breach of contract cases. Her intellectual property damages experience spans cases involving allegations of patent, copyright, and trademark infringement, as well as misappropriation of trade secrets. Ms. Mulhern has testified before the International Trade Commission on economic issues such as commercial success, domestic industry, remedy, bond, and public interest. In addition, she has worked with leading academic experts on commercial litigation matters including intellectual property, antitrust, and breach of contract cases.

Ms. Mulhern's project experience spans a variety of industries, including automotive, computer hardware and software, consumer products, entertainment, medical devices, pharmaceuticals, semiconductors, and telecommunications. She has assisted clients in all aspects of litigation projects, including development and review of pretrial discovery, development of economic and financial models to analyze damages, critique of analyses propounded by opposing experts, and preparation of testimony.

In non-litigation matters, Ms. Mulhern has assisted clients in valuing intellectual property and other business assets in the context of strategic alliances involving licensing and joint ventures. In addition, she has consulted on matters involving the application of economic principles to issues arising in the pharmaceutical and health care fields. Ms. Mulhern has been recognized as among the top economic experts for IP matters by Intellectual Asset Management (IAM) in the IAM Patent 1000, which identifies leading patent professionals around the globe.

EDUCATION

- | | |
|------|--|
| 1988 | M.Sc. in Economics, London School of Economics and Political Science, London, England |
| 1987 | Diploma with Merit in Economics, London School of Economics and Political Science, London, England |
| 1986 | B.S. in Mathematics, Bucknell University, Lewisburg, PA |

PROFESSIONAL EXPERIENCE

- | | |
|--------------|---|
| 1997–Present | Analysis Group, Inc., Washington, DC |
| 1994–1997 | Putnam, Hayes & Bartlett, Inc., Washington, DC |
| 1989–1994 | National Economic Research Associates, Inc., Washington, DC |

EXPERT REPORTS/TESTIMONY

- **In the Matter of Certain Vaporizer Devices, Cartridges Used Therewith, and Components Thereof (II), Investigation No. 337-TA-1460**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2026) and deposition testimony (2026) on behalf of complainant, JUUL Labs Inc, regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, bond, and commercial success in case involving e-cigarettes.
- **In the Matter of Certain Wearable Electroencephalogram Devices and Systems and Components Thereof, Investigation No. 337-TA-1458**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2026) and deposition testimony (2026) on behalf of complainant, Ceribell, Inc., regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, bond, public interest and commercial success in case involving point-of-care electroencephalogram (EEG) systems.
- **In the Matter of Certain Wi-Fi Routers, Wi-Fi Devices, Mesh Wi-Fi Network Devices and Components Thereof, Investigation No. 337-TA-1454**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2025 and 2026), deposition testimony (2026) and trial testimony (2026) on behalf of complainant, Estelgia LLC, regarding economic issues related to domestic industry, bond, commercial success, and public interest in case involving Wi-Fi routers, Wi-Fi devices, and mesh Wi-Fi network devices.
- **Janssen Pharmaceuticals, Inc. et al., v. Alkem Laboratories Ltd., et al.**
US District Court, District of New Jersey
Hatch-Waxman: expert report (2025), deposition testimony (2025) and trial testimony (2026) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product, Spravato, a nasal spray indicated for use in treatment resistant depression (TRD) and for treating depressive symptoms in adults with major depressive disorder with acute suicidal ideation or behavior.
- **Pfizer Inc., FoldRX Pharmaceuticals, LLC, PF Prism IMB B.V., Wyeth LLC, and The Scripps Research Institute, v. Dexcel Pharma Technologies Limited, et al.**
US District Court, District of Delaware
Hatch-Waxman: expert report (2025) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical products, Vyndamax and Vyndaqel, used in the treatment of transthyretin cardiac amyloidosis (ATTR-CM).
- **In the Matter of Certain TOPcon Solar Cells, Modules, Panels, Components Thereof, and Products Containing the Same, Investigation No. 337-TA-1422/1425**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2025) and deposition testimony (2025) on behalf of Respondents, Adani Solar USA Inc. and Mundra Solar PV Ltd., regarding economic issues related to domestic industry, in case involving solar cells and modules.

- **Maxell Ltd. v. TCL Electronics Holdings Ltd., et al.**
US District Court, Eastern District of Texas, Texarkana Division
Patent infringement: expert report (2025) and deposition testimony (2025) on behalf of plaintiff regarding reasonable royalty damages in case involving technology related to smart televisions.
- **OssiFi-Mab, LLC v. Amgen, Inc.**
US District Court, District of Massachusetts
Patent infringement: expert reports (2024 and 2025) and deposition testimony (2025) on behalf of plaintiff regarding reasonable royalty damages and commercial success in case involving treatment of osteoporosis.
- **In the Matter of Certain Semiconductor Devices and Products Containing the Same, Investigation No. 337-TA-1414**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2025) deposition testimony (2025), and trial testimony (2025) on behalf of complainants, Infineon Technologies Americas Corp. and Infineon Technologies Austria AG regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, and the appropriate amount of bond during the Presidential Review Period in case involving semiconductor chips.
- **Panasonic Holdings Corporation vs. Xiaomi Technology UK Ltd., et al. – UK, Claim No. HP-2023-000025**
In the High Court of Justice Business and Property Courts of England and Wales Intellectual Property List (ChD) Patents Court
Patent Infringement: expert reports (2024) on behalf of claimant, Panasonic Holdings Corporation, regarding analysis of FRAND licensing terms in European litigation involving standard essential patents related to 3G and 4G cellular technology.
- **In the Matter of Certain Surface Cleaning Devices and Components Thereof, Investigation No. 337-TA-1404**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2024) and deposition testimony (2024) on behalf of complainants, SharkNinja Operating LLC, Omachron Alpha, Inc., and Omachron Intellectual Property, Inc., regarding economic issues related to domestic industry and the appropriate amount of bond during the Presidential Review Period in case involving vacuum cleaners.
- **In the Matter of Certain Smart Wearable Devices, Systems, and Components Thereof, Investigation No. 337-TA-1398**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2024), deposition testimony (2024), and trial testimony (2024) on behalf of complainants, Ouraring, Inc. and Oura Health Oy, regarding economic issues related to domestic industry, cease and desist order, the amount and significance of inventory of accused products, the appropriate amount of bond during the Presidential Review Period, public interest, and commercial success in case involving smart ring wearable devices.

- **In the Matter of Certain Liquid Coolers for Electronic Components in Computers, Components Thereof, Devices for Controlling Same, and Products Containing Same, Investigation No. 337-TA-1394**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2024), deposition testimony (2024), and trial testimony (2024) on behalf of complainants, Cooler Master Co., Ltd., CMI USA, Inc., and CMC Great USA, Inc., regarding economic issues related to domestic industry in case involving liquid coolers for electronic components in computers.
- **Spinal Generations, LLC v. DePuy Synthes, Inc., Synthes USA, LLC, DePuy Synthes Sales, Inc., and Synthes USA Products, LLC**
US District Court, District of Delaware
Patent infringement: expert reports (2024) and deposition testimony (2024) on behalf of defendants regarding analysis of economic issues related to secondary considerations of non-obviousness and reasonable royalty damages in case involving orthopedic implants.
- **Charter Communications Operating, LLC v. Sonus Networks, Inc. and Ribbon Communications Operation Company, Inc.**
Supreme Court of the State of New York, New York County
Breach of contract: expert report (2024) and deposition testimony (2024) on behalf of plaintiff regarding damages arising from alleged breach of contract related to indemnification provisions for Voice-over-Internet-Protocol (“VoIP”) equipment resulting from an underlying patent infringement litigation.
- **Charter Communications Holding Company, LLC, Charter Communications Operating, LLC, and Bright House Networks, LLC v. Sonus Networks, Inc. and Ribbon Communications Operation Company, Inc.**
Superior Court of the State of Delaware, New Castle County
Breach of contract: expert report (2024) and deposition testimony (2024) on behalf of plaintiffs regarding damages arising from alleged breach of contract related to indemnification provisions for Voice-over-Internet-Protocol (“VoIP”) equipment resulting from an underlying patent infringement litigation.
- **Altor BioScience, LLC and NantCell, Inc. v. HCW Biologics and Hing C. Wong**
JAMS Arbitration, Los Angeles Resolution Center
Misappropriation of trade secrets and breach of contract: expert report (2024) and hearing testimony (2024) on behalf of Complainant, Altor BioScience, regarding unjust enrichment damages in case involving biopharmaceutical products.
- **UUSI, LLC d/b/a Nartron v. Samsung Electronics Co., Ltd. and Samsung Electronics American, Inc.**
US District Court, Western District of Michigan, Southern Division
Patent infringement: expert report (2024) and deposition testimony (2024) on behalf of defendants regarding reasonable royalty damages associated with patent related to capacitive touch sensor technology used in mobile tablets.

- **Actelion Pharmaceuticals Ltd. and Actelion Pharmaceuticals US, Inc. v. Bristol-Myers Squibb Co., Celgene Corp., and Celgene International II Sarl**
US District Court, District of New Jersey
Patent infringement: expert reports (2024) and deposition testimony (2024) on behalf of plaintiffs regarding reasonable royalty damages associated with patent related to a pharmaceutical product indicated for treatment of multiple sclerosis and ulcerative colitis.
- **The Trustees of the University of Pennsylvania v. Genentech, Inc.**
US District Court, District of Delaware
Patent infringement: expert reports (2024) and deposition testimony (2024) on behalf of defendants regarding reasonable royalty damages and equitable intervening rights associated with patents related to a method of treating breast cancer.
- **Dynamics, Inc. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Samsung Research America, Inc.**
US District Court, Southern District of New York
Patent infringement: expert report (2024) and deposition testimony (2024) on behalf of defendants regarding reasonable royalty damages associated with a patent allegedly directed to magnetic secure transmission (“MST”) technology included in Samsung Pay and incorporated in select model smartphones and smart watches.
- **In the Matter of Certain Vaporizer Devices, Cartridges Used Therewith, and Components Thereof, Investigation No. 337-TA-1368**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2024), deposition testimony (2024), and trial testimony (2024) on behalf of complainants, Juul Labs, Inc. and VMR Products LLC, regarding economic issues related to domestic industry, the appropriate amount of bond during the Presidential Review Period, and public interest in case involving vaporizer devices and cartridges.
- **In the Matter of Certain Wi-Fi Routers, Wi-Fi Devices, Mesh Wi-Fi Network Devices, and Hardware and Software Components Thereof, Investigation No. 337-TA-1361**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2023), deposition testimony (2023) and trial testimony (2024), on behalf of complainant, Netgear, regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, and the appropriate amount of bond during the Presidential Review Period, in case involving wireless networking equipment.
- **Puma Biotechnology, Inc. and Wyeth LLC v. Astrazeneca Pharmaceuticals LP and Astrazeneca AB**
US District Court, District of Delaware
Patent infringement: expert report (2023), deposition testimony (2023), and trial testimony (2024) on behalf of defendant regarding reasonable royalty damages associated with patents related to a pharmaceutical product used to treat non-small cell lung cancer (NSCLC).
- **RegenXBio, Inc. and The Trustees of the University of Pennsylvania v. Sarepta Therapeutics, Inc. and Sarepta Therapeutics Three LLC**
US District Court, District of Delaware
Patent infringement: expert report (2023) and deposition testimony (2023) on behalf of defendants regarding reasonable royalty damages for patents related to gene therapy used to treat muscular dystrophy.

- **Janssen Pharmaceuticals, Inc. and Janssen Pharmaceutica NV v. Tolmar, Inc.**
US District Court, District of Delaware
Hatch-Waxman: expert report (2023), deposition testimony (2023), and trial testimony (2023) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Invega Sustenna, an antipsychotic drug for the treatment of schizophrenia and schizoaffective disorder.
- **Continuous Composites, Inc. v. Markforged, Inc.**
US District Court, District of Delaware
Patent infringement: expert report (2023), deposition testimony (2023), and trial testimony (2024) on behalf of defendant regarding reasonable royalty damages for patents related to 3D printer technology.
- **California Safe Soil, LLC v. KDC Agribusiness, LLC, KDC Agribusiness Fairless Hills, LLC, KDC Agribusiness North Dakota, LLC, Do Good Foods LLC, Do Good Foods Managed Services LLC, Do Good Foods Facility Management LLC, Do Good Chicken LLC, Harold N. Kamine, Justin Kamine, Matthew Kamine, and Barry Starkman**
Delaware Chancery Court
Breach of contract, misappropriation of trade secrets, and other tortious conduct: expert reports (2022, 2024), deposition testimonies (2023, 2024), and trial testimony (2024) on behalf of plaintiffs regarding lost profits and unjust enrichment damages resulting from alleged unfair acts in case involving the manufacturing process for fertilizer and animal feed.
- **In the Matter of Certain Audio Players and Components Thereof (II), Investigation No. 337-TA-1330**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2023) and deposition testimony (2023) on behalf of complainant, Google LLC, regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, the appropriate amount of bond during the Presidential Review Period, and commercial success in a case involving connected smart home electronic devices.
- **Sparkman & Stephens Holdings, LLC and Sparkman & Stephens, LLC v. Mystic Seaport Museum, Inc.**
US District Court, District of Rhode Island
Copyright infringement and breach of contract: expert report (2022) and deposition testimony (2023) on behalf of plaintiffs regarding damages in a case involving architectural plans for building ships.
- **In the Matter of Certain Soft Projectile Launching Devices, Components Thereof, Ammunition, And Products Containing Same, Investigation No. 337-TA-1325**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2022), deposition testimony (2023), and trial testimony (2023) on behalf of complainants, Hasbro, Inc. and Spin Master, Inc., regarding economic issues related to domestic industry, the amount and significance of inventory of accused products, and the appropriate amount of bond during the Presidential Review Period in investigation involving toy guns with gel pellet ammunition.

- **In the Matter of Certain Graphics Systems, Components Thereof, and Digital Televisions Containing the Same, Investigation No. 337-TA-1318**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2022) and deposition testimony (2023) on behalf of complainants, Advanced Micro Devices, Inc. and ATI Technologies ULC, regarding economic issues related to domestic industry in investigation involving semiconductor chips.
- **In the Matter of Certain Movable Barrier Operator Systems and Components Thereof, Investigation No. 337-TA-1209 – Enforcement Proceeding**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2022), deposition testimony (2022), and hearing testimony (2023) on behalf of complainants, Overhead Door Corporation and GMI Holdings, Inc., to analyze economic issues related to the statutory maximum penalty associated with the alleged violation of the remedial order.
- **In the Matter of Certain Cellular Base Station Communication Equipment, Components Thereof, and Products Containing Same, Investigation No. 337-TA-1302**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2022), deposition testimony (2022), and hearing testimony (2022) on behalf of respondents, Ericsson AB, Telefonaktiebolaget LM Ericsson, and Ericsson Inc., regarding economic issues related to public interest in case involving 5G mmWave cellular communications equipment.
- **Janssen Pharmaceuticals, Inc., Janssen Pharmaceutica NV, Janssen Research & Development, LLC v. Mylan Laboratories Limited**
US District Court, District of New Jersey
Hatch-Waxman: expert report (2022), deposition testimony (2022), and trial testimony (2022) on behalf of plaintiffs regarding analysis of commercial success of patented dosing regimen of Invega Trinza (paliperidone palmitate), used to treat patients with schizophrenia.
- **Merck Sharp & Dohme B.V. and Merck Sharp & Dohme Corp. v. Aurobindo Pharma Ltd., Aurobindo Pharma USA, Inc., Cadila Healthcare Limited, Dr. Reddy's Laboratories, Inc., Dr. Reddy's Laboratories Ltd., Gland Pharma Ltd, Lek Pharmaceuticals d.d., LifeStar Pharma LLC, Mankind Pharma Limited, Mylan API US LLC, Mylan, Inc., Mylan Pharmaceuticals, Inc., Sandoz, Inc., Sun Pharmaceutical Industries, Inc., Sun Pharmaceutical Industries Ltd., Teva Pharmaceuticals USA, Inc., USV Private Limited, and Zydus Pharmaceuticals USA, Inc.**
US District Court, District of New Jersey
Hatch-Waxman: expert report (2022) and deposition testimony (2022) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Bridion, an injectable solution of sugammadex sodium used to reverse neuromuscular block in patients undergoing surgery.
- **In the Matter of Certain Smart Thermostat, Load Control Switches, and Components Thereof, Investigation No. 337-TA-1277**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2022), deposition testimony (2022), and trial testimony (2022) on behalf of respondents Alarm.com Holdings, Inc.; Alarm.com, Incorporated; EnergyHub, Inc.; ecobee, Inc.; Itron, Inc.; Resideo Smart Homes Technology (Tiajun); and Ademco, Inc. regarding economic issues related to domestic industry and the appropriate amount of bond during the Presidential Review Period in case involving load control switches, smart thermostats, and AMI meters.

- **Vifor Fresenius Medical Care Renal Pharma Ltd. and Vifor Fresenius Medical Care Renal Pharma France S.A.S. v. Teva Pharmaceuticals USA, Inc.**

US District Court, District of Delaware

Hatch-Waxman: expert report (2022) and deposition testimony (2022) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Velphoro, a phosphate binder used for patients with chronic kidney disease on dialysis.

- **In the Matter of Certain Fitness Devices, Streaming Components Thereof, and Systems Containing Same, Investigation No. 337-TA-1265**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2022), and trial testimony (2022) on behalf of respondents iFIT, Inc., FreeMotion Fitness, Inc., NordicTrack, Inc., lululemon athletica inc., Curiouser Products, Inc., and Peloton Interactive, Inc. regarding economic issues related to domestic industry and the appropriate amount of bond during the Presidential Review Period in case involving adaptive bit rate (ABR) technology used in video streaming components and systems including fitness devices.

- **In the Matter of Certain LTE-Compliant Cellular Communication Devices, Investigation No. 337-TA-1253**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021) and deposition testimony (2021) on behalf of respondents Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Motorola Mobility LLC regarding economic issues related to domestic industry, intervening rights, FRAND, and the appropriate amount of bond during the Presidential Review Period in case involving LTE-compliant cellular communication devices.

- **In the Matter of Certain High-Potency Sweeteners, Processes for Making Same, and Products Containing Same, Investigation No. 337-TA-1264**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021) and deposition testimony (2021) on behalf of complainants Celanese International Corporation, Celanese (Malta) Company 2 Limited, and Celanese Sales U.S., Ltd., regarding economic issues related to domestic industry and the amount and economic significance of inventory of the accused products in case involving Ace-K, a high-potency sweetener.

- **Rasmussen Instruments, LLC v. DePuy Synthes Products, Inc., DePuy Synthes Sales, Inc., and Medical Device Business Services, Inc.**

US District Court, District of Massachusetts

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2022) on behalf of defendants regarding reasonable royalty damages for patents related to instruments used in total knee arthroplasty (TKA).

- **In the Matter of Certain Smart Thermostat Systems, Smart HVAC Systems, Smart HVAC Control Systems, and Components Thereof, Investigation No. 337-TA-1258**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2021) on behalf of respondents ecobee Ltd., ecobee, Inc., and Google LLC regarding economic issues related to domestic industry, remedy, and the appropriate amount of bond during the Presidential Review Period in case involving smart thermostats and smart HVAC systems.

- **In the Matter of Certain Vehicle Control Systems, Vehicles Containing the Same, and Components Thereof, Investigation No. 337-TA-1235**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021) and deposition testimony (2021) on behalf of respondents Audi AG, Audi of America, LLC, Automobili Lamborghini S.p.A., Automobili Lamborghini America, LLC, Porsche AG, Porsche Cars North America, Inc., Volkswagen AG, and Volkswagen Group of America, Inc., regarding economic issues related to domestic industry, remedy, public interest, bond, and commercial success, in case involving vehicle control systems and vehicles that incorporate them.

- **In the Matter of Certain Automated Storage and Retrieval Systems, Robots, and Components Thereof, Investigation No. 337-TA-1228**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2021) on behalf of complainants AutoStore Technology AS, AutoStore AS, and AutoStore System Inc. regarding economic issues related to domestic industry, amount of inventory of accused products, and the appropriate amount of bond during the Presidential Review Period in case involving automated storage and retrieval systems and robots.

- **In the Matter of Certain Shingled Solar Modules, Components Thereof, and Methods for Manufacturing the Same, Investigation No. 337-TA-1223**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2021) on behalf of complainant The Solaria Corporation regarding economic issues related to domestic industry, amount of inventory of accused products, and the appropriate amount of bond during the Presidential Review Period in case involving solar panels.

- **In the Matter of Certain Video Processing Devices, Components Thereof, and Digital Smart Televisions Containing the Same, Investigation No. 337-TA-1222**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2021) on behalf of respondents Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Electronics HCMC CE Complex, Co., Ltd., LG Electronics Inc., LG Electronics U.S.A., Inc., TCL Technology Group Corporation, TCL Electronics Holdings Limited, TTE Technology, Inc., Shenzhen TCL New Technologies Co. Ltd., TCL King Electrical Appliances (Huizhou) Co. Ltd., TCL MOKA International Limited, TCL Smart Device (Vietnam) Co., Ltd, and Realtek Semiconductor Corp. regarding economic issues related to domestic industry and the appropriate amount of bond during the Presidential Review Period in case involving video processing devices in digital smart televisions.

- **Zimmer, Inc. d/b/a Zimmer Biomet and Zimmer US, Inc. v. Heraeus Medical LLC, Devin Childers, Robert Kolbe, James “Worth” Burns, Paul Cruz, and Kyle Kolbe**

Superior Court of the State of Indiana, Kosciusko County

Misappropriation of trade secrets, tortious interference, civil conspiracy, unfair competition, and breach of contract/breach of duty of loyalty: expert reports (2020, 2021) and deposition testimony (2021) on behalf of plaintiffs regarding damages resulting from alleged unfair acts in case involving bone cements.

- **In the Matter of Certain Mobile Electronic Devices and Laptop Computers, Investigation No. 337-TA-1215**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021) and deposition testimony (2021) on behalf of complainant Maxell, Ltd., regarding economic issues related to economic prong of domestic industry and remedy in case involving mobile devices.

- **In the Matter of Certain Movable Barrier Operator Systems and Components Thereof, Investigation No. 337-TA-1209**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2021), deposition testimony (2021), and trial testimony (2021) on behalf of complainants Overhead Door Corporation and GMI Holdings, Inc., regarding economic issues related to domestic industry, remedy, bond, and commercial success in case involving garage door operators and accessories.

- **ParkerVision, Inc. v. Qualcomm, Inc. and Qualcomm Atheros, Inc.**

US District Court, Middle District of Florida, Orlando Division

Patent infringement: expert report (2020) and deposition testimony (2020) on behalf of defendants regarding reasonable royalty damages for patents associated with RF receiver and transmitter technology used in semiconductor chips.

- **In the Matter of Certain Audio Players and Controllers, Components Thereof, and Products Containing Same, Investigation No. 337-TA-1191**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2020) and deposition testimony (2020) on behalf of respondent Google LLC regarding economic issues related to the appropriate amount of bond during the Presidential Review Period in case involving audio players and controllers.

- **In the Matter of Certain Bone Cements and Bone Cement Accessories, Investigation No. 337-TA-1175**

US International Trade Commission, Washington, DC

Misappropriation of trade secrets: expert report (2020), deposition testimony (2020), and trial testimony (2020) on behalf of complainants Zimmer, Inc. and Zimmer US, Inc. regarding economic issues related to domestic industry, injury, and the appropriate amount of bond during the Presidential Review Period in case involving bone cements.

- **In the Matter of Certain High-Density Fiber Optic Equipment and Components Thereof, Investigation No. 337-TA-1194**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2020), deposition testimony (2020), and trial testimony (2020) on behalf of respondents AFL Telecommunications LLC, FS.com Inc., Leviton Manufacturing Co., Inc., Panduit Corporation, and The Siemon Company regarding economic issues related to domestic industry, remedy, and bond in case involving high-density fiber optic equipment.

- **In the Matter of Certain Smart Thermostats, Smart HVAC Systems, and Components Thereof, Investigation No. 337-TA-1185**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2020), deposition testimony (2020), and trial testimony (2020) on behalf of respondents Alarm.com Incorporated, Alarm.com Holdings, Inc., ecobee Ltd., ecobee, Inc., Google LLC, and Vivint, Inc., regarding economic issues related to domestic industry, remedy, and bond in case involving smart thermostats and smart HVAC systems.
- **Maxell, Ltd. v. Apple, Inc.**
US District Court, Eastern District of Texas, Texarkana Division
Patent infringement: expert report (2020) and deposition testimony (2020) on behalf of plaintiff regarding reasonable royalty damages for patents associated with a variety of technologies related to features of mobile device products.
- **Sprint Communications Company, L.P. v. Charter Communications, Inc., Charter Communications Holdings, LLC, Spectrum Management Holding Company, LLC, Charter Communications Operation, LLC, Bright House Networks, LLC**
US District Court, District of Delaware
Patent infringement: expert reports (2020) and deposition testimony (2020) on behalf of defendants regarding analysis of damages in the form of lost profits and reasonable royalty in case involving VoIP (Voice over Internet Protocol) technology.
- **Hytera Communications Corp. Ltd. v. Motorola Solutions, Inc.**
US District Court, Northern District of Ohio, Eastern Division
Patent infringement: expert report (2020) and deposition testimony (2020) on behalf of defendants regarding analysis of economic issues related to secondary considerations of non-obviousness, i.e., commercial success and reasonable royalty damages, in case involving technology related to digital mobile radios.
- **In the Matter of Certain Mobile Devices with Multifunction Emulators, Investigation No. 337-TA-1170**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2020), deposition testimony (2020), and trial testimony (2020) on behalf of respondents Samsung Electronics Co., Ltd and Samsung Electronics America, Inc., regarding economic issues related to domestic industry, remedy and bond in case involving magnetic emulators used in secure payment cards and certain mobile devices.
- **Vifor Fresenius Medical Care Renal Pharma Ltd. and Vifor Fresenius Medical Care Renal Pharma France S.A.S. v. Lupin Atlantis Holdings SA, Lupin Pharmaceuticals, Inc., and Teva Pharmaceuticals USA, Inc.**
US District Court, District of Delaware
Hatch-Waxman: expert report (2019), deposition testimony (2020), and trial testimony (2021) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Velphoro, a phosphate binder used for patients with chronic kidney disease on dialysis.

- **Janssen Pharmaceuticals, Inc. and Janssen Pharmaceutica NV v. Teva Pharmaceuticals USA, Inc. and Teva Pharmaceuticals Industries Ltd.**

US District Court, District of New Jersey

Hatch-Waxman: expert report (2019), deposition testimony (2019), and trial testimony (2020) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Invega Sustenna, an antipsychotic drug for the treatment of schizophrenia and schizoaffective disorder.

- **In the Matter of Certain Bone Cements, Components Thereof and Products Containing the Same, Investigation No. 337-TA-1153**

US International Trade Commission, Washington, DC

Misappropriation of trade secrets: expert report (2019), deposition testimony (2019), and trial testimony (2020) on behalf of respondents Zimmer Biomet Holdings, Inc.; Biomet, Inc.; Zimmer Surgical, Inc.; Biomet France S.A.R.L.; Biomet Deutschland GmbH; Zimmer Biomet Deutschland GmbH; Biomet Global Supply Chain Center B.V.; Zimmer Biomet Nederland B.V.; Biomet Orthopedics, LLC; Biomet Orthopaedics Switzerland GmbH; Zimmer US, Inc.; Zimmer, GmbH; and Biomet Manufacturing, LLC, regarding economic issues related to domestic industry, injury, and the appropriate amount of bond during the Presidential Review Period in case involving bone cements.

- **In the Matter of Certain Botulinum Toxin Products, Processes for Manufacturing or Relating to Same, and Certain Products Containing Same, Investigation No. 337-TA-1145**

US International Trade Commission, Washington, DC

Misappropriation of trade secrets: expert report (2019), deposition testimony (2019), and trial testimony (2020) on behalf of respondents Daewoong Pharmaceuticals Co., Ltd., Daewoong Co., Ltd., and Evolus, Inc., regarding economic issues related to domestic industry, injury, and the appropriate amount of bond during the Presidential Review Period in case involving botulinum toxin products.

- **Adapt Pharma Operations Limited, Adapt Pharma, Inc., Adapt Pharma Limited, and Opiant Pharmaceuticals, Inc. v. Teva Pharmaceuticals USA, Inc. and Teva Pharmaceuticals Industries Ltd.**

US District Court, District of New Jersey

Hatch-Waxman: expert report (2019) and deposition testimony (2019) on behalf of plaintiffs regarding analysis of commercial success of patented pharmaceutical product Narcan Nasal Spray (naloxone hydrochloride), used in the treatment of opioid overdose.

- **In the Matter of Certain Electronic Nicotine Delivery Systems and Components Thereof, Investigation No. 337-TA-1139**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2019), deposition testimony (2019), and trial testimony (2019) on behalf of complainant Juul Labs, Inc., regarding economic issues related to domestic industry, amount of inventory of accused products, bond, and certain considerations of non-obviousness in case involving e-cigarettes.

- **AgroFresh, Inc. v. Essentiv LLC, Decco U.S. PostHarvest, Inc., and Cerexagri, Inc. d/b/a Deco Post-Harvest**

US District Court, District of Delaware

Patent infringement, unfair competition, and misappropriation of trade secrets: expert report (2018), deposition testimony (2019), and trial testimony (2019) on behalf of defendants, responding to plaintiff's claims of damages in case involving agricultural chemicals.

- **In the Matter of Certain Beverage Dispensing Systems and Components Thereof, Investigation No. 337-TA-1130**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2018), deposition testimony (2019), and trial testimony (2019) on behalf of complainants Heineken International B.V., Heineken Supply Chain B.V., and Heineken USA Inc. regarding economic issues related to domestic industry, amount of inventory of accused products, and bond in case involving draft beer dispensing systems.

- **Amgen, Inc., Amgen Manufacturing, Ltd., and Amgen USA, Inc. v. Sanofi, Sanofi-Aventis U.S., LLC, Aventisub, LLC, f/d/b/a/ Aventis Pharmaceuticals, Inc., and Regeneron Pharmaceuticals, Inc. ("Remand Proceeding")**

US District Court, District of Delaware

Patent infringement: expert report (2018) and deposition testimony (2018) on behalf of defendants regarding lost profits, price erosion, and reasonable royalty damages in case involving biologic product, PCSK9 inhibitor.

- **In the Matter of Certain Mobile Electronic Devices and Radio Frequency and Processing Components Thereof, Investigation No. 337-TA-1093**

US International Trade Commission, Washington, DC

Patent infringement: expert reports (2018), deposition testimony (2018), and trial testimony (2018) on behalf of complainant Qualcomm Incorporated regarding economic issues related to public interest in case involving smartphones and tablets that incorporate baseband processor modems.

- **In the Matter of Certain Road Construction Machines and Components Thereof, Investigation No. 337-TA-1088**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2018), deposition testimony (2018), and trial testimony (2018) on behalf of respondents Wirtgen GmbH, Joseph Vögele AG, Wirtgen Group Holding GmbH, and Wirtgen America, Inc., regarding economic issues related to domestic industry, commercial success, and bond in case involving road construction equipment.

- **In the Matter of Certain Thermoplastic-Encapsulated Electric Motors, Components Thereof, and Products and Vehicles Containing Same II, Investigation No. 337-TA-1073**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2018), deposition testimony (2018), and trial testimony (2018) on behalf of respondents Aisin Seiki Co., Ltd.; Aisin Holdings of America, Inc.; Aisin Technical Center of America, Inc.; Aisin World Corp. of America; Bayerische Motoren Werke AG; BMW of North America, LLC; BMW Manufacturing Co., LLC; Honda Motor Co., Ltd.; Honda North America, Inc.; American Honda Motor Co., Inc.; Honda of America Mfg. Inc.; Honda Manufacturing of Alabama, LLC; Honda R&D Americas, Inc.; Nidec Corporation; Nidec Automotive Motor Americas, LLC; Toyota Motor Corporation; Toyota Motor North America, Inc.; Toyota Motor Sales, U.S.A., Inc.; Toyota Motor Engineering and Manufacturing North America, Inc.; Toyota Motor Manufacturing, Indiana, Inc.; and Toyota Motor Manufacturing, Kentucky, Inc. regarding economic issues related to domestic industry in case involving fuel pumps, electric water pumps, power steering motors, and actuators.

- **In the Matter of Certain Mobile Electronic Devices and Radio Frequency and Processing Components Thereof, Investigation No. 337-TA-1065**

US International Trade Commission, Washington, DC

Patent infringement: expert reports (2018), deposition testimony (2018), and trial testimony (2018) on behalf of complainant Qualcomm Incorporated regarding economic issues related to public interest in case involving smartphones and tablet computers which incorporate baseband processor modems.

- **Takeda Pharmaceuticals USA, Inc. v. West-Ward Pharmaceutical Corporation, Hikma Americas, Inc., and Hikma Pharmaceuticals, PLC**

US District Court, District of Delaware

Patent infringement: expert report (2018) and deposition testimony (2018) on behalf of plaintiffs regarding lost profits related to temporary restraining order in case involving oral colchicine tablets.

- **In the Matter of Certain Microfluidic Devices, Investigation No. 337-TA-1068**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2018), deposition testimony (2018), and trial testimony (2018) on behalf of complainants Bio-Rad Laboratories, Inc. and Lawrence Livermore National Security, LLC, regarding economic issues related to domestic industry, bond, and amount and value of inventory of accused products, and public interest in case involving microfluidic devices used in analyzing biological samples for health care research.

- **In the Matter of Certain Road Milling Machines and Components Thereof, Investigation No. 337-TA-1067**

US International Trade Commission, Washington, DC

Patent infringement: expert report (2018), deposition testimony (2018), and trial testimony (2018) on behalf of complainant Wirtgen America, Inc., regarding economic issues related to domestic industry, bond, amount and value of inventory of accused products, and commercial success in case involving road milling machines.

- **Maxell, Ltd. v. ZTE Corporation and ZTE USA, Inc.**

US District Court, Eastern District of Texas, Texarkana Division

Patent infringement: expert report (2018), deposition testimony (2018), and trial testimony (2018) on behalf of plaintiff Maxell, Ltd., regarding reasonable royalty damages for patents associated with a variety of technologies related to features of mobile device products.

- **Maxell, Ltd. v. Huawei Device USA, Inc. and Huawei Device Co., Ltd.**
US District Court, Eastern District of Texas, Texarkana Division
Patent infringement: expert report (2018) and deposition testimony (2018) on behalf of plaintiff, Maxell, Ltd., regarding reasonable royalty damages for patents associated with a variety of technologies related to features of mobile device products.
- **AMAG Pharmaceuticals, Inc. v. Sandoz, Inc.**
US District Court, District of New Jersey
Hatch-Waxman: expert reports (2017) and deposition testimony (2017) on behalf of plaintiff regarding analysis of commercial success of patented pharmaceutical product Feraheme (ferumoxytol), which is used to treat iron deficiency anemia, and economic analysis regarding statutory permanent injunction factors.
- **In the Matter of Certain Digital Cable and Satellite Products, Set-Top Boxes, Gateways, and Components Thereof, Investigation No. 337-TA-1049**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2017) and deposition testimony (2017) on behalf of complainants Sony Corporation and Sony Electronics, Inc. regarding economic issues related to domestic industry in case involving digital televisions, set-top boxes, and media gateways.
- **In the Matter of Certain Semiconductor Devices and Consumer Audiovisual Products Containing the Same, Investigation No. 337-TA-1047**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2017), deposition testimony (2017), and trial testimony (2017) on behalf of complainant Broadcom Corporation regarding economic issues related to domestic industry, bond, and amount and value of inventory of accused products, in case involving SOC semiconductor devices and products that incorporate them including Blu-Ray players, DVD players, and digital televisions.
- **In the Matter of Certain Graphics Systems, Components Thereof, and Consumer Products Containing the Same, Investigation No. 337-TA-1044**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2017) and deposition testimony (2017) on behalf of complainants Advanced Micro Devices, Inc. and ATI Technologies ULC regarding economic issues related to domestic industry in case involving semiconductor components such as APUs and GPUs.
- **In the Matter of Certain Carbon and Alloy Steel Products, Investigation No. 337-TA-1002**
US International Trade Commission, Washington, DC
Unfair competition: expert report (2017) and deposition testimony (2017) on behalf of respondents Shanghai Baosteel Group Corporation; Baoshan Iron & Steel Co., Ltd.; Baosteel America, Inc.; Hebei Iron and Steel Group Co., Ltd.; Hebei Iron & Steel Group Hengshui Strip Rolling Co., Ltd.; Hebei Iron & Steel (Hong Kong) International Trade Co., Ltd.; Jiangsu Shagang Group; Jiangsu Shagang International Trade Co., Ltd.; Anshan Iron and Steel Group; Angang Group International Trade Corporation; Angang Group Hong Kong Co. Ltd.; Wuhan Iron and Steel Group Corp.; Wuhan Iron and Steel Co., Ltd.; WISCO America Co., Ltd.; Shougang Corporation; China Shougang International Trade & Engineering Corporation; Magang Group Holding Co. Ltd.; and Maanshan Iron and Steel Co. Ltd. regarding economic issues related to domestic industry and injury in case involving the importation of steel by respondents with a false designation of origin and misappropriation of trade secrets.

- **EndoEvolution, LLC v. Ethicon Endo-Surgery, Inc.**
American Arbitration Association, International Center for Dispute Resolution
Breach of contract, fraud, and misappropriation of trade secrets: expert report (2017) and hearing testimony (2017) on behalf of defendant Ethicon regarding monetary compensation resulting from alleged misappropriation of trade secrets relating to medical device used for minimally invasive surgery.
- **In the Matter of Certain Passenger Vehicle Automotive Wheels, Investigation No. 337-TA-1006**
US International Trade Commission, Washington, DC
Design patent and trademark infringement: expert report (2016) on behalf of complainant, Daimler AG, regarding economic issues related to domestic industry in case involving passenger vehicle automotive wheels.
- **Odyssey Wireless, Inc. v. Motorola Mobility LLC**
US District Court, Southern District of California, San Diego Division
Patent infringement: expert report (2016) and deposition testimony (2016) on behalf of defendants regarding reasonable royalty damages for patents associated with technology related to the LTE telecommunications standard.
- **In the Matter of Certain Radio Frequency Identification (“RFID”) Products and Components Thereof, Investigation No. 337-TA-979**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2016), deposition testimony (2016), and trial testimony (2016) on behalf of respondents Kapsch and Star regarding economic issues related to public interest, bond, and commercial success in case involving RFID technology used in electronic toll collection systems.
- **VideoShare, LLC v. Google, Inc. and YouTube, LLC**
US District Court, District of Delaware
Patent infringement: expert report (2016) and deposition testimony (2016) on behalf of defendants regarding reasonable royalty damages in case involving technology related to video sharing, uploading, and conversion.
- **Amgen, Inc. v. Sanofi, Sanofi-Aventis U.S., LLC, Aventisub, LLC, f/d/b/a/ Aventis Pharmaceuticals, Inc., and Regeneron Pharmaceuticals, Inc.**
US District Court, District of Delaware
Patent infringement: expert report (2015) and deposition testimony (2016) on behalf of defendants regarding lost profits, price erosion, and reasonable royalty damages in case involving biologic product, PCSK9 inhibitor.
- **CardiAQ Valve Technologies, Inc. v. Neovasc, Inc. and Neovasc Tiara, Inc.**
US District Court, District of Massachusetts
Breach of contract, fraud, and misappropriation of trade secrets: expert report (2015), deposition testimony (2016), and trial testimony (2016) on behalf of defendants in case involving mitral valve implants for the heart.

- **In the Matter of Certain Lithium Metal Oxide Cathode Materials, Lithium-Ion Batteries Containing Same, and Products with Lithium-Ion Batteries Containing Same, Investigation No. 337-TA-951**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2015) and deposition testimony (2015) on behalf of complainants BASF Corporation and UChicago Argonne LLC regarding economic issues related to domestic industry, remedy, bond, and public interest in case involving cathode materials used in lithium-ion batteries.
- **In the Matter of Certain Graphics Processing Chips, Systems on a Chip, and Products Containing the Same, Investigation No. 337-TA-941**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2015), deposition testimony (2015), and trial testimony (2015) on behalf of complainant Samsung regarding domestic industry, remedy, bond, and public interest in case involving GPU and SOC semiconductor devices.
- **The Washington University v. Wisconsin Alumni Research Foundation**
US District Court, District of Delaware
Breach of contract: expert report (2015), deposition testimony (2015), and trial testimony (2018) on behalf of defendant regarding damages arising from alleged breach of contract involving pharmaceutical product Zemplar.
- **In the Matter of Certain Windshield Wipers and Components Thereof, Investigation No. 337-TA-937**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2015), deposition testimony (2015), and trial testimony (2015) on behalf of complainants Valeo North America, Inc. and Delmex de Juarez S. de R.L. de C.V. regarding domestic industry in case involving windshield wipers.
- **In the Matter of Certain Dental Implants, Investigation No. 337-TA-934**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2015), deposition testimony (2015), and trial testimony (2015) on behalf of respondent Neodent regarding domestic industry in case involving dental implants.
- **In the Matter of Certain Formatted Magnetic Data Storage Tapes and Cartridges Containing the Same, Investigation No. 337-TA-931**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2015) and deposition testimony (2015) on behalf of respondents IBM and Oracle regarding public interest in case involving magnetic tape data storage systems.
- **In the Matter of Certain Sulfentrazone, Sulfentrazone Compositions, and Processes for Making Sulfentrazone, Investigation No. 337-TA-914**
US International Trade Commission, Washington, DC
Patent infringement: expert reports and trial testimony (2014 and 2015) on behalf of complainant FMC Corporation regarding domestic industry in case involving agricultural herbicide products.

- **In the Matter of Certain Devices Containing Non-volatile Memory and Products Containing the Same, Investigation No. 337-TA-922**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2015) on behalf of respondent Spansion Inc. regarding remedy, bond, and public interest in case involving non-volatile memory including NAND flash memory.
- **Comcast v. Sprint**
US District Court, District of Delaware
Patent infringement: expert reports (2014), deposition testimony (2014), and trial testimony (2014) on behalf of plaintiff Comcast regarding reasonable royalty damages in case involving telecommunications network technology.
- **In the Matter of Certain Antivenom Compositions and Products Containing the Same, Investigation No. 337-TA-903**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2014) and deposition testimony (2014) on behalf of complainant BTG regarding domestic industry, remedy, bond, and public interest in case involving antivenoms used in treating snake bites.
- **Louis Vuitton Malletier S.A. v. Sunny Merchandise Corp., et.al.**
US District Court, Southern District of New York
Trademark infringement: expert report (2014) and deposition testimony (2014) on behalf of plaintiff Louis Vuitton Malletier, S.A., regarding actual and unjust enrichment damages in case involving alleged use of Louis Vuitton trademarks on eyewear.
- **In the Matter of Certain Navigation Products, Including GPS Devices, Navigation and Display Systems, Radar Systems, Navigational Aids, Mapping Systems, and Related Software, Investigation No. 337-TA-900**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2014) and deposition testimony (2014) on behalf of complainant Furuno Electric Co. regarding domestic industry, remedy, bond, and public interest in case involving marine navigation devices.
- **Mylan v. GlaxoSmithKline**
US District Court, District of New Jersey
Breach of contract: expert reports (2011 and 2013), deposition testimony (2011 and 2014), and trial testimony (2014) responding to plaintiff's claims of damages resulting from breach of contract involving generic pharmaceutical product.
- **In the Matter of Certain Integrated Circuit Chips and Products Containing Same, Investigation No. 337-TA-859**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2013), deposition testimony (2013), and trial testimony (2014) on behalf of respondents LSI Corporation and Seagate Technology regarding domestic industry and bond in Section 337 case involving semiconductor technology.

- **Walker Digital, LLC v. Google, Inc.**
US District Court, District of Delaware
Patent infringement: expert report (2013) and deposition testimony (2014) regarding reasonable royalty damages for patent infringement case involving patents related to anonymous communication technology used in social networks.
- **In the Matter of Certain TV Programs, Literary Works for TV Production and Episode Guides Pertaining to Same, Investigation No. 337-TA-886**
US International Trade Commission, Washington, DC
Copyright infringement and unfair competition: expert report (2013) and deposition testimony (2014) on behalf of respondents The Walt Disney Company and Thunderbird Films, Inc., regarding the existence of a domestic industry and injury to a domestic industry in Section 337 case involving literary works related to a pilot television series.
- **In the Matter of Certain Microelectromechanical Systems (“MEMS Devices”) and Products Containing Same, Investigation No. 337-TA-876**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2013) and deposition testimony (2013) on behalf of complainant ST Microelectronics, Inc., regarding domestic industry, remedy, cease and desist order, and bond in Section 337 case involving microelectromechanical (MEMS) devices.
- **In the Matter of Certain Wireless Communications Equipment and Articles Therein, Investigation No. 337-TA-866**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2013), deposition testimony (2013), and trial testimony (2013) on behalf of complainants Samsung Electronics Co., Ltd. and Samsung Telecommunications America, LLC, regarding domestic industry, remedy, cease and desist order, and bond in Section 337 case involving wireless communications equipment.
- **Organogenesis, Inc. v. Vincent Ronfard and Healthpoint, Ltd.**
Commonwealth of Massachusetts Superior Court, Suffolk County
Breach of contract, unfair competition, and misappropriation of trade secrets: expert report (2013) and deposition testimony (2013) on behalf of plaintiff regarding damages arising from alleged violations relating to employment contract for companies in biopharmaceutical industry.
- **In the Matter of Certain Products Having Laminated Packaging, and Components Thereof, Investigation No. 337-TA-874**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2013), deposition testimony (2013), and trial testimony (2013) on behalf of respondents Remy Cointreau USA, Inc.; Pernod Ricard USA LLC; John Jameson Import Company; Moët Hennessy USA, Inc.; Champagne Louis Roederer; Maisons Marques & Domaines USA Inc.; Freixenet USA, Inc.; L’Oreal USA, Inc.; Hasbro, Inc.; Cognac Ferrand USA Inc.; Diageo North America, Inc.; WJ Deutsch & Sons Ltd.; and Beats Electronics LLC regarding domestic industry in Section 337 case involving products having laminated packaging.

- **In the Matter of Certain Projectors with Controlled-Angle Retarders, Components Thereof, and Products Containing Same, Investigation No. 337-TA-815**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2012) and deposition testimony (2012) on behalf of respondents Sony Corporation, Sony Corporation of America, and Sony Electronics Inc. regarding domestic industry in Section 337 case involving projectors.
- **In the Matter of Certain Dynamic Random Access Memory and NAND Flash Memory Devices and Products Containing Same, Investigation No. 337-TA-803**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2012) and deposition testimony (2012) on behalf of respondents Elpida Memory, Inc.; Elpida Memory (USA) Inc.; Hynix Semiconductor Inc.; Hynix Semiconductor America, Inc.; Acer Inc.; Acer America Corp.; ADATA Technology Co., Ltd; ADATA Technology (USA) Co., Ltd.; Asustek Computer Inc.; Asus Computer International; Dell, Inc.; Hewlett-Packard Company; Kingston Technology Co., Inc.; Logitech International S.A.; Logitech, Inc.; Best Buy Co., Inc.; and Wal-Mart Stores, Inc. regarding domestic industry, remedy, bond, and public interest in Section 337 case involving DRAM and NAND flash memory devices.
- **Genentech, Inc. v. UCB Celltech**
American Arbitration Association, International Center for Dispute Resolution
Breach of contract: expert reports (2012) and hearing testimony (2012) on behalf of plaintiff regarding damages arising from alleged breach of contract and fraud involving pharmaceutical products.
- **Apple Inc. and Next Software, Inc. (f/k/a Next Computer, Inc.) v. Motorola, Inc., and Motorola Mobility, Inc.**
US District Court, Northern District of Illinois
Patent infringement: expert report (2012) and deposition testimony (2012) on behalf of Motorola, Inc. and Motorola Mobility, Inc., regarding damages due Motorola associated with alleged infringement of patents directed to wireless communications functionality.
- **In the Matter of Certain Electronic Devices, Including Wireless Communication Devices, Portable Music and Data Processing Devices, and Tablet Computers, Investigation No. 337-TA-794**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2012), deposition testimony (2012), and trial testimony (2012) on behalf of complainants Samsung Electronics Co., Ltd. and Samsung Telecommunications America, LLC, regarding domestic industry in Section 337 case involving certain electronic devices.
- **Automated Merchandising Systems, Inc. v. Crane Co., and Seaga Manufacturing, Inc.**
US District Court, Northern District of West Virginia
Patent infringement: expert report (2011) regarding lost profits and reasonable royalty damages for patent infringement involving vending machines.
- **In the Matter of Certain Mobile Devices and Related Software, Investigation No. 337-TA-750**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2011) and deposition testimony (2011) on behalf of respondents Motorola Mobility, Inc., and Motorola Solutions, Inc., regarding domestic industry and appropriate amount of bond in Section 337 case involving mobile devices.

- **In the Matter of Certain Liquid Crystal Display Devices, Including Monitors, Televisions, and Modules, and Components Thereof, Investigation No. 337-TA-749C**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2011), deposition testimony (2011), and trial testimony (2011) on behalf of respondents AU Optronics Corporation; BenQ Corporation; Chimei InnoLux Corporation; MStar Semiconductor, Inc.; and Qisda Corporation regarding domestic industry in Section 337 case involving LCD displays.
- **In the Matter of Certain Wireless Communication Devices, Portable Music and Data Processing Devices, Computers and Components Thereof, Investigation No. 337-TA-745**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2011) and deposition testimony (2011) on behalf of complainant Motorola Mobility Inc., regarding domestic industry and certain secondary considerations of non-obviousness in Section 337 case involving wireless communication devices.
- **Paone v. Microsoft Corp.**
US District Court, Eastern District of New York
Patent infringement: expert reports (2008 and 2011) and deposition testimony (2009 and 2011) regarding reasonable royalty damages for patent infringement case involving patent related to encryption technology used in computer software.
- **Intex Recreation Corp. v. Team Worldwide Corp.**
US District Court, Washington, DC
Patent infringement: expert reports (2006 and 2011) and deposition testimony (2006) in a patent infringement case involving inflatable air mattresses. Provided testimony on lost profits and reasonable royalty damages.
- **B. Braun Melsungen et al. v. Terumo Medical Corp. et al.**
US District Court, District of Delaware
Patent infringement: expert report (2010), deposition testimony (2010), and trial testimony (2010) regarding commercial success of safety IV catheter and contribution of patented technology.
- **Touchcom v. Bereskin & Parr et al.**
US District Court, Eastern District of Virginia
Professional negligence: expert report (2010) and deposition testimony (2010) regarding damages due to plaintiff as a result of defendant's alleged malpractice in preparing and prosecuting patent application directed to interactive fuel pump system.
- **In the Matter of Certain DC-DC Controllers and Products Containing Same, Investigation No. 337-TA-698**
US International Trade Commission, Washington, DC
Patent infringement and misappropriation of trade secrets: expert report (2010) and deposition testimony (2010) on behalf of respondents uPI Semiconductor Corp. and Sapphire Technology regarding injury to domestic industry and scope of exclusion order in Section 337 case involving DC-DC controllers.

- **In the Matter of Certain MLC Flash Memory Devices and Products Containing Same, Investigation No. 337-TA-683**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2010), deposition testimony (2010), and trial testimony (2010) on behalf of respondent Samsung regarding domestic industry, scope of exclusion order, and appropriate amount of bond in Section 337 case involving patents directed to multi-level cell flash memory technology.
- **In the Matter of Certain Semiconductor Integrated Circuits and Products Containing Same, Investigation No. 337-TA-665**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2009), deposition testimony (2009), and trial testimony (2009) on behalf of respondents LSI and Seagate regarding domestic industry and scope of exclusion order in Section 337 case involving patents directed to semiconductor design and manufacturing processes.
- **McKesson v. Epic**
US District Court, Northern District of Georgia
Patent infringement: expert report (2009) and deposition testimony (2009) regarding reasonable royalty damages for patent infringement case involving patent related to health information software.
- **Samsung Electronics v. ON Semiconductor Corp.**
US District Court, District of Delaware
Patent infringement: expert reports (2008) and deposition testimony (2008) regarding reasonable royalty damages for patent infringement case involving patents directed to semiconductor manufacturing processes and products.
- **In the Matter of Certain Semiconductor Chips with Minimized Chip Package Size and Products Containing Same, Investigation No. 337-TA-630**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2008), deposition testimony (2008), and trial testimony (2008) on behalf of respondents Kingston, ProMOS, Elpida and Nanya regarding scope of exclusion order and appropriate amount of bond in Section 337 case involving semiconductor packaging technology.
- **In the Matter of Certain Baseband Processor Chips and Chipsets, Transmitter and Receiver (Radio) Chips, Power Control Chips and Products Containing Same, Including Cellular Telephone Handsets, Investigation No. 337-TA-543 – Enforcement Proceeding**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2008), deposition testimony (2008), and trial testimony (2008) on behalf of complainant Broadcom regarding claim that respondent violated cease and desist order. Provided testimony related to maximum value of penalty associated with alleged violations.
- **adidas America v. Wal-Mart**
US District Court, District of Oregon
Trademark infringement: expert report (2008) and deposition testimony (2008) regarding reasonable royalty damages for trademark infringement case involving striped footwear.

- **In the Matter of Certain Semiconductor Chips with Minimized Chip Package Size and Products Containing Same, Investigation No. 337-TA-605**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2008), deposition testimony (2008), and trial testimony (2008) on behalf of respondent ST Microelectronics regarding domestic industry, scope of exclusion order, and appropriate amount of bond to be set for products covered by exclusion order during Presidential Review Period in Section 337 case involving semiconductor packaging technology.
- **JDB Medical, Inc. and James D. Beeton, Flint Medical, Inc. and Keith Flint v. The Sorin Group, S.p.A. and ELA Medical, Inc.**
US District Court, District of Colorado
Breach of contract damages: expert report (2007) and deposition testimony (2008) responding to plaintiff's claim of damages resulting from breach of a sales agreement involving cardiac rhythm management devices.
- **TruePosition v. Andrew Corp.**
US District Court, District of Delaware
Patent infringement: expert report (2006), deposition testimony (2007), and trial testimony (2007) on behalf of plaintiff TruePosition in a patent infringement case involving cellular telephone location equipment. Provided testimony on lost profits damages.
- **In the Matter of Certain NOR and NAND Flash Memory Devices and Products Containing Same, Investigation No. 337-TA-560**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2006), deposition testimony (2006), and trial testimony (2006) on behalf of respondent ST Microelectronics regarding domestic industry, scope of exclusion order, and appropriate amount of bond to be set for products covered by exclusion order during Presidential Review Period in Section 337 case involving NAND and NOR flash memory products.
- **In the Matter of Certain Flash Memory Devices and Components Thereof, and Products Containing Such Devices and Components, Investigation No. 337-TA-552**
US International Trade Commission, Washington, DC
Patent infringement: expert report (2006) and deposition testimony (2006) on behalf of respondent Hynix regarding certain secondary considerations, domestic industry, scope of exclusion order, and appropriate amount of bond to be set during Presidential Review Period in Section 337 case involving NAND flash memory products.
- **In the Matter of Certain Baseband Processor Chips and Chipsets, Transmitter and Receiver (Radio) Chips, Power Control Chips and Products Containing Same, Including Cellular Telephone Handsets, Investigation No. 337-TA-543**
US International Trade Commission, Washington, DC
Patent infringement: expert reports (2005 and 2006), deposition testimony (2006), and trial testimony (2006) on behalf of complainant Broadcom regarding scope of exclusion order associated with baseband and radio chips used in cellular telephones and other handheld devices in Section 337 case.

- **In the Matter of Certain NAND Flash Memory Circuits and Products Containing the Same, Investigation No. 337-TA-526**
US International Trade Commission, Washington, DC
Patent infringement: rebuttal expert report and deposition testimony (2005) on behalf of respondent ST Microelectronics regarding domestic industry, scope of exclusion order, and appropriate amount of bond to be set for products covered by exclusion order during Presidential Review Period in Section 337 case involving NAND flash memory products.
- **PDI, Inc. v. Cellegy Pharmaceuticals, Inc.**
US District Court, Northern District of California, San Francisco
Commercial damages: expert report and deposition testimony (2005) regarding analysis of damages arising from claims of fraud and breach of contract in case involving male testosterone hormone replacement therapy.
- **Minuteman Trucks, Inc. v. HN80 Corporation and Sterling Truck Corp.**
Superior Court of Massachusetts, Suffolk County
Commercial damages: rebuttal expert report and trial testimony (2004) regarding issues related to calculation of damages associated with alleged violation of Massachusetts statute regarding dealer pricing in case involving heavy trucks.
- **Miltope Corporation and IV Phoenix Group Inc. v. DRS Technologies et al.**
US District Court, Eastern District of New York
Misappropriation of trade secrets: expert report and deposition testimony (2003) on reasonable royalty damages in case involving rugged personal computers.
- **John W. Evans, et al. v. General Motors Corp.**
Superior Court of Connecticut, Judicial District of Waterbury at Waterbury
Misappropriation of trade secrets: expert report and deposition testimony (2002) and trial testimony (2003) on reasonable royalty damages in case involving automotive engine technology.
- **SRAM Corporation v. AD-II Engineering, Inc.**
US District Court, Northern District of Illinois
Patent infringement: expert report and deposition testimony (2002) on reasonable royalty damages due patent holder and lost profits due alleged infringer arising from allegedly false accusations involving patents related to bicycle twist shifters.
- **Qwest Communications International et al. v. WorldQuest Networks, Inc.**
US District Court, Eastern District of Virginia
Trademark infringement and unfair competition: expert report on damages in case involving sales of prepaid calling cards.
- **The Quigley Corporation v. GumTech International, et al.**
US District Court, Eastern District of Pennsylvania
Patent infringement: expert report and deposition testimony (2001) on damages in case involving patent related to the use of zinc gluconate to reduce the duration of the common cold.
- **Cytoc Corporation v. Autocyte, Inc.**
US District Court, District of Delaware
Patent infringement: expert report and deposition testimony (2000) on damages in case involving liquid based cervical cancer screening tests.

- **Hearthstone, Inc. v. Ronald M. Hawes, et al.**

US District Court, Eastern District of Virginia

Copyright and trademark infringement, breach of contract, and unfair competition: expert report on damages involving architectural plans for log homes.

SELECTED LITIGATION CONSULTING ASSIGNMENTS

Commercial Damages/Intellectual Property

- **Cubist v. Teva**

US District Court, District of Delaware

Hatch-Waxman: analysis of commercial success of patented pharmaceutical product Cubicin (daptomycin).

- **Cubist v. Hospira**

US District Court, District of Delaware

Hatch-Waxman: analysis of commercial success of pharmaceutical product Cubicin (daptomycin).

- **Nokia v. Apple**

US District Court, District of Delaware

Patent infringement: analysis of issues associated with determination of FRAND royalty for patents incorporated in wireless telecommunications products on behalf of Nokia.

- **Novartis v. Teva**

US District Court, District of Delaware

Hatch-Waxman: analysis of commercial success of patented pharmaceutical products Zometa and Reclast (zoledronic acid).

- **Nokia v. Qualcomm**

Delaware Chancery Court

Commercial litigation: analysis of issues associated with determination of FRAND royalty for patents incorporated in wireless communications products on behalf of Nokia.

- **O2 Micro v. Samsung Electronics**

US District Court, Eastern District of Texas

Patent damages: analysis of plaintiff's claims with respect to reasonable royalty damages for patents directed to technology used in LCD screens.

- **GlaxoSmithKline v. Ranbaxy**

US District Court, District of New Jersey

Hatch-Waxman: analysis of commercial success of patented pharmaceutical product Valtrex (valacyclovir hydrochloride).

- **Medinol Ltd. v. Boston Scientific Corp.**

US District Court, Southern District of New York

Breach of contract damages: analysis of damages resulting from allegations of breach of contract and misappropriation of trade secrets in case involving coronary stents.

- **Burst.Com v. Microsoft Corp.**
US District Court, District of Maryland
Patent and trade secrets damages: analysis of reasonable royalty damages in case involving software used for streaming media. Respond to plaintiff's claim of lost profits damages and unjust enrichment arising from misappropriation of trade secrets.
- **Medtronic AVE v. Advanced Cardiovascular Systems and Guidant Corp.**
US District Court, District of Delaware
Patent and trade secrets damages: analysis of lost profits and reasonable royalty damages in case involving coronary stents. Response to plaintiff's claim of damages arising from misappropriation of trade secrets.
- **Titan Sports, Inc., etc. v. Turner Broadcasting Systems, Inc. et al.**
US District Court, District of Connecticut
Copyright and trademark damages: unjust enrichment due to misappropriation of intellectual property involving two popular wrestling characters.
- **Stairmaster Sports/Medical Products, a Limited Partnership v. Groupe Procycle, Inc. et al.**
US District Court, District of Delaware
Patent damages: lost profits, reasonable royalty, and prejudgment interest involving patent directed to stair-climbing fitness equipment.
- **BTG v. Magellan Corp./BTG v. Trimble Navigation**
US District Court, Eastern District of Pennsylvania
Patent damages/intervening rights: reasonable royalty, prejudgment interest, value of inventory on hand, preparation and investments made, and business commenced (as of patent reissue) involving a patent directed to secret or secure communications technology employed in global positioning system products.
- **Joint Medical Products Corp. v. DePuy Orthopaedics, Inc. et al.**
US District Court, District of Connecticut
Patent damages: lost profits and reasonable royalty for patents directed to orthopedic implants.
- **Cordis Corp. v. SciMed Life Systems, Inc.**
US District Court, District of Minnesota
Patent damages: lost profits, reasonable royalty, and prejudgment interest for patent directed to balloon catheters used in Percutaneous Transluminal Coronary Angioplasty (PTCA).
- **Nexstar Pharmaceuticals, Inc. v. The Liposome Company**
US District Court, District of Delaware
Patent damages/intervening rights: valuation of inventory on hand, preparations and investments made, and business commenced (as of patent reissuance) involving patents directed to lipid formulations of an anti-fungal pharmaceutical.
- **Autonation, Inc. v. Acme Commercial Corp., at al. (CarMax)**
US District Court, Southern District of Florida
Trademark infringement/unfair competition: reasonable royalty associated with trademark infringement and unfair competition in the auto superstore business.

- **General Motors (GM) v. Lopez**

US District Court, Eastern District of Michigan

Trade secrets: analysis of GM damages and Volkswagen unjust enrichment due to alleged theft of trade secrets by former GM employee.

Antitrust Litigation

- **Joe Comes et al. v. Microsoft Corp.**

Iowa District Court for Polk County

Analysis of economics of computer software industry and resulting implications for market structure and firm profitability.

- **Daniel Gordon et al. v. Microsoft Corp.**

Minnesota District Court for Hennepin County

Analysis of economics of computer software industry and resulting implications for market structure and firm profitability.

- **Burst.Com, Inc. v. Microsoft Corp.**

US District Court, District of Maryland

Examination and evaluation of plaintiff's business strategy and likely implications with respect to plaintiff's claims of actual damages due to alleged antitrust violations.

- **In Re Microsoft Corp. Antitrust Litigation – All Purchaser Actions**

US District Court, District of Maryland

Analysis of economics of computer software industry and resulting implications for market structure and firm profitability.

- **Microsoft I-V Cases**

Superior Court of The State of California, for The City and County of San Francisco

Analysis of economics of computer software industry and resulting implications for market structure and firm profitability.

- **Vitamin Antitrust Litigation**

US District Court, District of Columbia

Preliminary analysis of factors affecting supply and demand for Vitamin C.

- **Industrial Silicon Antitrust Litigation**

US District Court, Western District of Pennsylvania

Analysis of issues related to likelihood of successful cartelization in production of industrial ferrosilicon, magnesium ferrosilicon, and silicon metal.

- **Independent Service Provider v. IBM**

Texas State Court, Corpus Christi

Analysis of issues related to liability and damages including definition of relevant market, assessment of market concentration, and evaluation of antitrust damages.

SELECTED NON-LITIGATION CONSULTING ASSIGNMENTS

- **Intellectual Property Valuation**
Assist clients with interpretation and/or negotiation of license terms for patented technology in a variety of fields, including consumer products, pharmaceuticals, and semiconductors.
- **Corporate Committee of the American College of Nuclear Physicians**
Analyses of the clinical and economic value of nuclear medicine in cardiology and oncology.
- **Lincoln General Hospital**
Business valuation of two health care providers for use in determining relative shares of the parties in a joint venture.
- **Pharmaceutical Partners for Better Health Care**
Comprehensive study of the Canadian health care system with particular emphasis on the effects of potential reforms on the pharmaceutical industry.

PUBLICATIONS

“Patent Damages in US Courts: Overview of Current State of Play,” (with J. Jarosz, J. McLean, and R. Vigil), *IAM Yearbook 2019: Building IP Value in the 21st Century*, 2018.

“Recently Released FDA Guidance and Biosimilar Development: Implications for the Litigation Environment,” (with G. Long), *Update: Food and Law, Regulation and Education*, March 2012, pp. 19-21.

“The 25% Rule Lives On,” (with J. Jarosz and M. Wagner), *IP Law 360*, September 8, 2010.

“Licensing in the Presence of Technological Standards,” (with J. Browning), *The Licensing Journal*, Volume 29 No. 7, August 2009, pp. 18-29.

“Use of the 25 Per Cent Rule in Valuing IP,” (with R. Goldscheider and J. Jarosz), *les Nouvelles*, Volume XXXVII No. 4, December 2002, pp. 123-133.

“Clinical and Economic Value of Cardiovascular Nuclear Medicine,” monograph published by *Meniscus Health Care Communications*, (with K. Neels), 1996.

“The Health Care System in Canada,” (with R. Rozek), Chapter 4, *Financing Health Care*, edited by U. Hoffmeyer and T. McCarthy, Kluwer Academic Publishers, 1994.

“Discounted Cash Flow Analysis in Patent Infringement Litigation,” (with R. Rozek), *Licensing Economics Review*, Volume 1, August 1991, pp. 7-10.

PRESENTATIONS AND SPEAKING ENGAGEMENTS

“Maximizing Returns on Patent Enforcement,” Certified Patent Valuation Analyst Virtual Webinar, May 2025 (with Thomas McGahee, John Jarosz, and Yuan Tian).

“Why Has the U.S. Biosimilar Experience Not Yet Lived Up to the Hype?” Analysis Group Law & Economics Symposium: Current Topics in Life Sciences, May 2019 (Anthony LoSasso, Richard Mortimer, and Paul Greenberg).

“ITC Section 337 Investigations: Trends and Significant Court Rulings in 2018,” The Knowledge Group Webinar, January 2019 (with Jordan Coyle, Stephen Akerley, and Daniel McGavock).

“ITC Section 337 Litigation & Enforcement: A Look Back and a Sneak Peek to the Future,” The Knowledge Group Webinar, January 2018 (with Gary Hnath and Eric Schweibenz).

“ITC Section 337 Litigation: Tips & Strategies for your Firm,” The Knowledge Group Webinar, September 2017 (with Jordan Coyle, Edward Donovan, Eric Schweibenz, and Pallavi Seth).

“ITC Section 337 Investigations: Your Thorough Guide for 2017 and Beyond,” The Knowledge Group Webinar, May 2017 (with Edward Donovan, Blaney Harper, and P.J. McCarthy).

“IP Valuation: Practical Guide on Effective Methods and Approaches for 2015 and Beyond,” The Knowledge Group Webinar, October 2015 (with Chuck Faunce and Philip Green).

“Trolls, NPEs, and PAEs ‘Oh My’ – Sorting out Remedies in Patent Infringement Litigation,” American Law Institute Continuing Legal Education, September 2013 (with Krish Gupta and Peter Strand).

“The Evolution of License Comparability in the Estimation of Reasonable Royalty Damages,” West Legal Education Center Webinar, July 2013 (with John Jarosz).

“Taking and Defending Expert Depositions,” Practicing Law Institute, Taking and Defending Depositions Seminar, March 2013 (with Kimo S. Peluso and Jack G. Stern).

“Taking and Defending Expert Depositions,” Practicing Law Institute, Taking and Defending Depositions Seminar, May 2012 (with Kimo S. Peluso and Jack G. Stern).

“Calculating Reasonable Royalty Damages after *Uniloc v. Microsoft*: An Economic Perspective,” Intellectual Property Law Section of the DC Bar, July 2011 (with Peter Strand).

“Patent Infringement: Calculating Royalty Damages in a Post-*Uniloc* World,” Strafford Publications Webinar, March 2011 (with Paul Michel, George Pappas, and John Jarosz).

“Clearing the Way for Biosimilars: New Complexities Around Competition and Consumer Harm,” ABA Antitrust Section Teleseminar, February 2011 (Moderator, with Panelists: Seth Silber, Iain Cockburn, Julie McEvoy, and Matt Cantor)

“Damages Apportionment After *Lucent*,” The 10th Anniversary Wilmer Hale Intellectual Property Conference, May 2010 (with Michael R. Heyison and Dominic E. Massa).

“Licensing in the Presence of Technological Standards,” Licensing Executives Society, Annual Meeting, October 2008.

“Reasonable Royalty Determination in the Presence of Standards and University Licensing,” Law Seminars International, Calculating and Providing Patent Damages Workshop, October 2006.

“Providing Effective Royalty Testimony,” Licensing Executives Society/Association of University Technology Managers Spring Meeting (Workshop 4-H), May 2006 (with John Jarosz and Lisa Pirozzolo).

“Meeting the Standards for Price Erosion and Convoyed Sales,” Law Seminars International, Calculating and Providing Patent Damages Workshop, February 2006.

“Intellectual Property Damages from an Economist’s Perspective,” DC Bar Association, Trade Secret Section, November 2005 (with John Jarosz and Abram Hoffman).

“Factors affecting Royalties,” Licensing Executive Society Annual Meeting (Workshop 2-M), October 2005 (with Robert Vigil).

“Trade Secrets Damages: What Can A Successful Claimant Expect to Recover?” Trade Secrets Committee of the Intellectual Property Law Section of the DC Bar, February 2005.

“Economics of Price Erosion and Lost Conveyed Sales,” Law Seminars International, Calculating and Proving Patent Damages Workshop, March 2004.

“An Economist’s Perspective on Reach-Through Royalties,” Law Seminars International, Calculating and Proving Patent Damages: Recent Developments and New Tools for Success, June 2003.

“Trade Secrets Damages and Recent Developments,” Trade Secrets Committee of the Intellectual Property Law Section of the DC Bar, May 2002.

“Industry Royalty Rates and Profitability: An Empirical Test of the 25% Rule,” Licensing Executives Society Annual Meeting (Workshop 3-L), October 2001 (with John Jarosz and Robert Vigil).

“Estimating the Economic Value of Trade Secrets,” U.S. Sentencing Commission (USSC) Symposium on Federal Sentencing Policy for Economic Crimes and New Technology Offenses, October 13, 2000.

“Estimating Economic Recovery in Trade Secrets Cases,” Trade Secrets Committee of the Intellectual Property Law Section of the DC Bar, September 1999.

“Industry Royalty Rates and Profitability: An Empirical Test of the 25% Rule,” Licensing Executives Society Annual Meeting, (Workshop 3-11), October 1998 (with John Jarosz).

“Royalty Rates and Awards in Patent Infringement Cases: 1916–1996,” Licensing Executives Society Annual Meeting, November 1997 (with John Jarosz).

PROFESSIONAL AFFILIATIONS

American Economic Association (AEA)

Licensing Executives Society (LES)

IAM Patent 1000 (2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2024): The World’s Leading Patent Practitioners – Economic Experts

EXHIBIT F

CURRICULUM VITA

MASSOUD PEDRAM

Personal Information

Charles Lee Powell Chair in Electrical Engineering and Computer Science
University of Southern California
Ming Hsieh Department of Electrical and Computer Engineering
3740 McClintock Ave – EEB 300B
Los Angeles CA 90089-2562
Email: pedram@usc.edu
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Fax: (213) 740-9803
URL: <http://www.mpedram.com/>
Wiki: https://en.wikipedia.org/wiki/Massoud_Pedram

Impact on the Field

- Advancing the theory and practice of energy-efficient computing and introducing methodologies, hardware/software solutions, and electronic design automation algorithms and tools for increasing computation speed and reducing power dissipation in CMOS circuits and systems, custom computing architectures, power electronics including power conversion and regulation, semiconductor and superconductor manufacturing, modeling and analysis of semiconductor devices including bulk and FinFet devices, superconductor electronics and superconductive computing, and quantum computing,

Research Interests

- Integrated circuit design, Computer-aided design methodologies and tools for CMOS circuits and systems, high-performance and energy-efficient computing, power/thermal management, power conversion and regulation, domain-specific accelerator design, semiconductor device design and manufacturing, electrical energy storage systems, embedded systems, cloud computing and data center resource management, homomorphic computing, post-CMOS and beyond-CMOS devices and circuits including 3D integrated circuits, superconductor electronics, and quantum computing hardware.

Education

- University of California at Berkeley, Ph.D. in Electrical Engineering and Computer Sciences, 1991
- University of California at Berkeley, M.S. in Electrical Engineering and Computer Sciences, 1989
- California Institute of Technology, B.S. in Electrical Engineering, 1986

Employment History

- Charles Lee Powell Chair in Electrical Engineering and Computer Science in the Viterbi School of Engineering, University of Southern California, 2017-present
- Stephen and Etta Varra Professor in the Viterbi School of Engineering, University of Southern California, 2013-2017

- Professor, Department of Electrical Engineering, University of Southern California, 2000-2012
- Associate Professor, Department of Electrical Engineering, University of Southern California, 1996-2000
- Assistant Professor, Department of Electrical Engineering, University of Southern California, 1991-1996
- Part-time Research Position, Xerox Palo Alto Research Center, 1987-1989
- Graduate Student Researcher, Department of Electrical Engineering and Computer Sciences, University of California at Berkeley, 1986-1991

Honors and Awards

- Elected a *Fellow of the National Academy of Inventors (Class of 2025)*, an honor that recognizes academic inventors whose work has generated significant societal and economic impact.
- Recipient of the 2023 *IEEE Trans. on Applied Superconductivity Van Duzer Prize Award*
- "For contributions to low power design of VLSI circuits and contributions to energy efficient computing," *ACM Fellow*, 2023
- "For leadership in low-power design of VLSI circuits and contributions to energy-efficient computing resulting in sustainable computing infrastructure," Fellow of the *American Association for the Advancement of Science (AAAS)*, 2023
- Best Paper Award of the *Int'l Symp. on Quality Electronic Design*, Mar. 2020
- Best Paper Award of the *Asia and South Pacific Design Automation Conference*, Jan. 2019
- *USC Viterbi School of Engineering Senior Research Award*, 2017
- "For contributions to modeling and design of low power VLSI circuits and systems, and energy efficient computing," *IEEE Circuits and Systems Society Charles A. Desoer Technical Achievement Award*, 2015
- Recognized as the *Second Most Prolific and Second Most Cited Author Awards* from the 20th Anniversary *Int'l Symp. on Low Power Electronics and Design*, Rome, Italy, Jul. 2015
- *Frequent Author Award* (Top Three Author Award) at the 20th Anniversary *Asia and South Pacific Design Automation Conference*, Chiba/Tokyo, Japan, Jan. 2015
- *ACM SIGDA Distinguished Service Award*, 2014
- Best Paper Award of the *ACM/IEEE Int'l Symp. on Low Power Design and Electronics*, Aug. 2014
- Best Paper Award of the *IEEE Computer Society Annual Symp. on VLSI*, Jul. 2014
- Recognized as one of the four DAC Prolific Authors (with 50+ papers) and the DAC Bronze Cited Author at the 50th anniversary of the Design Automation Conf., Austin, TX, Jun. 2013
- 2013 DAC Volunteer Service Award sponsored by IEEE CEDA
- Best Paper Awards of the 1989, 2000, and 2012 IEEE Int'l Conference on Computer Design, VLSI in Computers and Processors
- ACM Distinguished Scientist, 2008
- ACM SIGDA Service Award, 2004
- "For contributions to the theory and practice of low-power design and CAD," IEEE Fellow, 2001
- ACM SIGDA Distinguished Service Award, 2000
- Presidential Early Career Award for Scientists and Engineers, 1996
- NSF's Young Investigator Award, 1994
- NSF's Research Initiation Award, 1992
- 2007 IEEE Trans. on Circuits and Systems Guillemain-Cauer Best Paper Award
- Honorable Mention Award at the 2008 Int'l Symp. on Low Power Electronics and Design
- Best Paper Awards of the 33rd ACM/IEEE Design Automation Conference, Jun. 1996 and the 42nd ACM/IEEE Design Automation Conference, Jun. 2005
- Distinguished Paper Citation of the IEEE Int'l Conference on Computer Aided Design, 1990

- 1996 IEEE Circuits and Systems Society VLSI Systems Trans. Best Paper Award

University Service

- Co-organized and co-chaired the USC Symposium on the Future of Computing: A 25-Year Vision, March 13-14, 2025--see <https://sac.usc.edu/usc-symposium-on-the-future-of-computing/>.
- Member of the Computer Engineering Faculty Recruitment Committee, 2010-2011, 2013/2014, 2014/2015, 2016-2017, 2019-2025.
- Member of the VSoE Research Committee at USC, Sept. 2013 - 2017
- Member of the VSoE Transformative Faculty Hire Committee, Sept. 2013 - 2016
- Member of the Ming Hsieh Institute's Advisory Council, Jul. 2013 – Dec. 2015
- Chair of a faculty Promotion Committee in the EE department, 2014/2015, 2016-2017
- Member of the EE Dept. Faculty Recruitment Committee, 2014/2015
- Served on the Best Dissertation Selection Committee in the EE Dept., 2013
- Internal reviewer for an NSF Career Program proposal, 2013
- Chaired committee for evaluating Janet Roveda's Recruitment as an associate professor, 2012
- Division Director of USC Computer Engineering Division, Dec. 2005-Jun. 2010, and Acting Division Director at the end of 2012 and early 2013
- Served as the Chair of the 2005 Distinguished Lecture Series of the EE Dept.
- Served as the Associate Director of the CENG division of the EE Department at USC from 1998 to 2001
- Served on various departmental committees including the Faculty Merit Review Committee, Faculty Recruitment Committee, and the Promotion Committees in the EE department
- Chaired the Outstanding Research Paper Award Committee of the EE-Systems Department at USC in 1999
- Served on the Graduate Student Admissions Committee for entering M.S. and Ph.D. students in the CENG division of the EE-Systems Department at USC

Professional Service

- Served as technical area lead and panel moderator for the NSF Workshop on Micro/Nano Circuits and Systems Design in Dec 2020
- Participated in and contributed to the Computing Community Consortium (CCC) "Next Steps in Quantum Computing: Computer Science's Role," May 22-23, 2018
- Associate Editor of *Research, A Science Partner Journal*, 2018-present
- Served on the IEEE CASS Fellow (Selection) Committee, 2018
- Executive Committee Chair of the *Int'l Symp. on Low Power Electronics and Design*, July 2012 - present
- Editor-in-Chief of the *ACM Trans. on Design Automation of Electronic Systems*, Jun. 2008–May 2014
- Inaugural Editor-in-Chief of the *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, Jan. 2010–Dec. 2013
- Served as a member of the 2011 Selection Committee for the ACM SIGDA Outstanding New Faculty Award
- Served as the Chair of the Selection Committee for the IEEE CEDA and ACM SIGDA 2010 A. *Richard Newton Technical Impact Award in Electronic Design Automation*
- General Co-chair of the 2010 *IEEE Int'l Symp. on Circuits and Systems*, Paris, France
- Associate Editor
 - Foundations and Trends in Electronic Systems and Circuits Journal, Now Publishers, Inc. of Delft, 2006-2014
 - Journal of Low Power Electronics, 2005-2014

- IEEE Trans. on Computer Aided Design, 1997-2006
- IEEE Trans. on Circuits and Systems I, 2002-2003
- ACM Trans. on Design Automation of Electronic Systems, 1998-2004
- Distinguished Lecturer/Speaker
 - Association for Computing Machinery (ACM), 2007-2015
 - IEEE Solid State Circuits Society, 2001-2007
- VP of Publications of the IEEE Circuits and Systems Society, 2005-2006
- Chair of the Distinguished Lecturer Program, IEEE Circuits and Systems Society, 2003-2004
- Member of the Board of Governors, IEEE Circuits and Systems Society, 2000-2003
- Member of the ACM-SIGDA Advisory Board, managing the ACM Outstanding Ph.D. Dissertation Award in Electronic Design Automation, the DAC Young Student Support Program, the Multimedia Monograph Series, and the SIGDA Technical Committees, 1996-2009
- Chair of the Best Paper Award Committee for the IEEE Trans. on Computer Aided Design, 1999
- Member of the Research or Technical Advisory Board
 - Atrenta Inc., 2003-2015
 - Envis Inc., 2006-2009
 - PwrLite Inc., 2007-2009
 - Fulcrum Microsystems, 2001-2004
 - Aplus Design Technologies Inc., 2000-2003
 - Magma Design Automation Inc., 1997-2002
 - EPIC Design Technology, 1994-1997
- General Chair
 - *Int'l Symp. on Circuits and Systems*, June 2010
 - *Int'l Symp. on Quality of Electronic Design*, San Jose, CA, Apr. 2007
 - *Int'l Symp. on Physical Design*, Monterey, CA, Apr. 2003
 - *Int'l Symp. on Low Power Electronics and Design*, Monterey, CA, Aug. 1997
 - *Int'l Symp. on Low Power Design*, Laguna, CA, Apr. 1995
- Technical Chair
 - *Int'l Symp. on Physical Design*, Del Mar, CA, Apr. 2002
 - *Int'l Symp. on Low Power Electronics and Design*, Monterey, CA, Aug. 1996
 - *Int'l Workshop on Low Power Design*, Napa Valley, CA, Apr. 1994
- Executive Committee Member
 - *Int'l Symp. on Low Power Electronics and Design*, since 1996
 - *Int'l Conference on Computer Aided Design*, 2006-2010
- Technical Program Committee Member
 - *The 18th CSI International Symposium on Computer Architecture & Digital Systems*, Tehran, Iran, Oct. 7-8, 2015
 - *Int'l Conference on Hardware/Software Codesign and System Synthesis*, 2013-2015
 - *Symp. on Embedded Systems for Real-Time Multimedia (ESTIMedia)*, 2007-2012
 - *Symp. on VLSI Design, Automation and Test (VLSI-DAT)*, 2012
 - *2009 Great Lakes Symp. on VLSI*
 - *2008 IEEE Int'l Design and Test Workshop*, Tunisia
 - *Symp. on VLSI Design, Automation and Test (VLSI-DAT)*, Hsinchu, Taiwan, 2009
 - *NanoArch*, 2007-2008
 - *The 13th CSI Computer Conference (CSICC'08)*, Kish Island, Iran, March 9-11, 2007
 - *Conference on VLSI – Design Automation Technologies*, Taipei, Taiwan, 2006-2007
 - *1998-2000 Int'l Conference on Computer Aided Design*
 - *1998-2001 Int'l Symp. on Physical Design*
 - *1997-2000 Int'l Workshop on Logic Synthesis*
 - *2000 Asia and South Pacific Design Automation Conference*
 - *1998-2003 Design Automation and Test in Europe Conference*
 - *1994-2000 Int'l Symp. on Low Power Electronics and Design*
 - *2001-2003 Int'l Symp. on Quality of Electronic Design*

US Patents

- [S1] Nazemi; Mahdi (Los Angeles CA); Fayyazi; Arash (Los Angeles CA); Esmaili; Amirehossein (Los Angeles CA); Pedram; Massoud (Beverly Hills CA), "System and Method for Hybrid Arithmetic and Logic Processing of Neural Networks," United States Patent filed on Dec. 22, 2022. EFS ID: 47261286. Application Number: 20230281432, Publication Date: Sep 7, 2023.
- [S2] Zhang; Bo (Los Angeles CA); Chen; Zeming (Los Angeles CA); Pedram; Massoud (Beverly Hills CA); "Design of High-Performance and Scalable Montgomery Modular Multiplier Circuits," US Patent filed on Apr 23, 2021, US Application Publication number: 20230185537, US Application Publication Date: Jun 15, 2023, International Patent filed on May 7, 2021, International Application Number: PCT/US2021/028896.
- [S3] Pasandi; Ghasem (Los Angeles, CA); Pedram; Massoud (Beverly Hills CA), "Efficient pipelined architecture for superconducting single flux quantum logic circuits utilizing dual clocks," United States Patent **11,303,281**, Issued: Apr. 4, 2022, Filed: Apr. 5, 2021.
- [S4] Katam; Naveen (Los Angeles, CA); Mukhanov; Oleg (Elmsford, NY); Pedram; Massoud (Los Angeles, CA), "Superconductive magnetic field programmable gate array," United States Patent **10,707,873**, Issued: Jul. 7, 2020, Filed: Mar. 21, 2019.
- [S5] Pedram; Massoud (Beverly Hills, CA); Iranli; Ali (Los Angeles, CA), "Dynamic backlight scaling for power minimization in a backlit TFT-LCD," United States Patent **8,094,118**, Issued: Jan 10, 2012, Filed: Mar 2, 2006.
- [S6] Fallah; Farzan (San Jose, CA); Pakbaznia; Ehsan (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Sizing and placement of charge recycling (CR) transistors in multithreshold complementary metal-oxide-semiconductor (MTCMOS) circuit," United States Patent **7,834,684**, Issued: Nov 16, 2010, Filed: Oct 31, 2008.
- [S7] Fallah; Farzan (San Jose, CA); Abdollahi; Afshin (Los Angeles, CA); Pedram; Massoud (Los Angeles, CA); "Power mode transition in multi-threshold complementary metal oxide semiconductor (MTCMOS) circuits," United States Patent **7,613,942**, Issued: November 3, 2009, Filed: May 31, 2006.
- [S8] Fallah; Farzan (San Jose, CA); Amelifard; Behnam (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Setting threshold voltages of cells in a memory block to reduce leakage in the memory block," United States Patent **7,573,775**, Issued: August 11, 2009, Filed: February 9, 2007.
- [S9] Fallah; Farzan (San Jose, CA); Amelifard; Behnam (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "PG-Gated Data Retention Technique for Reducing Leakage in Memory Cells," United States Patent **7,447,101**, Issued: November 4, 2008, Filed: December 22, 2006.
- [S10] Fallah; Farzan (San Jose, CA); Pakbaznia; Ehsan (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Recycling Charge to Reduce Energy Consumption During Mode Transition in Multi-Threshold Complementary Metal-Oxide-Semiconductor (MTCMOS) Circuits," United States Patent **7,400,175**, Issued: July 15, 2008, Filed: May 30, 2007.
- [S11] Fallah; Farzan (San Jose, CA), Aghaghiri; Yazdan (Los Angeles, CA), Pedram; Massoud (Los Angeles, CA). "System and method for identifying optimal encoding for a given trace," United States Patent **7,236,107**, Issued: June 26, 2007, Filed: September 20, 2004.
- [S12] Fallah; Farzan (San Jose, CA); Aghaghiri; Yazdan (Los Angeles, CA); Pedram; Massoud (Los Angeles, CA). "Reducing transitions on address buses using instruction-set-aware system and method," United States Patent **6,907,511**, Issued: June 14, 2005, Filed: January 14, 2003.
- [S13] Fallah; Farzan (San Jose, CA); Aghaghiri; Yazdan (Los Angeles, CA); Pedram; Massoud (Los Angeles, CA). "System and method for reducing transitions on address buses," United States Patent **6,834,335**, Issued: December 21, 2004, Filed: August 9, 2002.
- [S14] Fallah; Farzan (San Jose, CA); Aghaghiri; Yazdan (Los Angeles, CA); Pedram; Massoud (Los Angeles, CA), United States Patent **6,813,700** "Reduction of bus switching activity using an encoder and decoder," Issued: November 2, 2004, Filed: June 3, 2002.

Citation Statistics and Top 20 Cited Papers as of June 13, 2026

Google Scholar: citations: 40,844, h-index=102, i10-index = 607

1. M. Pedram. "Power minimization in IC design: principles and applications," *ACM Trans. on Design Automation of Electronic Systems*, Vol. 1, No. 1, 1996, pages 3-56. [842 citations]
2. M. Pedram and S. Nazarian. "Thermal modeling, analysis and management in VLSI circuits: Principles and methods," *Proc. of IEEE, Special Issue on Thermal Analysis of ULSI*, Vol. 94, No. 8, Aug. 2006, pages 1487-1501. [695 citations]
3. P. Rong and M. Pedram. "An analytical model for predicting the remaining battery capacity of Lithium-ion batteries," *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, Vol. 14, No. 5, May 2006, pages 441-451. [531 citations]
4. Q. Wu, M. Pedram, and X. Wu. "Clock-gating and its application to low power design of sequential circuits," *IEEE Trans. on Circuits and Systems—I*, Vol. 47, No. 3, Mar. 2000, pages 415-420. [484 citations]
5. JM. Chang and M. Pedram. "Energy minimization using multiple supply voltages," *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, Vol. 5, No. 4, Dec. 1997, pages 436-443. [484 citations]
6. A. E. Eshratifar, M. S. Abrishami and M. Pedram. "JointDNN: An Efficient Training and Inference Engine for Intelligent Mobile Cloud Computing Services," in *IEEE Trans. on Mobile Computing*, 20(2): 565-576 (2021). [474 citations]
7. E. Macii, M. Pedram, and F. Somenzi. "High level power modeling, estimation and optimization," *IEEE Trans. on Computer Aided Design*, Vol. 17, No. 11, Nov. 1998, pages 1061-1079. [445 citations]
8. A. Abdollahi, F. Fallah, and M. Pedram. "Leakage current reduction in CMOS VLSI circuits by input vector control," *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, Vol. 12, No. 2, Feb. 2004, pp.140-154. [396 citations]
9. F. Fallah and M. Pedram. "Standby and active leakage current control and minimization in CMOS VLSI circuits," *IEICE Trans. on Electronics*, Special Section on Low-Power LSI and Low-Power IP, Vol. E88–C, No. 4 Apr. 2005, pages 509-519. [379 citations]
10. O. Akbari, M. Kamal, A. Afzali-Kusha, and M. Pedram. "Dual-quality 4:2 compressors for utilizing in dynamic accuracy configurable multipliers," *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, Vol. 25, No. 4, Apr. 2017, pp. 1352-1361. [374 citations]
11. Q. Qiu and M. Pedram. "Dynamic power management based on continuous-time Markov decision processes," *Proc. of 36th Design Automation Conference*, Jun. 1999, pages 555-561. [364 citations]
12. M. Maleki, K. Dantu, and M. Pedram. "Power-aware source routing protocol for mobile ad hoc networks," *Proc. of Symp. on Low Power Electronics and Design*, Aug. 2002, pages 72-75. [352 citations]
13. E. Pakbaznia and M. Pedram. "Minimizing data center cooling and server power costs," *Proc. of Symp. on Low Power Electronics and Design*, Aug. 2009, pages 145-150. [335 citations]
14. K. Choi, R. Soma and M. Pedram. "Dynamic Voltage and Frequency Scaling based on Workload Decomposition," *Proc. of Symp. on Low Power Electronics and Design*, Aug. 2004, pages 174-179. [319 citations]
15. X. Lin, Y. Wang, Q. Xie, and M. Pedram. "Task Scheduling with Dynamic Voltage and Frequency Scaling for Energy Minimization in the Mobile Cloud Computing Environment," *IEEE Trans. on Services Computing*, Vol. 8, No. 2, March-April 2015, pp. 175-186. [315 citations]
16. WC. Cheng, Y. Hou, and M. Pedram. "Power minimization in a backlit TFT-LCD display by concurrent brightness and contrast scaling," *IEEE Trans. on Consumer Electronics*, Vol. 50, No. 1, Feb. 2004, pages 25-32. [315 citations]
17. K. Choi, R. Soma, and M. Pedram. "Fine-grained dynamic voltage and frequency scaling for precise energy and performance trade-off based on the ratio of off-chip access to on-chip

computation times,” *IEEE Trans. on Computer Aided Design*, Vol. 24, No. 1, Jan. 2005, pp.18-28. [314 citations]

18. A. H. Ajami, K. Banerjee, and M. Pedram. “Modeling and analysis of non-uniform substrate temperature effects in high performance VLSI,” *IEEE Trans. on Computer Aided Design*, Vol. 24, No. 6, Jun. 2005, pages 849-861. [282 citations]
19. H. Goudarzi and M. Pedram. “Multi-dimensional SLA-based resource allocation for multi-tier cloud computing systems,” *Proc. of IEEE Conf. on Cloud Computing*, Jul. 2011, pages 324-331. [279 citations]
20. A. E. Eshratifar, A. Esmaili, and M. Pedram. “BottleNet: A Deep Learning Architecture for Intelligent Mobile Cloud Computing Services,” *Proc. of Int’l Symp. on Low Power Electronics and Design*, Jul. 2019. [267 citations]

Publications

Books

- [B1] M. Pedram and J. Rabaey (editors). *Power-aware Design Methodologies*, Kluwer Academic Publishers, Boston, Jun. 2002.
- [B2] JM. Chang and M. Pedram. *Power Optimization and Synthesis at Behavioral and System Levels Using Formal Methods*, Kluwer Academic Publishers, Boston, Jun. 1999.
- [B3] S. Iman and M. Pedram. *Logic Synthesis for Low Power VLSI Designs*, Kluwer Academic Publishers, Boston, Nov. 1997.
- [B4] J. Rabaey and M. Pedram (editors). *Low Power Design Methodologies*, Kluwer Academic Publishers, Boston, Oct. 1995.
- [B5] M. Pedram. *ACM-SIGDA Multimedia Monograph Series*, Volumes 1-12, ACM Publications.

Book Chapters

- [BC1] E. Khatami Zenoian, M. Kamal, A. Afzali-Kusha, and M. Pedram. “Enabling Memory-Augmented Neural Networks for Efficient Edge Applications,” In *AI-Enabled Electronic Circuit and System Design: From Ideation to Utilization*, Springer Nature Switzerland, 2025, pp. 565-605.
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- [O2] Mustafa Munir, Arash Fayyazi, Shahin Nazarian, Massoud Pedram. "qVSim: Verification and Simulation Frameworks Targeting Post-layout SFQ Circuits," Presented at Applied Superconductivity Conference, Oct. 2022.
- [O3] Arash Fayyazi, Mustafa Munir, Shahin Nazarian, Massoud Pedram. "Logic Verification of Sequential Circuits in Superconducting Single Flux Quantum Technologies," Presented at Applied Superconductivity Conference, Oct. 2022.
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- [O6] Mingye Li, Bo Zhang, Massoud Pedram. "Striking a Good Balance Between Area and Throughput of RSFQ Circuits Containing Feedback Loops," Presented at Applied Superconductivity Conference, Oct. 2022.
- [O7] Haolin Cong, Mingye Li, Mustafa Altay Karamuftuoglu and Massoud Pedram. "Striking a Good Balance Between Area and Throughput of RSFQ Circuits Containing Feedback Loops," Presented at Applied Superconductivity Conference, Oct. 2022.
- [O8] N. Katam and M. Pedram. "Timing Characterization and Enabling Static Timing Analysis in Rapid Single Flux Quantum Circuits," Presented at *13th Biennial European Conference on Applied Superconductivity (EUCAS)*, Geneva, Switzerland, Sept. 2017.
- [O9] N. Katam, A. Shafaei, and M. Pedram. "Design of Complex Rapid Single Flux Quantum Cells with Application to Logic Synthesis," Presented at *16th International Superconductive Electronics Conference (ISEC)*, Sorrento, Italy, Jun. 2017.
- [O10] S. Nazar Shahsavani, A. Shafaei, C. J. Fourie, and M. Pedram. "An Integrated Cell Placement and Interconnect Synthesis Tool for Large-scale SFQ Logic Circuits," Presented at *Applied Superconductivity Conference (ASC)*, Denver, United States, Sept. 2016.
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- [O21] J. Lou, A. Salek, and M. Pedram. "An exact solution to simultaneous technology mapping and linear placement problem for trees," *Proc. of Int'l Workshop on Logic Synthesis*, May 1997, session 4, paper 4.
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- [O23] CS. Ding and M. Pedram. "A comparative study of switching activity estimators," *Proc. of Int'l Workshop on Power and Timing Modeling, Optimization, and Simulation*, Oct. 1995, pages 377-396.
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- [O27] K. R. Pan and M. Pedram. "LUT-based FPGA synthesis for low power," *Proc. of Int'l Workshop on Power and Timing Modeling, Optimization, and Simulation*, Oct. 1994, pages 124-131.
- [O28] D. Mukherjee, M. Pedram, and M. A. Breuer. "CONSYST - an integrated test controller synthesis system," Presented at the *Int'l Test Synthesis Workshop*, May 1994.
- [O29] J. Monteiro, S. Devadas, B. Lin, CY. Tsui, M. Pedram, and A. M. Despain. "Exact and approximate methods of switching activity estimation in sequential logic circuits," *Proc. of 1994 Int'l Workshop on Low Power Design*, Apr. 1994, pages 117-122.
- [O30] C. Ding and M. Pedram. "Efficient symbolic simulation under the extended bounded delay model," *Proc. of Tau'93: Int'l Workshop on Timing Issues in the Specification and Synthesis of Digital Systems*, Sept. 1993.
- [O31] S. Iman, M. Pedram, C. Fabian, and J. Cong. "Finding uni-directional cuts based on physical partitioning and logic restructuring," *Proc. of 4th ACM/IEEE Physical Design Workshop*, Apr. 1993, pages 187-198.
- [O32] M. Pedram, B. S. Nobandegani, and B. T. Preas. "On the segmentation design and routability analysis for row-based FPGAs," *Proc. of 4th ACM/IEEE Physical Design Workshop*, Apr. 1993, pages 213-214.
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EXHIBIT G

Udit Sharma, M.S.

Education

Master of Science in Materials Engineering (2000), Dept. of Chemical and Materials Engineering, University of Alberta, Canada

Bachelor of Technology in Materials Engineering (1997), Dept. of Materials and Metallurgical Engineering, Indian Institute of Technology, India

Diploma in Biotechnology (2003), Vitesse, National Research Council Canada in partnership with Univ. of Ottawa and Carleton University.

Experience

2022 - Present

Scientific Fellow

2016 - 2022

Project Manager, Materials Sciences

Eurofins EAG Laboratories

- Consult, plan and manage multi-technique analysis projects.
- Discuss new projects with the clients. Provide recommendations on suitability of various materials characterization techniques.
- Write summary reports, communicate results to the clients.
- Technical account management for selected clients/accounts.
- Design technical plans to address customer's needs and discuss pricing
- Responsible for technical accuracy of projects and managed accounts.

2008 -2016

Senior Engineer

EAG (Evans Analytical Group)

- Performed TEM/STEM/EELS/EDS/SEM/EBSD characterization of a wide variety of advanced materials systems and devices, nano-structures, MEMS, thin film coatings, displays (LEDs, OLEDs), GaN, GaAs and InP based optical devices, coatings on metals, polymers, image sensors, microelectronic devices, FinFETs, MRAMs, Si devices, CdTe, CIGS photovoltaics.
- The focus of the work was to interface with clients to understand the goals of analysis, set up a plan of experiment, perform TEM/STEM/EELS/EDS/SADP/CBED/NBD/FFT characterization, interpret the results and provide a summary report of the data and its interpretation. The position involved frequent pre-and post-analysis discussions with the clients.
- Position involved understanding complex analytical goals, performing analysis in a timely manner and presenting high quality results in an easy and clearly interpretable manner,
- Supported a broad spectrum of clients from different industries/academia with work ranging from the front end, viz design and synthesis of new types of materials, to new product or device development, process monitoring, yield enhancement, fab support, reliability to back end, viz failure analysis.
- Interfaced with many other analytical groups within EAG for multi technique analyses like SIMS, Auger, XPS, SEM, DB-FIB, RBS, FTIR, RAMAN, AFM, XRD.
- Trained other team members.

2004-2007

Electron Microscopist & Member of R&D Group

Semiconductor Insights Inc.

- Set up and operate TEM analysis. Involved ownership of the instrument, development of imaging & sample preparation techniques and training of lab personnel.
- Develop imaging and analysis protocols for TEM, SEM, AFM, SCM for applications in competitive intelligence and Intellectual Property support.
- Analysis of various Si & SiGe-based devices.

2000-2003

TEM Analyst, Materials and Device Analysis Group**Nortel Networks/Bookham Inc.**

- Primary responsibilities were process characterization for yield improvement and process reliability qualification on InP and GaAs-based devices through TEM analysis and FIB microsurgery; for: a) MOCVD and MBE growth defects, layer thickness uniformity, and quality of interfaces, b) characterization of ohmic contacts and other processing parameters, such as photolithography through study of grating quality in DFB/DBR lasers, c) life tested and field breakdowns.
- Comprehensive analysis report writing. Understanding of processing of various InP and GaAs optoelectronic devices.

1998-2000

Graduate Research Assistant**University of Alberta**

- Conducted microstructural characterization of microalloyed steels using TEM and SEM.
- Investigations into the structure-property relationship of X70, X80 and X90 steel's microstructure and thermo-mechanical processing.
- Teaching assistant for course 'Introduction to Materials Science'

Publications

- New approaches for characterizing Lithium Ion Battery Materials. S.Patel, B.Burrow, D. Susnitzky and U. Sharma. The 15th International Meeting on Lithium Batteries - IMLB 2010, © 2010 The Electrochemical Society. Poster Session.
- Microstructural Characterization of CdTe and CIGS Thin Film Solar Cells by Analytical Transmission Electron Microscopy. Microscopy and Microanalysis / Volume 15 / Supplement S2 / July 2009, pp 148-149
- Analytical Transmission Electron Microscopy characterization of nano-structured materials. Nanotech Conference May 2009, Poster Session.
- FIB-TEM Method of root cause analysis of InP ridge laser devices. E. Griswold, U. Sharma, M. Phaneuf. J. Vac. Sci. Technol. A 22, 930 (2004)
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- U. Sharma and D.G. Ivey, Microstructure of Microalloyed Linepipe Steels, presented at IPC2000 Conference, Calgary, Alberta, October 1-5, 2000, published in Proceedings of the International Pipeline Conference 2000, Volume 1, ASME, New York, N.Y., 193-201

Udit Sharma, M.S.

Depositions and Testimonies

Orexo AB et al. v. Sun Pharmaceutical Industries Ltd. et al., No. 20-cv-12588 (consolidated):

Deposition Jan 2023, US District Court for the District of New Jersey (virtually).

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EXHIBIT H



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Professional Profile

Dr. Sourì's expertise is in microelectronics and computing systems. His professional activities involve advising industrial and legal clients as well as government entities on science and technology matters addressing issues related to intellectual property, product reliability, and failure analysis.

Dr. Sourì has led complex investigations involving electronics, computer communications, and software controls for safety-critical applications in the medical device, automotive, aviation and process controls industries.

Dr. Sourì's experience in the area of microelectronics systems includes technical investigations into computer memories, device physics, integrated circuit (IC) fabrication processes, and esoteric semiconductor material systems for solid-state lighting. He also has broad experience with power electronic devices, circuit protection, IC packaging, and printed circuit board assemblies and display technologies as used in various industries including consumer appliances.

In computing systems, Dr. Sourì has worked extensively with embedded controls systems for automotive applications, computer hard disk drives and for drug delivery and implantable devices. He also has expertise in telephony, secure mobile and Internet communications, microprocessor architectures particularly for mobile computing and has analyzed audio/video/image processing software and developed digital asset delivery technologies.

Dr. Sourì received his Ph.D. in Electrical Engineering at Stanford University on 3-Dimensional integration of ICs and has taught several courses on IC fabrication, optical fiber communications, TCP/IP networking, and communications protocols and implementation of systems on 3-D chips. He also read Engineering Science at Balliol College, Oxford, where he specialized in photoreflectance microscopy of semiconductor materials. His current research interests include: cloud computing, information security and privacy; financial portfolio optimization, and trading algorithms and software.

Academic Credentials & Professional Honors

M.A., University of Oxford, UK, 2007

Ph.D., Electrical Engineering, Stanford University, 2003

M.S., Electrical Engineering, Stanford University, 1994

B.A., Engineering Science, University of Oxford, UK, 1992

Prior Experience

Founder, Merenga Inc., 2000-2002

Co-Founder and Engineering Manager, arcadiaOne, Inc., 1999-2000

Research Engineer, Circuit Protection Division, Raychem, 1996-1997

Research Scientist, Corporate Research & Development, Raychem, 1994-1996

Professional Affiliations

Institution of Engineering and Technology—IET (member)

Institute of Electrical and Electronic Engineers (senior member)

Oxford University Society (life member)

Oxford Union (member)

Patents

Patent 6,188,556: Two-Terminal Transistor PTC Circuit Protection Devices, WO0024126, 2001 (with C. McCoy, H. Duffy, A. Cogan, and R. Bommakant).

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Subramanian V, Toita M, Ibrahim N, Souri S, Saraswat K. Low-leakage germanium-seeded laterally-crystallized single-grain 100nm TFTs for vertical integration applications. IEEE Electron Device Letters, Vol. 20, pp. 341-343, 1999.

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Chu A, Souri S, Melfand M, Kinsman K, Dupon R. Superior electrical performance of Raychem ZnO Varistor through advanced processing. Proceedings, 1st Asian Meeting on Ferroelectricity, Xian, China, October 5-8, 1995.

Books

A. Arora, S. Arun Lele, N. Medora, S. Souri. Lithium-Ion Battery Failures in Consumer Electronics. Artech House, 2019.

Souri S, Chiang T, Kapur P, Banerjee K, Saraswat KC. 3-D ICs deep submicron interconnect performance modeling and analysis. In: Interconnect Technology and Design for Gigascale Integration. Davis J and Meindl J (eds), Kluwer Academic Publishers, October 2003.

Shukri J. Sourì, Ph.D.

List of Testimony

Longitude Licensing Limited, et al. v BOE Technology Group Co. Ltd. Et al. in the United States District Court For The Eastern District Of Texas, Marshall Division; Case Nos. 2:25-CV-00440, 2:25-CV-00358. Deposition 2026. Retained by Defendants.

Hyundai Motor Company, et al. v Hyundai Technology Group, Inc., et al. in the United States District Court For The Central District Of California; Case No. 8:23-CV-01709. Deposition 2025, Trial 2026. Retained by Plaintiffs.

Longitude Licensing Limited, et al. v BOE Technology Group Co. Ltd. in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:23-CV-00515. Deposition 2025, Trial 2026. Retained by Defendants.

Entropic Communications, LLC. v DirecTV, LLC. et al. in the United States District Court For The Central District Of California; Case No. 2:22-CV-07775. Deposition 2025. Retained by Plaintiffs.

Entropic Communications, LLC. v Dish Network Corp. et al. in the United States District Court For The Central District Of California; Case No. 2:22-CV-07959. Deposition 2025. Retained by Plaintiffs.

Adaptive Spectrum And Signal Alignment, Inc. v Charter Communications, Inc. et al. in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:24-CV-00124. Deposition 2025. Retained by Plaintiffs.

ParkerVision, Inc. v RealTek Semiconductor Corp. in the United States District Court For The Western District Of Texas, Waco Division; Case No. 6:22-CV-001162; 6:23-CV-00347. Depositions 2025. Retained by Defendants.

Time Warner Cable Enterprise LLC et al. v Nokia of America Corp in the Supreme Court of The State of New York, County of New York. Index No. 654900/2022. Deposition 2025. Retained by Defendants.

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RightQuestion LLC v Verizon Business Network Services, LLC et al. and AT&T Corp, et al. in

the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:24-CV-00091 and No. 2:24-CV-00094. Deposition 2025. Retained by Plaintiffs.

Hanshow Technology Co., Ltd. v SES-Imagotag GMBH et al. in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:23-CV-0174. Depositions 2024, 2025. Retained by Plaintiffs.

Time Warner Cable Enterprises LLC v. Nokia of America Corporation, in the Supreme Court of The State of New York, County of New York. Index No. 650748/2022. Deposition 2024. Retained by Defendants.

Advanced Micro Devices, Inc. v. Realtek Semiconductor Corporation, in the United States Patent And Trademark Office before the Patent And Trial Board. Case No. IPR2023-00920. Deposition 2024. Retained by Plaintiffs.

Aspen Specialty Insurance Company, et al., v. Nucor Corporation and Nucor Steel Louisiana, LLC in the General Court of Justice, Superior Court Division, In The State of North Carolina, County of Mecklenburg; Case No. 19-CVA-19887. Deposition 2024. Retained by Defendants.

In the Matter of Certain Integrated Circuits, Components Thereof, And Products Containing The Same, Investigation No. 337-TA-1350, before the United States International Trade Commission. Deposition 2023, Trial 2023. Retained by Complainant, RealTek Semiconductor.

Ollnova Technologies, Ltd v ecobee Technologies, ULC in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:22-CV-00072. Deposition 2023, Trial 2023. Retained by Defendant.

Entropic Communications, LLC v Charter Communications, Inc. in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:22-CV-00125. Deposition 2023. Retained by Plaintiffs.

Ecofactor, Inc. v Ecobee, Inc. in the United States District Court For The Western District Of Texas, Waco Division; Case No. 6:22-CV-00033. Deposition 2023. Retained by Defendant.

Ecofactor, Inc. v Ecobee, Inc. in the United States District Court For The District Of Delaware; C.A. No. 1:21-CV-00323-MN. Deposition 2023. Retained by Defendant.

Ecofactor, Inc. v Ecobee, Inc. in the United States District Court For The Western District Of Texas, Waco Division; Case No. 6:21-CV-00428. Deposition 2022, Trial 2023. Retained by Defendant.

Orange Electronic Co. Ltd v Autel Intelligent Technology Corp., Ltd in the United States District Court For The Eastern District Of Texas, Marshall Division; Case No. 2:21-CV-0240. Deposition 2023, Trial 2023. Retained by Defendant.

In the Matter of Certain Smart Television, Investigation No. 337-TA-1338, before the United States International Trade Commission. Depositions 2023. Retained by Complainant, Maxell.

In the Matter of Certain Mobile Electronic Devices, Investigation No. 337-TA-1324, before the United States International Trade Commission. Deposition 2023. Retained by Complainant, Maxell.

MaxPower Semiconductor, Inc. v. ROHM Company, Ltd. And ROHM Semiconductor USA, LLC. in JAMS; Arbitration No. 1100110602. Trial 2022. Retained by Defendant.

In the Matter of Certain Power Inverters And Converters, Vehicles Containing The Same, And Components Thereof, Investigation No. 337-TA-1267, before the United States International Trade Commission. Deposition, Trial 2022. Retained by Respondents, BMW, Volkswagen.

In the Matter of Silicon Photovoltaic Cells And Modules With Nanostructures, And Products Containing The Same, Investigation No. 337-TA-1271, before the United States International Trade Commission. Deposition, Trial 2022. Retained by Complainant, ASG Technology.

In the Matter of Certain Smart Thermostat Systems, Smart HVAC Systems, Smart HVAC Control Systems, And Components Thereof, Investigation No. 337-TA-1258, before the United States International Trade Commission. Deposition, Trial 2021. Retained by Respondents, ecobee.

In the Matter of Certain Electrical Connectors And Cages, Components Thereof, And Products Containing The Same, Investigation No. 337-TA-1241, before the United States International Trade Commission. Deposition, Trial 2021. Retained by Complainant, Amphenol.

Ecofactor, Inc. v Ecobee, Inc. in the United States District Court For The Western District Of Texas, Waco Division; Case No. 6:20-CV-00078. Deposition 2021. Retained by Defendant. A-3

In re: Joint Venture and Contract Disputes Between INTEL CORPORATION and MICRON TECHNOLOGY, INC., and MICRON UTAH, AMERICAN ARBITRATION ASSOCIATION, Case No. 01-20-0014-3443. Deposition, Trial 2021. Retained by Intel.

HLFIP Holding, Inc. v York County, Pennsylvania in the United States District Court For The Middle District Of Pennsylvania; Civil Action No. 1:20-cv-00186. Deposition 2021. Retained by Plaintiff.

Xilinx, Inc. v. Arbor Global Strategies, LLC, in the United States Patent And Trademark Office before the Patent And Trial Board. Case No. IPR2020-01567/68/70/71. Deposition 2021. Retained by Defendant.

Peter Romans et al. v Ford Motor Company in the United States District Court For The Southern District Of Ohio, Eastern Division; Case No. 2:16-CV-00068. Deposition 2017. Trial 2021. Retained by Defendant.

Estech Systems, Inc. v. Target Corp. et al. in the United States District Court for the Eastern District of Texas, Marshall Division; Case No.: 2:20-cv-00123 (Lead Case). Deposition 2021 Retained by Defendants.

Arbor Global Strategies LLC v. Samsung Electronics Co. Ltd et al. in the United States District Court for the Eastern District of Texas, Marshall Division; Case No.: 2:19-cv-00333. Deposition 2021. Retained by Plaintiff.

Louisiana Citizens Property Insurance Corporation, et al. v. Arceneaux Ford, Inc., et al. in the 16th Judicial District Court, Parish of St. Mary, State of Louisiana, No. 129069 Division "A". Deposition 2021. Retained by Defendants.

Matthews International Corp. v. IR Environmental Solutions, LLC et al. in the United States District Court for the Western District of Pennsylvania; Case No.: 02-20-cv-00089. Deposition, Hearing 2020. Retained by Plaintiff.

State of Ohio v Peter Romans in the Madison County Common Pleas Court. Trial 2020. Retained by Plaintiff.

In the Matter of Certain Smart Thermostats, Smart HVAC Systems, and Components Thereof, Investigation No. 337-TA-1185, before the United States International Trade Commission. Deposition, Trial 2020. Retained by Respondent, ecobee.

EXHIBIT I

Vivek Subramanian

Education

3 / 96 - 6 / 98

Ph.D. in Electrical Engineering, Stanford University
Stanford, California

Honors earned: Graduate Fellowship, Eastman Kodak Company

9 / 94 - 3 / 96

MS in Electrical Engineering, Stanford University
Stanford, California

8 / 90 - 5 / 94

BS in Electrical Engineering, Louisiana State University
Baton Rouge, Louisiana

Honors earned: Summa cum laude, College Honors, Outstanding Senior, Junior and Sophomore, Honors College

Professional Experience

2018 - present

École polytechnique fédérale de Lausanne, Switzerland

2025 -present, Director, Institute of Electrical and Micro Engineering (IEM)

2021 – present: Director, Micromanufacturing Science and Engineering Center (M2C)

2021 – 2025: Associate Director, Institute of Electrical and Micro Engineering (IEM)

2021 – present: Scientific Board, Swiss National Strategic Focus Area – Advanced Manufacturing

8/18 – present: Professor

2000 - present

Department of Electrical Engineering & Computer Sciences, University of California
Berkeley, CA

7/20 to 12/25 : Adjunct Professor

7/11 – 6/20: Professor

7/05 – 7/11: Associate Professor

7/00-7/05: Assistant Professor

Honors Earned:

2015 IEEE Kiyo Tomiyasu Award

Best in session and best in track awards, 2013 IMAPS Microelectronics Conference

Outstanding paper award, 2012 IMAPS Microelectronics Conference

2008 Printed Electronics Champion, Printed Electronics USA Conference

Outstanding Teaching Award, EECS Department, UC Berkeley, 2005

Best paper award, 2004 IEEE Device Research Conference

Nominated to MIT's Technology Review top 100 young innovators list (TR100), 2002

National Science Foundation Young Investigator (CAREER) Award, FY2002,

Nominated to National Academy of Engineering's Frontiers of Engineering, 2002

2002 Paul Rappaport Award for best paper in an IEEE EDS Journal

2009 - 2018

Professor, World Class University Program, Sunchon National University
Sunchon, Korea

Principal Investigator leading large research program focused on developing a new department in the area of printed electronics including 9 faculty covering multiple disciplines and 3 different countries

2000 - present

Independent Consultant

Consultant to the semiconductor industry and its associated fields in the following areas:

Memory technology and design

Silicon Process Technology

Display / Imager and Flexible Electronics Technology

RFID Technology

Intellectual Property Consulting

Technology Evaluation / Venture Capital due diligence

2014 – 2022	Founder and CTO, Locix Inc. San Bruno, CA Senior leadership role in venture-funded startup company working on wireless networking
2004 – 2013	Founding Scientific Advisor, Kovio, Inc. Sunnyvale, CA Scientific advising to printed electronics startup company
2008 – 2011	Chief Technical Advisor, QuSwami, Inc. San Francisco, CA Scientific advising to energy conversion device startup company
1998-2000	Consulting Assistant Professor, Electrical Engineering Department, Stanford University. Stanford, CA Advisory role for research group of Prof. Krishna C. Saraswat
1998-2000	Visiting Research Engineer, Electrical Engineering Department, University of California. Berkeley, CA Research into 25nm MOSFET technologies for giga-scale integration
1998 - 2000	Founder, Matrix Semiconductor, Inc. Santa Clara, CA Co-founder and technical advisor of startup company working on high-density memory technology Honors Earned: Nominated to Scientific American's SA50 List for Visionary Technology Finalist for 2003 World Technology Award for Information Technology Hardware Winner, 2005 EDN Innovation Award
1998	Co-instructor, Electrical Engineering Department, Stanford University. Stanford, CA Co-teaching of EE311, Advanced Integrated Circuit Fabrication Processes
1997	Intern, Advanced Product Research and Development Laboratory, Motorola Inc. Austin, TX Research into process development issues affecting SiGe SEMFET devices
1996	Head Teaching Assistant EE410: IC Fabrication Laboratory, Stanford University Stanford, CA Coordination and instruction of EE410, graduate level laboratory course.
1994-1998	Research Assistant, Electrical Engineering, Stanford University Stanford, CA Research into crystallization of amorphous Si and SiGe films using low thermal budget processes. Professional Affiliations and Activities Program Committee, IEEE Electronic Components and Technology Conference, 2014 to 2016 Chair, Scientific Board, iPACK, Royal Institute of Technology (KTH), Sweden, 2011 to 2016 Tampere Institute of Technology Faculty Search Committee, Finland, 2013 University of Oulu Faculty Search Committee, Finland, 2013 Served as external thesis committee member for several universities world-wide, including University of Cape Town (South Africa), Tampere Institute of Technology (Finland), Technical University of Eindhoven (Netherlands), Indian Institute of Science (India), Indian Institute of Technology (India), 2009 to date Associate Editor, IEEE Journal of Display Technology, 2008 to date Technical Program Chair, Large Area, Organic, and Printed Electronics Conference, 2012-2013 Scientific Committee, Large Area, Organic, and Printed Electronics Conference, 2009-2011 Scientific Committee, International Conference on Printed and Flexible Electronics, 2009-2013 IEEE Electron Devices Society Organic Electronics Committee, 2003 - 2005 Executive Committee, International Electron Device Meeting, 2003 to 2009 Technical Program Committee, International Electron Device Meeting, 2001-2002 Technical Program Committee member, Device Research Conference, 2000-2002 Technical Program Committee member, VLSI-TSA Conference, 2005 Member, Institute of Electrical and Electronic Engineers

List of Patents, Publications and Presentations

Patents

1. “Systems and methods for precise radio frequency localization in the presence of multiple communication paths”, M Seth, L Kong, T Ylamurto, V Subramanian, US Patent 12,108,362
2. “Systems and methods for providing wireless asymmetric network architectures of wireless devices with power management features”, V Subramanian, E Alon, V Pavate, US Patent 11,924,757
3. “Systems and methods for precise radio frequency localization in the presence of multiple communication paths”, M Seth, L Kong, T Ylamurto, V Subramanian, US Patent 11,683,779
4. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate, T. Ylamurto, US Patent 11,327,147
5. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate, T. Ylamurto, US Patent 11,276,308
6. “Systems and methods for coarse and fine time of flight estimates for precise radio frequency localization in the presence of multiple communication paths”, M. Seth, L. Kong, T. Ylamurto, V. Subramanian, US Patent 11,228,999
7. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate, T. Ylamurto, US Patent 11,030,902
8. “Systems and methods for precise radio frequency localization in the presence of multiple communication paths”, M. Seth, L. Kong, T. Ylamurto, V. Subramanian, US Patent 10,757,675
9. “Systems and methods for using ranging to determine locations of wireless sensor nodes based on radio frequency communications between the nodes and various RF-enabled devices”, T. Ylamurto, M. Seth, L. Kong, V. Subramanian, US Patent 10,551,479
10. “Systems and methods for providing communications within wireless sensor networks based on a periodic beacon signal”, T. Ylamurto, V. Pavate, E. Alon, V. Subramanian, US Patent 10,536,901
11. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate T. Ylamurto, US Patent 10,514,704.
12. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate T. Ylamurto, US Patent 10,504,364
13. “Systems and methods for coarse and fine time of flight estimates for precise radio frequency localization in the presence of multiple communication paths”, M. Seth, L. Kong, T. Ylamurto, V. Subramanian, US Patent 10,470,156
14. “Methods and apparatus for monitoring wound healing using impedance spectroscopy”, M. Maharbiz, V. Subramanian, A. C. Arias, S. Swisher, A. Liao, M. Lin, F. Pavinatto, Y. Khan, D. Cohen, E. Leeftang, S. Roy, M. Harrison, D. Young, US Patent 10,463,293
15. “Systems and methods for using radio frequency signals and sensors to monitor environments”, T. Bakhishev, V. Subramanian, V. Pavate T. Ylamurto, US Patent 10,156,852.
16. “Systems and methods for determining locations of wireless sensor nodes in a tree network architecture having mesh-based features”, L. Kong, T. Bakhishev, T. Ylamurto, V. Subramanian, M. Seth, US Patent 10,104,508.
17. “Systems and methods for providing wireless asymmetric network architectures of wireless devices with power management features”, V. Subramanian, E. Alon, V. Pavate, US Patent 10,028,220.
18. “Random delay generation for thin-film transistor based circuits”, V Subramanian, M Mao, Z Wang, US Patent 9,985,664
19. “Systems and methods for determining locations of wireless sensor nodes in a network architecture having mesh-based features for localization”, L. Kong, T. Bakhishev, T. Ylamurto, V. Subramanian, M. Seth, US Patent 9,846,220
20. “Systems and methods for determining locations of wireless sensor nodes in a tree network architecture having mesh-based features”, L. Kong, T. Bakhishev, T. Ylamurto, V. Subramanian, M. Seth, US Patent 9,763,054
21. “Through silicon vias and thermocompression bonding using inkjet-printed nanoparticles”, V. Subramanian and J. Sadie, US Patent 9,717,145

22. "Systems and methods for providing wireless asymmetric network architectures of wireless devices with anti-collision features", V. Subramanian, E. Alon, V. Pavate, US Patent 9,706,489
23. "Systems and methods for determining locations of wireless sensor nodes in an asymmetric network architecture", V. Subramanian, E. Alon, V. Pavate, US Patent 9,529,076
24. "Systems and methods for providing wireless sensor networks with an asymmetric network architecture", V. Subramanian, E. Alon, V. Pavate, US Patent 9,380,531
25. "Transparent metal oxide nanoparticle compositions, methods of manufacture thereof and articles comprising the same", V. Subramanian, S. K. Volkman, US Patent 9,343,202
26. "Three-dimensional nonvolatile memory and method of fabrication", M. G. Johnson, T. H. Lee, V. Subramanian, P. M. Farmwald, J. M. Cleeves, US Patent 9,214,243
27. "Dense arrays and charge storage devices", T. H. Lee, V. Subramanian, J. M. Cleeves, I. G. Kouznetsov, M. G. Johnson, P. M. Farmwald, US Patent 9,171,857
28. "Print compatible designs and layout schemes for printed electronics", Z. Wang, V. Subramanian, L. Cleveland, US Patent 9,155,202
29. "Wireless devices including printed integrated circuitry and methods for manufacturing and using the same", P. Smith, C. Choi, V. Pavate, J. M. Cleeves, V. Subramanian, R. Young, V. Biviano, US Patent 9,004,366.
30. "Dense arrays and charge storage devices", T. H. Lee, V. Subramanian, J. M. Cleeves, M. G. Johnson, P. M. Farmwald, I. Kouznetsov, US Patent, 8,981,457.
31. "High reliability surveillance and/or identification tag/devices and methods of making and using the same", V. Subramanian, P. Smith, V. Pavate, A. Kamath, C. Choi, A. Chandra, J. M. Cleeves, US Patent, 8,933,806.
32. "Surveillance devices with multiple capacitors", P. Smith, C. Choi, J. M. Cleeves, V. Subramanian, A. Kamath, S. Molesa, US Patent, 8,912,890.
33. "Pillar-shaped nonvolatile memory and method of fabrication", M. G. Johnson, T. H. Lee, V. Subramanian, P. M. Farmwald, J. M. Cleeves, US Patent 8,897,056
34. "Dense arrays and charge storage devices", T. H. Lee, V. Subramanian, J. M. Cleeves, I. G. Kouznetsov, M. G. Johnson, P. M. Farmwald, US Patent 8,853,765
35. "Dense arrays and charge storage devices", T. H. Lee, V. Subramanian, J. M. Cleeves, I. G. Kouznetsov, M. G. Johnson, P. M. Farmwald, US Patent 8,823,076
36. "Random delay generation for thin-film transistor based circuits", V. Subramanian, M. Mao, Z. Wang, US Patent 8,810,298
37. "Vertically stacked field programmable nonvolatile memory and method of fabrication", M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleeves, US Patent 8,503,215
38. "Process-variation tolerant series-connected NMOS and PMOS diodes, and standard cells, tags, and sensors containing the same", V. Subramanian, P. Smith, US Patent 8,471,308
39. "Printed compatible designs and layout schemes for printed electronics", Z. Wang, V. Subramanian, L. Cleveland, US Patent 8,383,952
40. "Method for making surveillance devices with multiple capacitors", P. Smith, C. Choi, J. M. Cleeves, V. Subramanian, A. Kamath, S. Molesa, US Patent 8,296,943
41. "High reliability surveillance and/or identification tag/devices and methods of making and using the same", V. Subramanian, P. Smith, V. Pavate, A. Kamath, C. Choi, A. Chandra, and J. M. Cleeves, US Patent 8,264,359
42. "High reliability surveillance and/or identification tag/devices and methods of making and using the same", V. Subramanian, P. Smith, V. Pavate, A. Kamath, C. Choi, A. Chandra, and J. M. Cleeves, US Patent 8,227,320
43. "Vertically stacked field programmable nonvolatile memory and method of fabrication", M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleeves, US Patent 8,208,282
44. "Reliable tag deactivation", J. M. Cleeves, V. Subramanian, US Patent 8,138,921
45. "Combined static and dynamic frequency divider chains using thin film transistors", V. Subramanian, US Patent 8,085,068

46. "Process-variation tolerant diode, standard cells including the same, tags and sensors containing the same, and methods for manufacturing the same", V. Subramanian, P. Smith, US Patent 7932537
47. "Three terminal nonvolatile memory Device with vertical gated diode", T. H. Lee, V. Subramanian, J. M. Cleaves, M. G. Johnson, P. M. Farmwald, I. G. Kouznetsov, US Patent 7825455
48. "Vertically stacked field programmable nonvolatile memory and method of fabrication", V. Subramanian, J. M. Cleaves, US Patent 7816189
49. "Multi-mode tags and methods of making and using the same", P. Smith, J. M. Cleaves, V. Pavate, V. Subramanian, US Patent 7750792
50. "Method of manufacturing complementary diodes", V. Subramanian, P. Smith, US Patent 7528017
51. "Vertically stacked field programmable nonvolatile memory and method of fabrication", V. Subramanian, J. M. Cleaves, US Patent 7319053
52. "Vertically stacked field programmable nonvolatile memory and method of fabrication", V. Subramanian, J. M. Cleaves, US Patent 7265000
53. "Vertically stacked field programmable nonvolatile memory and method of fabrication", V. Subramanian, J. M. Cleaves, US Patent 7160761
54. "Vertically stacked field programmable nonvolatile memory and method of fabrication", V. Subramanian, J. M. Cleaves, US Patent 7157314
55. "Dense arrays and charge storage devices", T. H. Lee, V. Subramanian, J. M. Cleaves, A. J. Walker, C. Petti, I. Kouznetzov, M. G. Johnson, P. M. Farmwald, B. Herner, US Patent 7129538
56. "Patterning three dimensional structures", C. K. Li, J. N. Knall, M. A. Vyvoda, J. M. Cleaves, V. Subramanian, US Patent 7071565
57. "Monolithic three dimensional array of charge storage devices containing a planarized layer", T. H. Lee, V. Subramanian, J. M. Cleaves, A. J. Walker, C. J. Petti, I. G. Kouznetzov, M. G. Johnson, P. M. Farmwald, and B. Herner, US Patent 6881994
58. "Thermal processing for three dimensional circuits", V. Subramanian, J. M. Cleaves, J. N. Knall, C. K. Li, and M. A. Vyvoda, US Patent 6,770,939
59. "Vertically stacked field programmable nonvolatile memory and method of fabrication", M. G. Johnson, T. H. Lee, V. Subramanian, and P. M. Farmwald, US Patent 6780711
60. "Multigate semiconductor device with vertical channel current and method of fabrication", J. M. Cleaves and V. Subramanian, US Patent 6677204
61. "Thermal processing for three dimensional circuits", V. Subramanian, J. M. Cleaves, J. N. Knall, C. K. Li, and M. A. Vyvoda, US Patent 6624011
62. "Multigate semiconductor device with vertical channel current and method of fabrication", J. M. Cleaves and V. Subramanian, US Patent 6580124
63. "Patterning three dimensional structures", C. K. Li, J. N. Knall, M. A. Vyvoda, J. M. Cleaves, and V. Subramanian, US Patent 6627530
64. "Low cost three-dimensional memory array", M. Johnson, T. Lee, V. Subramanian, P. Farmwald, J. Knall, US Patent 6515888
65. "Integrated circuit structure including three-dimensional memory array", M. Johnson, T. Lee, V. Subramanian, P. M. Farmwald, J. M. Cleaves, US Patent 6385074.
66. "FINFET transistor structures having double gate channel extending vertically from a substrate and methods of manufacture", C. Hu, T-J. King, V. Subramanian, L. Chang, X. Huang, Y-K. Choi, J. T. Kedzierski, N. Lindert, J. Bokor, W-C. Lee, US Patent 6413802
67. "Vertically stacked field programmable nonvolatile memory and method of fabrication", M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleaves, US Patent 6185122.
68. "Vertically Stacked Field Programmable Nonvolatile Memory and Method of Fabrication", M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleaves, US Patent 6034882.

69. “Vertically Stacked Field Programmable Nonvolatile Memory and Method of Fabrication”, M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleaves, US Patent 6351406.
70. “Vertically Stacked Field Programmable Nonvolatile Memory and Method of Fabrication”, M. Johnson, T. Lee, V. Subramanian, M. Farmwald, and J. M. Cleaves, US Patent 6483736.

Invited Magazine Articles, Books, Chapters, and Monographs

1. Chapter in “Inkjet Technology for Digital Fabrication”, Editors: Ian M. Hutchings, Graham D. Martin, Wiley, ISBN: 978-0470681985
2. Chapter in “Applications of Organic and Printed Electronics: A Technology-Enabled Revolution (Integrated Circuits and Systems), Editor: E. Cantatore, ISB: 978-1-461-43159-6
3. Chapter in “Inkjet-based Micromanufacturing”, Editors: Korvink, Smith, Shin, Wiley, ISBN: 978-3-527-31904-6
4. Chapter in “Organic Electronics II. More Materials and Applications”, Editor: H. Klauk, Wiley, ISBN: 978-3-527-32647-1
5. Chapter in “Transparent Electronics: From Synthesis to Applications”, Editors: A. Facchetti and T. Marks, Wiley, ISBN: 978-0-470-99077-3, 2010.
6. Chapter in “The chemistry of inkjet inks”, Editor: S. Magdassi, World Scientific Publishing Company, ISBN: 978-9812818218, 2009
7. Chapter in “Organic Field-Effect Transistors”, Editors: Z. Bao, and J. Locklin, CRC Press, ISBN: 978-0849380808, 2007
8. “Developments in printed RFID”, V. Subramanian, Pira Publishing, UK, 2006.
9. “3D Chips: Future Possibilities”, V. Subramanian, Silicon India, Feb 2003, pp. 24-25

Invited Conference Presentations

1. **Invited**, “Materials, Process, and Device Innovations for High-Performance Printed Electronics”, PRINSE’18 – The 5th Printed Intelligence Industry Seminar on Manufacturing Solutions for Customer Needs and Vision to the Future, Oulu, Finland, Jan 31-Feb 1, 2018.
2. **Invited**, “Highly scaled gravure printed organic transistors for high-performance printed organic electronics”, High-frequency printed and direct-written organic-hybrid integrated circuits (HEROIC) Workshop, Milan, Italy, Nov 8-10, 2017.
3. **Keynote**, “Printed Electronics: Innovations in Materials, Processes, and Devices”, Association of International Metallizers, Coaters, and Laminators (AIMCAL) R2R Conference USA 2017, Tampa, FL, Oct-15-18, 2017.
4. **Invited**, “High-performance gravure printed organic transistors”, 2017 European Materials Research Society Meeting, Strasbourg, France, May 22-26, 2017
5. **Invited**, “Printed Electronics: Innovations in Materials, Processes, and Devices”, 2017 International Thin Film Transistor Conference (ITC 2017), Austin, TX, February 23-24, 2017.
6. **Invited**, “High-speed Printing of Transistors: From Inks to Devices”, 2016 International Society of Coating Science and Technology Symposium (ISCST), Pittsburgh, PA, September 18-21, 2016.
7. **Keynote**, “Printed Electronics: Innovations in Materials, Processes, and Devices”, 2016 Canadian Printable Flexible Wearable electronics Symposium (CPES2016), April 19-20, Oakville, Canada
8. **Keynote**, “Printed Electronics: Innovations in Materials, Processes, and Device”, 2016 Society of Vacuum Coaters Techcon, Indianapolis, IN, May 10-13, 2016.
9. **Invited**, Subramanian, Vivek, Chung, Seungjun, Grau, Gerd and Scheideler, William J., “71-2: Invited Paper: Printed Transistors and MEMS for Large-Area Electronics”, 2016 Society for Information Display Symposium, San Francisco, May 20-27, 2016.
10. **Keynote**, “Printed Electronics: Innovations in Materials, Processes, and Devices”, 31st Philippine Chemistry Congress, Iloilo, Philippines, April 13-15, 2016.
11. **Keynote**, “Advanced Processes, Materials, and Devices for Emerging Printed Electronics Applications”, 3rd Swiss Conference on Printed Electronics and Functional Materials, Neuchatel, Switzerland, Oct 1-2, 2015.

12. **Invited**, “High-performance printed organic transistors: Materials, processes, and devices”, V. Subramanian, G. Grau, H. Kang, and R. Kitsomboonloha, 2015 American Chemical Society National Meeting and Exposition, Boston, MA, Aug 17th, 2015.
13. **Invited**, Subramanian, Vivek, Jaewon Jang, William Scheideler, and Sarah Swisher. “39.1: Invited Paper: Printed Inorganic Transistors Based on Transparent Oxides.” In SID Symposium Digest of Technical Papers, vol. 46, no. 1, pp. 587-590. 2015.
14. **Plenary**, “Printed Electronics: Advanced Technologies Enabling New Applications”, 2015 Flexible and Printed Electronics Conference, Monterey, CA, February 23-26, 2015.
15. **Invited**, “High-resolution gravure printing of organic thin film transistors for high-performance printed electronic systems”, V. Subramanian, R. Kitsomboonloha, J. Cen, H. Kang, G. Grau, and W. J. Scheideler, Materials Research Society Fall Meeting, Boston, MA, Dec 3, 2014.
16. **Invited**, “Advanced printed electronics technologies for novel ubiquitous human interactive systems”, 2014 CMOS Emerging Technologies Research Symposium, Grenoble, France, July 6-8, 2014.
17. **Invited**, “Printed Electronics: A pathway to functionally-rich systems”, 64th annual IEEE Electronic Components and Technology Conference, Orlando, FL, May 27, 2014.
18. **Invited**, “Printed Electronics: The Confluence of Printing and Semiconductors”, Canadian Printed Electronics Symposium, Montreal, Canada, April 9, 2013.
19. **Invited** “Modeling, scaling, and integration of gravure printing for fast switching Organic FETs”, V. Subramanian, S. J. S. Morris, H. Kang, Materials Research Society Fall 2012 Meeting, Boston, MA, Nov 25-30, 2012
20. **Invited**, “Printed Nanoparticles as routes to high-performance printed conductors: Synthesis, Printing Processes, and Device Applications”, V. Subramanian, H. Kang, R. Kitsomboonloha, and S. K. Volkman, The 2012 International Conference on Flexible and Printed Electronics ICFPE2012, Tokyo, Japan, September 6th - 8th, 2012
21. **Invited** “Advanced Printing Processes for High-Performance Printed Transistors”, V. Subramanian, Flextech Workshop on Printing Electronics - Ink and Substrate Interactions, Kalamazoo, MI, August 1-2, 2012
22. **Invited** “High-performance Printed Transistors: Materials, Processes, and Devices”, V. Subramanian, 2nd CPEM International Symposium “Organic Semiconductors and Printed Electronics”, Gyeonggi, South Korea, May 10th, 2012.
23. **Invited** “Highly-Scaled Gravure and Inkjet Printed Organic Transistors: Tools, Processes, and Devices”, Hongki Kang & Vivek Subramanian, 2012 Flextech Flexible Electronics and Display Conference, Phoenix, AZ, Feb 5-9, 2012.
24. **Plenary** “Nanomaterials for printed electronics: synthesis, design, and applications”, V. Subramanian, International Conference on Nano Science and Nano Technology, Suncheon, Korea, November 11th, 2011.
25. **Invited** “Advances in scaling of printed transistors”, V. Subramanian, International Seminar on Printed Electronics, Seoul, Korea, June 8th, 2011.
26. **Invited** “High-Performance Fully Printed Transistors: Materials, Processes, and Device Characteristics”, V. Subramanian, H. Tseng, R. Kitsomboonloha, and A. de la Fuente Vornbrock, 2011 Electrochemical Society Meeting, Montreal, Canada, May 2011
27. **Invited** “From Droplets to Devices: Printed Transistor Processes, Integration, and Characteristics”, Vivek Subramanian, Daniel Soltman, Huai-Yuan Tseng, Rungrot Kitsomboonloha and Alejandro de la Fuente Vornbrock, Materials Research Society Spring Meeting, San Francisco, CA, April 2011
28. **Keynote** “Printed electronics: Innovations in tools, materials, devices, and applications”, V. Subramanian, 2011 Korea Printed Electronics Association (KoPeA) meeting, Seoul, Korea, March 2011
29. **Keynote** “Printed RF tags and sensors: the confluence of printing and semiconductors”, V. Subramanian, F. Liao, and H-Y. Tseng, The European Microwave Integrated Circuits Conference 2010, Paris, France, September 2010.
30. **Invited** “Droplet-on-demand direct patterning of active materials: materials, modeling, and integration”, V. Subramanian, The 2010 IEEE Lithography Workshop, Maui, Hawaii, November 2010
31. **Plenary** “Printed electronics: where are we, and where are we going”, V. Subramanian, 2010 Large Area Organic, and Printed Electronics Convention (LOPE-C), Frankfurt, Germany, June 2010.
32. **Keynote** “Printed RFID: Technology Trends and Outlook”, V. Subramanian, IEEE International Conference on RFID 2010, Orlando, Florida, April 2010.

33. **Invited** “Mechanistic studies on sintering of nanoparticles for formation of solution-processed thin films”, 2010 Spring meeting of the Materials Research Society, San Francisco, CA, April 2010.
34. **Invited** “Printed Electronics: the confluence of printing and semiconductors”, Vivek Subramanian, 2009 International Conference on Flexible and Printing Electronics, Jeju Island, Korea, Nov 2009.
35. **Invited** “Organic Transistor Vapor and Biosensors – Technology Status and Implementation Issues”, Vivek Subramanian, Frank Liao, Lakshmi Jagannathan, Electrochemical Society Meeting, Vienna, Austria, October 2009.
36. **Invited** “Printed transistors for low-cost electronics: the confluence of printing and printable electronic materials”, Vivek Subramanian, Alejandro de la Fuente Vornbrock, Steven Molesa, Daniel Soltman, Huai-Yuan Tseng, and Steven K. Volkman, SPIE Optics + Photonics, August 2-6, 2009, San Diego, CA
37. **Invited** “Printed Zinc Oxide Based Electronics: Materials, Devices, and Outlook”, V. Subramanian, S. K. Volkman, and D. R. Redinger, LOPE-C, June 23-25, 2009, Frankfurt, Germany
38. **Invited** “Printed Devices for Low-Cost Electronics: Technology Status and Outlook”, V. Subramanian, PETEC Inaugural Event, March 17, 2009, Durham, United Kingdom
39. **invited** “Printed Electronic Tags and Sensors for Smart Packaging Applications”, Vivek Subramanian, Josephine Chang, Lakshmi Jagannathan, Frank Liao, Steve Molesa, David Redinger, Daniel Soltman, Huai-Yuan Tseng, Steve Volkman, Shong Yin, 2009 Flexible Electronics and Display Conference, Phoenix, AZ, Feb 2-5, 2009.
40. **invited** “Electronic Nose Sensors for Consumer Packaging”, V. Subramanian, IDTechEx Printed Electronics USA, San Jose, CA, Dec 2-5, 2008.
41. **invited** “Solution-Processed Transparent Transistors for Low-Cost, Flexible Displays”, V. Subramanian, S. K. Volkman, and D. R. Redinger, LEOS 2008: The 21st Annual Meeting of the IEEE Lasers and Electro-Optics Society, Newport Beach, CA, 9-13 November, 2008.
42. **invited** “Organic chemical and biosensors for smart packages”, V. Subramanian, IDTechEx Printed Electronics Asia, Tokyo, Japan, Oct 8-9, 2008.
43. **invited** “Printed Electronics for Low-Cost Tags, Displays, and Sensors”, V. Subramanian, 1st International Conference on R2R Printed Electronics, Seoul, Korea, April 30-May 2, 2008.
44. **invited** “Printed organic RFID tags: technology, roadmaps, and challenges”, V. Subramanian, 1st Plastic Electronics Asia Conference and Showcase, Seoul, Korea, June 24-26, 2008
45. **invited** “Power supply considerations and solutions for printed electronic tags and sensors”, V. Subramanian, IDTechEx Printed Electronics Europe, Dresden, Germany, April 8-9, 2008
46. **invited** “Printed Chemical and Biosensors: Technology, Design and Challenges”, GOSPEL Workshop on Plastic Chemical Sensors, Dresden, Germany, April 9, 2008
47. **Plenary** “Printed Electronics For Low-Cost Electronic Systems: Technology Status and Application Development”, Vivek Subramanian, Josephine B. Chang, Alejandro de la Fuente Vornbrock, Daniel C. Huang, Lakshmi Jagannathan, Frank Liao, Brian Mattis, Steven Molesa, David R. Redinger, Daniel Soltman, Steven K. Volkman, Qintao Zhang, 38th European Solid State Device Research Conference, Edinburgh, Scotland, September 15-19, 2008
48. **invited** “Printed Organic Transistors for Low-cost Tags, Displays, and Sensors: Technology, Modeling and Challenges”, Vivek Subramanian, Alejandro de la Fuente Vornbrock, Huai-Yuan Tseng and Shong Yin, Materials Research Society Fall Meeting, Boston, MA, Dec 1-4, 2008.
49. **invited** “Printed Electronics for Low-Cost Tags and Sensors”, Vivek Subramanian, Sixth Annual Workshop on Microelectronics and Electron Devices, April 18, 2008.
50. **invited** “Printed organic RF tags, chemical sensors and biosensors”, Vivek Subramanian, International Symposium for Flexible Electronics and Display, Hsinchu, Taiwan, Nov 18, 2007.
51. **invited** “Printed Tags and Sensors for RFID: Opportunities and Challenges”, Vivek Subramanian, Printed Electronics USA 2007, San Francisco, CA, Nov 14, 2007.
52. **invited** “Printed RFID Tags: Materials, Processes, and Devices”, Vivek Subramanian, Daniel Soltman, Daniel Huang, Steven Molesa, 2007 AiChE Annual Meeting, Salt Lake City, UT, Nov. 8, 2007.
53. **invited** “Printed organic RFID tags: technology, roadmaps, and challenges”, Vivek Subramanian, 3rd Global Plastic Electronics Conference and Showcase, Frankfurt, Germany, Oct 30, 2007.

54. **invited** "Printed organic transistors: technology, characteristics, and modeling", Vivek Subramanian, Alejandro de la Fuente Vornbrock, Daniel Soltman, Huai-Yuan Tseng, and Steve Molesa, 212th meeting of the Electrochemical Society, Washington, DC, Oct 8, 2007.
55. **invited** "Printed electronic tags and sensors - device, materials, and circuit issues", Vivek Subramanian, 2007 Organic Electronics Conference and Exhibition, Frankfurt, Germany, Sept 25, 2007.
56. **invited** "Inkjet printing of transistors, diodes, and passive components for smart tags", Vivek Subramanian, Innovations in Inkjet: 3rd DPI Inkjet Workshop, Eindhoven Institute of Technology, Netherlands, June 28-29, 2007.
57. **invited** "Printed organic transistors", A. de la Fuente and V. Subramanian, Industry Forum on Printed Electronics, Chicago, IL, April 23-24, 2007
58. **invited** "From Materials to Circuits: Implications of printed materials on printed RFID", V. Subramanian, 2007 Pira printed RFID conference, Frankfurt, Germany, June 4-6, 2007.
59. **keynote** "Droplet on demand lithography: An enabling technology for low-cost electronics", V. Subramanian, A. de la Fuente Vornbrock, S. Molesa, D. Redinger, Q. Zhang, and S. K. Volkman, SPIE Advanced lithography, 27th February, 2007
60. **invited** "Printed RFID tags: performance needs and technology trends", V. Subramanian, 2007 IDTechEx Smart labels USA , Boston, MA, February 20-23, 2007.
61. **keynote** "Printed Organic Transistors for low-cost tagging and sensing applications", V. Subramanian, J. Chang, A. de la Fuente Vornbrock, S. Molesa, D. Soltman, and Q. Zhang. 6th International IEEE Conference on Polymers and Adhesives in Microelectronics and Photonics, Tokyo, January 15th-18th, 2007.
62. **invited** "Nano Inks for Printed RFID", V. Subramanian, Pira Nanoscale Inks and Pigment Technologies Conference, Chicago, IL, Sept 27-28, 2006.
63. **invited** "Toward Printed RFID: Materials, Processes, and Devices", V. Subramanian, 5th Annual Printed Electronics and Displays Conference and Trade Fair, Las Vegas, Oct 11-13, 2006.
64. **invited** "Inkjetted Organic Transistors for Smart Tagging Applications", Vivek Subramanian, Jean M. J. Frechet, Steven Molesa, Josephine Chang, Amanda R. Murphy, Alejandro de la Fuente Vornbrock, Steven Volkman, 2006 AiChE Annual Meeting, Nov 17th, 2006
65. **invited focal paper** "All-printed electronics: Materials, Devices, and Circuit Implications", V. Subramanian, J. Chang, S. Molesa, D. Redinger, and S. Volkman, Digital Fabrication 2006, Denver, CO, Sept 17-22, 2006.
66. **invited** "Printed transistors and passive components for low-cost electronics applications", V. Subramanian, J. B. Chang, S. E. Molesa, S. K. Volkman, and D. R. Redinger, International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA) April, 2006, pp. 68-9.
67. **invited** "Advanced printed materials and devices for RFID applications", V. Subramanian, IDTechEx Printed Electronics Europe, April 19-21, 2006.
68. **invited** "Printed Electronic Nose Vapor Sensors for Consumer Product Monitoring", V. Subramanian, J. Lee, V. Liu, S. Molesa, IEEE International Solid State Circuits Conference, Paper 15.3, February, 2006.
69. **invited** "Nanotechnology in Printed Electronics: Nanoparticles, Nanorods, and Nanowhy", V. Subramanian, Pira Pack Electronics Conference, January 25-26, 2006.
70. **invited** "All-printed RFID Tags: Materials, Devices, and Circuit Implications", Vivek Subramanian, Paul C. Chang, Daniel Huang, Josephine B. Lee, Steven E. Molesa, David R. Redinger, and Steven K. Volkman, VLSI Design, 2006, Hyderabad India, Jan 2006.
71. **invited** "Printed organic transistors: materials and technology", Vivek Subramanian, Pacificchem 2005: American Chemical Society Pacific Basin meeting, December 15-20, 2005, Honolulu, HI,
72. **invited** "Low-cost arrayed gas sensors for environmental monitoring". Vivek Subramanian, Josephine Lee and Vincent Liu, Materials Research Society Fall Symposium, Paper S2.1, Boston, MA, December 2005.
73. **invited** "Prospects for All-printed RFID: Materials, Devices, Circuits", V. Subramanian, International Workshop on Radio Frequency Identification (RFID) and Wireless Sensors, 11-13 November, 2005, Kanpur, India.
74. **invited** "Printed electronics for low-cost tagging applications", V. Subramanian, 2005 International Symposium on Flexible Electronics and Display, Hsinchu, Taiwan, October 2005.

75. **invited** "Printed electronic nose gas sensors", Vivek Subramanian, Plastic Electronics 2005, Frankfurt, Germany, Oct 4-5, 2005
76. **invited** "Uses of nanotechnology in printed electronics", Vivek Subramanian, Printed Electronics 205, Pira International, Thistle Marble Arch, London, UK, Sept 14-15, 2005
77. **invited** "Printed organic transistors for low-cost RFID applications", Vivek Subramanian, Proceedings of SPIE-The International Society for Optical Engineering vol. 5940, (594013/1-594013/9) , pp. 170-178, 2005.
78. **invited** "Advanced printed materials and devices for ultra-low-cost RFID", Vivek Subramanian, Printed RFID: Achieving low cost tags through printed electronics – Pira International, Gatwick, Surrey, UK, May 10-11, 2005.
79. **Plenary session invited talk** "Printed Organic Transistors for Ultra-low-cost RFID Applications", Vivek Subramanian, Paul C. Chang, Josephine B. Lee, Steven E. Molesa, and Steven K. Volkman, Polytronic 2004: 4th International IEEE Conference on Polymers and Adhesives in Microelectronics and Photonics, Portland, Oregon, USA, 12 September 2004.
80. **invited** "Materials, Device, and Circuits Issues for low-cost printed RFID", V. Subramanian, 2nd Cintelliq Organic Semiconductor Conference, 2004 (OSC-2004), Cambridge, England, October 2004.
81. **invited** "All-printed flexible organic thin film transistors: Current status and outlook for the future", V. Subramanian, J. M. J. Fréchet, P. C. Chang, D. C. Huang, J. B. Lee, F. Liao, B. A. Mattis, S. Molesa, A. R. Murphy, D. R. Redinger, and S. K. Volkman, 206th meeting of the Electrochemical Society, Honolulu, HI, October 2004.
82. **invited** "Nanoscale organic transistors: towards gain in molecular devices", V. Subramanian, P. C. Chang, J. B. Lee, T. Le, A. Murphy, J. M. J. Frechet, and A. Liddle, 2004 IEEE Nanoscale Devices and Systems Integration Conference, Miami, FL, 2004.
83. **invited** "Towards Printed Low-Cost RFID Tags: Device, Materials and Circuit Technologies", V. Subramanian, 2nd Advanced Technology Workshop on Printing an Intelligent Future: Printed Organic and Molecular Electronic Technologies, Boston, MA, March 16-19, 2003.
84. **invited** "Prospects for organic semiconductor based devices and circuits: applications, performance, and reliability considerations", V. Subramanian, 2002 Topical Research Conference on Reliability, Oct 2002.
85. **invited** "FinFET - quasi-planar double-gate MOSFET", S. H. Tang, L. Chang, N. Lindert, Y-K. Choi, W-C. Lee, X. Huang, V. Subramanian, J. Bokor, T-J. King, C. Hu, 20001 IEEE International Solid-State Circuits Conference Digest of Technical Papers, pp. 118-119, 437, 2001.
86. **invited** "3-D ICs with Multiple Si Layers: Performance Analysis, and Technology", K. Saraswat, K. Banerjee, A. Joshi, P. Kalavade, S. Sourì, and V. Subramanian, *presented at* The Fifth International Symposium on Low and High Dielectric Constant Materials: Materials Science, Processing, and Reliability Issues, 197th meeting of the Electrochemical Society, Toronto, Ontario, Canada, May 14-18, 2000.
87. **invited** "Novel 3-D Structures", K. C. Saraswat, S. J. Sourì, V. Subramanian, A. R. Joshi, and A. W. Wang, 1999 IEEE International SOI Conference Proceedings, pp. 54-55.
88. **invited** "Seeding Technology for High Performance TFTs", K. C. Saraswat and V. Subramanian, Proceedings of the Electronic Display Forum 98, EIAJ/SEMI, Yokohama, April 1998, pp. 2.7-2.15.
89. **invited** "A Low-Temperature Polycrystalline Si TFT Technology for Large-Area AMLCD Drivers", K. C. Saraswat, V. Subramanian, and S. Jurichich, Polycrystalline Thin Films III - Structure, Texture, Properties, and Applications, MRS Symposium Proceedings pp 439-449, 1997.
90. **invited** "A low temperature polycrystalline SiGe CMOS TFT technology for large area AMLCD drivers", K. C. Saraswat, S. Jurichich, T.-J. King, V. Subramanian, and A. Wang, Proceedings of Thin Film Transistor Technologies III, ECS Symposium, 1996, pp. 186-196.

Journal Papers

1. Manoharan, Sanjay, and Vivek Subramanian. "Magnetically-guided, stimuli-responsive microdevices for endovascular therapy." *Biomedical Microdevices* 28, no. 1 (2026): 6.
2. Lemecho, Biruktait Ayele, Jordi Espín, Pattaraphon Rodlamul, Florian Kiefer, Wendy Lee Queen, and Vivek Subramanian. "A Sustainable Multi-Zeolite Synthetic Framework from a Single Natural Clay: CO₂/H₂O Adsorption Performance and Life Cycle Assessment Benefits." *Sustainable Energy & Fuels* (2026).
3. "Tumbleweed-inspired robots with hybrid mobility for terrestrial exploration", S Manoharan, B Lemecho, MM Fadlelmula, V Subramanian, *Nature Communications*, 2025

4. "A Pill That Prints-An Ingestible Bioprinter for Non-Invasive Structured Bioink Deposition", S Manoharan, V Subramanian, *Advanced Science*, e12411, 2025.
5. "Highly Stable Low-Temperature Phosphate Glass as a Platform for Multimaterial 3D Printing of Integrated Functional Microfluidic Devices", B Mazinani, MM Fadlelmula, V Subramanian, *Advanced Engineering Materials*, 2501603
6. CC Li, J Toombs, V Subramanian, HK Taylor, "Intensity-coupled multi-beam phase retrieval for holographic volumetric additive manufacturing", *Optics Express* 32 (26), 47620-47637, 2024
7. MM Fadlelmula, B Mazinani, V Subramanian, "Towards personalized microfluidics: 3D printing of high-performance micropumps by control and optimization of fabrication-induced surface roughness", *Additive Manufacturing* 94, 104468, 2024
8. Haq, Muhammad Refatul, Babak Mazinani, Vivek Subramanian, and Helmut Schift. "Room temperature imprinting of water-based microparticulate inks for realizing glass microfluidic channels." *Materials & Design* 242 (2024): 112982.
9. Biaou, Carlos, Matthew Mcphail, Kazutaka Eriguchi, Vivek Subramanian, and Oscar Dubon. "Tracking deep-level defects in degrading perovskite solar cells with a multifactorial approach." *Organic Electronics* 130 (2024): 107074.
10. Hodaei, Amin, and Vivek Subramanian. "Sub-mm 3 dimensional scaling of fully-integrated additively-fabricated microsupercapacitors for embedded energy storage applications." *Journal of Materials Chemistry A* 12, no. 17 (2024): 10229-10241.
11. Hodaei, Amin, and Vivek Subramanian. "Additive fabrication of fully-integrated high-performance millimeter-scale microsupercapacitors: Fine-tuning chemistry to maximize performance". *Journal of Power Sources* 588 (2023): 233738.
12. Sels, Annelies, Remo Blum, Rajasundar Chandran, Enrica Montinaro, Jan Schildknecht, Mickael Chabart, and Vivek Subramanian. "Digital LIGA: Exploitation of droplet-on-demand inkjet printing to fabricate complex mechanical structures by electroforming." *Journal of Micromechanics and Microengineering* (2023).
13. Annelies Sels and Vivek Subramanian, "Printed Platinum Nanoparticle Thin-Film Structures for Use in Biology and Catalysis: Synthesis, Printing, and Application Demonstration", *ACS Omega Article ASAP*, DOI: 10.1021/acsomega.2c04687
14. **Invited** Scheideler, W. J., & Subramanian, V. (2022). How to print high-mobility metal oxide transistors—Recent advances in ink design, processing, and device engineering. *Applied Physics Letters*, 121(22), 220502.
15. Deng, Mengmeng, and Vivek Subramanian. "Screen-Printable Cu–Ag Core–Shell Nanoparticle Paste for Reduced Silver Usage in Solar Cells: Particle Design, Paste Formulation, and Process Optimization." *ACS Applied Electronic Materials* (2022).
16. **invited** Bonnassieux, Yvan, Christoph J. Brabec, Yong Cao, Tricia Breen Carmichael, Michael L. Chabinye, Kwang-Ting Cheng, Gyoujin Cho et al. "The 2021 flexible and printed electronics roadmap." *Flexible and Printed Electronics* 6, no. 2 (2021): 023001.
17. Deka, Nishita, and Vivek Subramanian. "On-chip fully integrated field emission arrays for high-voltage MEMS applications." *IEEE Transactions on Electron Devices* 67, no. 9 (2020): 3753-3760.
18. **invited** Gerd Grau and Vivek Subramanian, "Dimensional scaling of high-speed printed organic transistors enabling high-frequency operation", *Flex. Print. Electron.* 2020, Vol. 5 014013
19. **invited** William Scheideler and Vivek Subramanian, "Printed flexible and transparent electronics: enhancing low-temperature processed metal oxides with 0D and 1D nanomaterials", *2019 Nanotechnology* 30 272001
20. R. Kumar, K. M. Johnson, N. X. Williams, V. Subramanian, "Scaling Printable Zn–Ag₂O Batteries for Integrated Electronics", *Adv. Energy Mater.* 2019, 1803645.
21. William J. Scheideler, Matthew W. McPhail, Rajan Kumar, Jeremy Smith, and Vivek Subramanian, "Scalable, High-Performance Printed InOx Transistors Enabled by Ultraviolet-Annealed Printed High-k AlOx Gate Dielectrics", *ACS Applied Materials & Interfaces* 2018 10 (43), 37277-37286.
22. J. Smith, S. Chung, J. Jang, C. Biao, V. Subramanian, "Solution-processed complementary resistive switching arrays for associative memory", *IEEE Transactions on Electron Devices* 64 (10), pp 4310, 2017
23. A. Zeumault, V. Subramanian, "Use of high-k encapsulation to improve mobility in trap-limited metal oxide semiconductors", *physica status solidi (b)* 254 (10), 1700124, 2017
24. A. Zeumault, W. Scheideler, V. Subramanian, "Electrostatic tuning of spray-deposited ZnO for controlled mobility enhancement", *Advanced Functional materials*, 27 (30), pp. 1701021, 2017

25. W. J. Scheideler, R. Kumar, A. R. Zeumault, V. Subramanian, "Low-Temperature-Processed Printed Metal Oxide Transistors Based on Pure Aqueous Inks", *Advanced Functional Materials* 2017, 1606062.
26. Jaewon Jang, Vivek Subramanian, "Effect of electrode material on resistive switching memory behavior of solution-processed resistive switches: Realization of robust multi-level cells", *Thin Solid Films*, Volume 625, Pages 87-92, 1 March 2017.
27. S. Chung, M. A. U. Karim, H. J. Kwon, W. Scheideler and V. Subramanian, "A High-Speed Inkjet-Printed Microelectromechanical Relay With a Mechanically Enhanced Double-Clamped Channel-Beam," *Journal of Microelectromechanical Systems*, vol. 26, no. 1, pp. 95-101, Feb. 2017.
28. G. Grau, E. J. Frazier, and V. Subramanian, "Printed unmanned aerial vehicles using paper-based electroactive polymer actuators and organic ion gel transistors", *Nature Microsystems & Nanoengineering*, vol 2, pp. 16032, 2016
29. *invited* Gerd Grau, Jialiang Cen, Hongki Kang, Rungrot Kitsomboonloha, William J Scheideler, and Vivek Subramanian "Gravure-printed electronics: recent progress in tooling development, understanding of printing physics, and realization of printed devices", *IOP Journal of Flexible and Printed Electronics*, vol. 1, No. 2, pp. 023002, 2016
30. Scheideler, William J., Jeremy Smith, Igal Deckman, Seungjun Chung, Ana Claudia Arias, and Vivek Subramanian. "A robust, gravure-printed, silver nanowire/metal oxide hybrid electrode for high-throughput patterned transparent conductors." *Journal of Materials Chemistry C* 4, no. 15 (2016): 3248-3255.
31. Karim, Muhammed Ahosan Ul, Seungjun Chung, Elad Alon, and Vivek Subramanian. "Fully Inkjet-Printed Stress-Tolerant Microelectromechanical Reed Relays for Large-Area Electronics." *Advanced Electronic Materials* (2016).
32. Khan, Yasser, Felipe J. Pavinatto, Monica C. Lin, Amy Liao, Sarah L. Swisher, Kaylee Mann, Vivek Subramanian, Michel M. Maharbiz, and Ana C. Arias. "Bioelectronic Interfaces: Inkjet-Printed Flexible Gold Electrode Arrays for Bioelectronic Interfaces (*Adv. Funct. Mater.* 7/2016)." *Advanced Functional Materials* 26, no. 7 (2016): 981-981.
33. Jang, Jaewon, Seungjun Chung, Hongki Kang, and Vivek Subramanian. "P-type CuO and Cu₂O transistors derived from a Sol-Gel Copper (II) acetate monohydrate precursor." *Thin Solid Films* (2016).
34. Grau, Gerd, and Vivek Subramanian. "Fully High-Speed Gravure Printed, Low-Variability, High-Performance Organic Polymer Transistors with Sub-5 V Operation." *Advanced Electronic Materials* (2016).
35. Zeumault, Andre, and Vivek Subramanian. "Mobility Enhancement in Solution-Processed Transparent Conductive Oxide TFTs due to Electron Donation from Traps in High-k Gate Dielectrics." *Advanced Functional Materials* (2016).
36. Zeumault, Andre, William Scheideler, Gerd Grau, Jeremy Smith, and Vivek Subramanian. "Patterning of Solution-Processed, Indium-Free Oxide TFTs by Selective Spray Pyrolysis." *Advanced Electronic Materials* (2015).
37. Jang, Jaewon, Hongki Kang, Himamshu C. Nallan Chakravarthula, and Vivek Subramanian. "Fully Inkjet-Printed Transparent Oxide Thin Film Transistors Using a Fugitive Wettability Switch." *Advanced Electronic Materials* 1, no. 7 (2015).
38. Grau, Gerd, Rungrot Kitsomboonloha, Hongki Kang, and Vivek Subramanian. "High performance printed organic transistors using a novel scanned thermal annealing technology." *Organic Electronics* 20 (2015): 150-157.
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Conference Presentations

1. A. Hodaei, V. Subramanian, "Sub-Mm3 Dimensional Scaling of Additively Fabricated Microsupercapacitors for Embedded Energy Applications", 2023 IEEE NANO Conference, Jeju, Korea, July 2-5, 2023.
2. B. Mazinani, V. Subramanian, "Synthesis and characterization of highly durable P2O5-ZnO-Na2O/CaO-Fe2O3 glasses for low-temperature sealing applications", 2023 European Materials Research Society Spring Meeting, Strasbourg, France, May 29-June 2, 2023
3. M. Fadlelmula, V. Subramanian, "A Sol-gel inkjet printable PZT ink for additively fabricated mechanical transducers for energy harvesting, sensing, and mechanical actuation", 2023 European Materials Research Society Spring Meeting, Strasbourg, France, May 29-June 2, 2023
4. A. Hodaei, V. Subramanian, "All-Additively-Fabricated Microsupercapacitors: Fine-Tuning Chemistry to Maximize Performance", 2023 European Materials Research Society Spring Meeting, Strasbourg, France, May 29-June 2, 2023
5. M. Deng, V. Subramanian, "Understanding and insight of paste formulation and curing behavior with sub-100-nanometer Ag-Cu core-shell nanoparticles", 10th Metallization and Interconnection Workshop, Genk, Belgium, Nov 14-15, 2022.
6. McPhail, Matt, Rajan Kumar, and Vivek Subramanian. "Slurry Design for a Low Cost, High Capacity Magnesium Semi-Solid Flow Battery." In ECS Meeting Abstracts, no. 2, p. 300. IOP Publishing, 2020.
7. R. Kumar, E. Kedzie, B. D. McCloskey, and V. Subramanian, "Characterizing Zn Corrosion in Printed Zn-Air batteries Via Operando Dens, XRD, and XAS", Electrochemical Conference on Energy and the Environment: Bioelectrochemistry and Energy Storage, MA2019-04 July 21-26, 2019, Glasgow, Scotland
8. Carlos Biaou, Matt McPhail, Vivek Subramanian and Oscar Dubon, "Comprehensive Multifactorial Studies on the Degradation of Perovskite Solar Cells in Operation", 2019 Spring Materials Research Society Meeting, Phoenix, AZ, April 22-26, 2019.
9. Rajan Kumar, Jessica E. Nichols, Bryan D. McCloskey, and Vivek Subramanian, "Low-Temperature-Processing of Printed Cathodes for Aqueous Metal-Air Batteries", ECS AiMES 2018 Meeting, September 30, 2018 - October 4, 2018, Cancun, Mexico
10. N. Deka and V. Subramanian, "First demonstration of vacuum-sealed fully integrated BEOL-compatible field emission devices for Si integrated high voltage applications," 2018 76th Device Research Conference (DRC), Santa Barbara, CA, 2018, pp. 1-2.
11. "Inkjet-printed MEM relays for active solar cell routing", S Patel, WJ Scheideler, MAU Karim, V Subramanian, Micro Electro Mechanical Systems (MEMS), 2018 IEEE, 616-619
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13. "Morphological Control of Gravure Printed Perovskite Films", M. McPhail and V. Subramanian, 2017 Materials Research Society Fall meeting, Boston, MA, Nov 29, 2017.
14. "Improving high-speed nanomaterial printing with sub-process-decoupled gravure printer design", W. Scheideler, V. Subramanian, ASME 2017 Conference on Smart Materials, Adaptive Structures, and Intelligent Systems, Snowbird, UT, Sept 18-20, 2017
15. "Development of a printed cathode and catalyst layer for printed zinc-air batteries, R. Kumar, N. Williams, and V. Subramanian, 232nd ECS Meeting, National Harbor, MD, Oct 1-5, 2017.
16. "UV-annealing-enhanced stability in high-performance printed InOx transistors, W. Scheideler, V. Subramanian, 2017 IEEE Electron Devices Technology and Manufacturing Conference (EDTM), Toyama, Japan, Feb 28-Mar 2, 2017.
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EXHIBIT J

SCOTT EMMET THOMPSON

Prof. Electrical Engineering and Computer Science Department / Univ. of Florida
IEEE Fellow for contributions on advancing MOSFETs
Former Intel Corporation Fellow and Director of Computer Chip Technology

A. Education and Training

Education

- 1992 Ph.D. (Electrical Engineering), University of Florida
- 1988 Master of Science in Electrical Engineering, University of Florida
- 1987 Bachelor of Electrical Engineering with Highest Honors, University of Florida

B. Professional Experience

- ❑ Tenured Full Professor and Faculty/ Electrical and Computer Engineering, University of Florida (2/2004 to present).
- ❑ Founder and CTO of SuVolta (a low power chip company) (2009 - 2/2015).
 - Technology sold to Fujitsu and used in Mibeaut image sensor processor
- ❑ Researcher on advanced chip technology at University of Florida totaling \$4M worth of research contracts.
 - 2019-2021 Research funding from Semiconductor Research Corporation (SRC)
- ❑ Consultant (non litigation) last 5 years: Xperi Corporation
- ❑ **Intel Corporation** 1992 to 2004. Intel Fellow (highest technical level) and Director of computer chip technology (semiconductor manufacturing) including transferring and bringing up new manufacturing facilities worldwide.
- ❑ Intel Corporation experience and industry consulting on semiconductor manufacturing, semiconductor technology, yield and quality.
- ❑ Developed ultra low power chip technology for image sensors, SOC, RFIC, Memory, CPU, image sensors processors and other chips.
- ❑ Developed graduate level courses at University of Florida on advanced semiconductor chip, semiconductor design and packaging technology (chip stacking) and computer chip technology including CPU, DRAM, GPU, image sensors, 2D and 3D image sensor processing.
- ❑ National Science Foundation Panel Reviewer for semiconductor/computer chip funding.

C. Legal cases

- ❑ Bold is the client
 - Godo Kaisha IP Bridge v/ **Xilinx/TSMC**
 - Case No. 2:17-CV-00100-JRG-RSP (E.D. Tex.) - filed 1/31/2017

- **KAIST IP v. Apple**
 - March /2019 Report to Intellectual Property Trial and Appeals board regarding Korean Patent No. 10-0458288
 - Case in Korea : Intellectual Property Trial and Appeal Board (“IPTAB”) regarding Korean Patent No. 10-0458288 (hereinafter, “KR ‘288 patent”).
- **Godo Kaisha IP Bridge, Inc. v. NVIDIA**
 - NVIDIA Singapore Pte Ltd. as a technical expert in the customs proceeding initiated by petitioner Godo Kaisha IP Bridge 1
- **Chinese Academy of Sciences v. Intel**
 - IPR report Sept / 2018 in case IPR2018-01574 - filed 9/18/2018
 - IPR report Mar / 2019 in case IPR2019-00834 - filed 3/19/2019
 - Beijing Court Testimony IME vs Lenovo 11/21/2021
 - Beijing court Testimony IME vs Intel China 1/12/2022
- **IFT v. TSMC (ITC case)**
 - Case 1:19-cv-00308-UNA (D. Del.) - filed 2/13/2019
 - Inv. No. 337-TA-1149 - initial notice 4/3/2019
- **NXP v. Impinj**
 - 9,495,631, IPR2020-00543 - filed 2/10/2020
 - 9,633,302, IPR2020-00516 - filed 2/5/2020
 - 6,680,523, IPR2020-01630 - filed 9/17/2020
 - 7,538,444, IPR2021-00003 - filed 10/5/2020
 - Oakland District Court Case No. 4:19-cv-03161-YGR
- **Katana Silicon Technology LLC v. GlobalFoundries 2022**
 - IPR2022-01083
- **Ericsson v. Apple** Brazil Patent BRPI 9811639-8 filed 2022
- **Greenthread v. Intel** Case 6:22-cv-105-ADA filed 12/2022
- **ITC Daedalus v. TSMC** Case 337-TA-1336 filed 10/2022
- **Oasis Tooling, Inc v. GlobalFoundries** C.A. No. 22-312-CJB 3/2023
- **BiTMICRO LLC v. Intel** IPR2023-00781 filed 3/31/2023
- **Longitude Licensing LTD, v Western Digital Technologies, Inc.** IPR2023-01286 filed 8/08/2023
- **TSMC IPRs**
 - 8,907,425, IPR2025-01090 filed 6/06/2025
 - 8,198,686, IPR2025-01091 filed 6/06/2025
- **Samsung IPRs**
 - 7,772,673, IPR2024-01403 filed 9/2024
 - 9,245,826 IPR 2024-01405 filed 9/2024
 - 7,745,886 IPR2024-01402 filed 9/2024
 - 9,147,609, IPR 2024-01404 filed
- **Inpria v. Lam 2025**
- **ITC Longitude Licensing Ltd. and Marlin Semiconductor v. TSMC** Case 337-TA-1443 filed 02/2025

- MOSAID Vs Intel 2026

Honors and Awards

- ❑ 100 granted US patents, 60 publications and 1 book on semiconductors
- ❑ Elected to **IEEE Fellow** in 2006 for Advancing Fabrication of nanoscale MOSFET
- ❑ Elected to Intel Fellow in 2003
- ❑ Thompson and co-workers were the first to publish at the International Electron Device Meeting (IEDM) in 2002 on a 90nm logic technology which introduced high levels of strain for significant mobility enhancement using SiGe
- ❑ 2008 SEMI award North American Recipient
- ❑ 2014 TiE 50: Top Startup (SuVolta)
- ❑ 2012 ACE Award IEEE Spectrum: IEEE Emerging Technology: SuVolta
- ❑ 2011 EuroAsia IC Industry Award for start-up
- ❑ 2008 Semi Award North America (for Develop of Strained Silicon Transistors)

D. Publications/Patents

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- [3] S. E. Thompson et al., "Digital Circuits Having Improved Transistors, and Methods therefor" US Patent 10,250,257, 2019.
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EXHIBIT K

Jeff (Shaojie) Wang, Ph.D.

Education

Ph.D. Materials Science and Engineering, Lehigh University
M.S. Physics, Indiana University
B.S. Physics, University of Science and Technology of China

Experience

2012 – Present

**Sr. Principal Scientist,
TEM Group**

Eurofins Nanolab Technologies

- Interface with new customers in various industries in the development of new materials test plan; Responsible for coordinating different characterization groups to provide comprehensive analysis to meet customer requirements
- Perform the special customer-requested / demonstration projects, and responsible for resolving customer complaints.
- Conduct analytical services using various TEM techniques (TEM/STEM/PED/EDS/EELS/EFTEM) to support a wide range of customers in semiconductor and consumer electronics.
- Incorporate new techniques into business application and turn them into revenue. Established and expanded the business application of the new Precession Electron Diffraction (PED) system for the crystal grain orientation mapping and strain analysis business.
- Perform new tools and attachments upgrades evaluation to support company's strategic decision.
- Responsible for tool hardware and software issues and new engineer training
- Develop DigitalMicrograph scripts and Python codes for data acquisition/processing/report automation.

2012 – 2012

**Sr. Process Engineer, TD-Contamination
Free Manufacturing Group**

GlobalFoundries

- Developed Bright Field Inspection (BFI), SEM Review, Automatic Process Inspection (API) and VisEdge recipes; Conducted BFI tool and recipe comparison.
- Conducted Defect Source Analysis (DSA).

2006 – 2012

**Research Assistant,
Glass Research Group**

Lehigh University

- Fabricated nano-macroporous bioactive glass (BG) scaffolds by combining the sol-gel process with polymer sponge replication or spinodal decomposition method; Tailored nanopore size and surface area by solvent exchange or sintering method; studied the role of nanoporosity on their performance.
- Utilized SEM and mercury intrusion porosimetry to characterize BG scaffold macrostructures, and TEM and nitrogen adsorption porosimetry to characterize nanostructures.
- Investigated in vitro cell and in vivo animal responses to BG scaffolds using PCM and confocal microscope.

2003 – 2006

Research Assistant, Neuroscience

Indiana University

- Conducted research on the computational properties of neural networks via modeling and experimentation; Extracted significant neuronal signal using multivariate statistical analysis method; Modeled learning behavior in neural networks.

2001 – 2003

Research Assistant, Thin Films

University of Science and Technology of China

- Executed research on the properties of Zinc Oxide thin films grown on Silicon substrates bombarded with electron beam; proposed model explaining preferential growth results.

Skills

- Materials Characterization: TEM (FIB/TEM/STEM/EDS/EELS/PED/EFTEM/TEM-Tomography) • SEM (EDS/EBSD) • XPS • FTIR • UV-VIS • DSC • TGA • XRD • SIMS • Viscometer • Laser scattering particle analyzer • BET porosimetry • Hg porosimetry • Confocal / Fluorescence microscopy
- Materials Processing: Sol-gel process • Melt-quench glass making • Cellular ceramics fabrication by polymer sponge replication / spinodal decomposition • Thin film growth by sputtering • Attrition / Ball milling • Slurry mixing and coating
- Computer Skills: DigitalMicrograph scripting • Python coding • ImageJ • Adobe Photoshop • Multivariate Statistical Analysis • MATLAB • Mathematica • C • Fortran • Origin
- Related Courses/Training: Engineering Materials & Processes • Mechanical behavior of Materials • Structure & Properties of Materials • Thermodynamics & Kinetics • Electron microscopy & microanalysis • Advanced TEM • Lehigh Microscopy School • Advanced surface chemistry • Semiconductor physics • Thin film physics

Patents & Publications

- S. J. Wang and H. Jain, "Fabrication of porous glass bioscaffolds by sol-gel and polymer sponge methods", U.S. Patent No. 9,321,675 (2016).
- S. J. Wang, M. M. Falk, H. Jain, et al. "Nanoporosity Significantly Enhances the Biological performance of Engineered Glass Tissue Scaffolds", Tissue Eng. Part A. 19, 1632 (2013).
- R. Y. Golovchak, S. J. Wang, A. Ingram, and H. Jain, "Positron annihilation lifetime spectroscopy of nano-macroporous bioactive glasses", J. Mater. Res., 27, 2561 (2012).
- S. J. Wang, M. M. Falk, H. Jain, et al. "Evaluation of 3D nano-macro porous bioactive glass scaffold for hard tissue engineering", J. Mater. Sci.: Mater. Med. 22, 1195 (2011).
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